Finite Observer Consensus as a Reconstruction Principle: Normal Forms, the Standard Model Lie Type, and a Route to the Einstein Field Equation

Bernhard Mueller[1,2,*] Alexander Osika[3] Mario Poneder Kai Xue
Ben Cassie Peter Nguyen Jinwook Kim[4] David Matscheko
Jonathan Hill William T. Glynn Maarten Antonie Visser Kale Arnav Anirudha
Brieuc de La Fournière

September 9, 2026

Paper release: r2040

Author affiliations:
[1] Pragma Research Inc.
[2] Information Physics Institute
[3] EtherWorks
[4] Oraclizer Labs
[*]Corresponding author: bernhard@floatingpragma.ai.

Abstract

Observer Patch Holography proposes that public physics is reconstructed by finite self-reading patches whose records agree across typed overlaps. Twelve-port carriers federate over spherical support. A signed-record response branch has an exact rank-three quotient, and the celestial S^2 supports Lorentz-frame kinematics. Convergent repair selects protected public normal forms, while authenticated provenance generates finite causal posets with canonical source height. Across their stated branches, the constructions give density-operator quantum probability, the exact Tsirelson bound and no-cloning, the Standard Model gauge Lie type, an anomaly-free fifteen-state generation, and the conditional Einstein equation up to a cosmological constant.

A supplied charged-scalar/Maxwell action provides a calculable matter model. For positive mass squared, nonnegative quartic coupling, smooth Neumann references and Ritz-projected initial data, its real zero-current sector has first-order spatial trajectory convergence to nonlinear quartic-wave solutions on uniform conforming refinements. On the fixed volume mesh, the full reduced interacting kinetic metric is complete. Its declared Laplace–Beltrami Hamiltonian has a unique self-adjoint closure, a neutral Hilbert sector and explicit normalized states.

Conservative source-record populations with a declared local read law converge to flat $(3+1)$ causal order and volume; interval counts recover proper-time ratios. A declared scalar action on the same source addresses has classical and Fock-space continuum comparisons, with certified compact-detector errors smaller than the induced signal throughout a nonzero time window.

Keywords: foundations of physics; Observer Patch Holography; finite reconstruction; causal sets; quantum records; Einstein field equation; gauge structure.

1 Introduction and claim boundary

Quantum probability, Lorentzian kinematics, gravitational dynamics, and gauge structure enter standard descriptions through distinct mathematical inputs [19, 20, 21, 58, 67]. Observer Patch Holography asks whether they can instead arise as different public readouts of one finite observer architecture.

The primitive object is an observer patch: a finite subsystem with local state, an observable boundary, readback, durable records, and authorized repair moves. At each regulator, local carriers realize observer patches and federate through typed overlaps. Each carrier has twelve primitive boundary ports with oriented icosahedral incidence. The screen is a federation-level construction: its overlap nerve maps with degree one to a refinement tower with oriented S^2 support. Local carrier boundary, federation nerve, and spherical support are distinct parts of the construction.

OPH identifies an objective fact with a record that survives every authorized comparison. A patch begins with a private, repairable record. Patches compare their readbacks on shared boundaries and accept only repairs that preserve protected data. When the named repair relation terminates, satisfies the local diamond, and is complete, every maximal schedule from a fixed initial quotient state reaches the same consistent normal form. The resulting protected record is idempotent, schedule independent, and insensitive to presentation data hidden by the quotient. OPH interprets this public normal form as emergent objective reality. Authenticated read-after-write provenance generates a finite informational poset. Its canonical source height is zero at roots and one plus the maximum direct-parent height otherwise, and equals the attained longest authenticated-parent-chain length. Independently, the exact rank-three positive source quotient defines the four-dimensional ambient target carrier $\mathbb{R} \oplus V_{\mathrm{src}}$, with a Lorentz form and S^2 null-direction space. A positive `timeScale` times source height enters only as the temporal coordinate of an event placement in that carrier. A finite enumeration along one source axis and scale larger than its diameter constructs an injective forward-causal placement for every finite log. This auxiliary placement is neither source-selected nor order-reflecting. This is not an intrinsic dimension estimator for the finite poset. It removes a freely supplied rank-four chart from the finite precursor. A source-selected spatial event map and physical causal identification require compatible refinement. The direct causal-set route also requires faithful order embedding, calibrated count density, independent dimension tests, manifoldlikeness and topology. Smooth Einstein promotion further requires either same-family tensor-curvature convergence or the independent continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic.

Observer freedom is constrained before a state is selected. A1 fixes the accessible hardware, including local port incidence, overlap interfaces, the complete reversible-response tangent, and the spherical support. A2 fixes agreement on shared operational meaning and requires proper carrier rechartings to be implemented from inside that response. A3 selects the least informative state within the feasible set left by those constraints. Observers use authorized local operations and coordinate descriptions within fixed quotient-visible hardware and protocol. Authorized operations and rechartings preserve its invariants. On the signed-record response branch, the normalized repair response has an intrinsic rank-three Gram quotient. Its public carrier-position readout leaves observers exactly three independent directions to recover, and authorized rechartings preserve their rank. A faithful physical-position realization must preserve this dimension.

Within the closure hypothesis, the twelve-port carrier and S^2 support are fixed facts of one self-consistent universe. Internal observers encounter them through completed records. Within their record history, they can infer the hardware and protocol and later construct a realization that returns the same invariant readouts. The complete structure has no beginning at which the architecture is chosen. Inhabiting the closed solution, discovering its specification, and constructing it are compatible relations within one fixed point. Closure requires fidelity among the inferred specification, its later implementation, and the inhabited world.

The equation $T(\mathfrak{U}) = \mathfrak{U}$ states the universe-level closure hypothesis. It extends beyond the finite normal-form theorem and places the simulating description, the recovered architecture, and the inhabited system in one mathematical object. The structure carries two internal orders. Observer time orders completed records, while well-founded repair descent proves termination. No global clock orders these components. Reconstruction followed by construction in observer record order is therefore compatible with timeless closure and sends no information toward earlier records.

The physical thesis is that, in one inhabited common realization, these protected public readouts are the observables of our universe. The supporting results have different logical types. Finite theorems recover density-operator probability, Lüders conditioning, the Tsirelson ceiling, no-cloning, the intrinsic rank-three position carrier, the local Standard Model gauge Lie type, and an anomaly-free fifteen-state generation. Reconstruction contracts give Lorentz-frame kinematics and the conditional Einstein equation when their stated geometric and physical antecedents hold. Closure diagnostics land close to the measured fine-structure and cosmological constant coordinates, and a frozen prospective propagation branch fixes its first anisotropic harmonic at angular rank six. The full physical thesis requires one common realized tower carrying all of these readouts, including four-dimensional event geometry.

The case for OPH is the reach of one observer architecture. Protected normal forms define shared records; complete reversible response with endogenous transport fixes the local gauge Lie type; and one supplied matter action supports both real-sector continuum trajectories and an interacting Hilbert space. These results connect objects that ordinary physical models introduce separately. Their common physical realization is the empirical content of the proposed unification.

Each link has an explicit input and an independently inspectable output. The closure hypothesis supplies the timeless self-referential reading. A1 fixes patch hardware, A2 fixes agreement and endogenous transport, and A3 selects the least-informative compatible state. Named repair and propagation laws then support finite theorems and certified computations. Termination, confluence and a physical kinetic operator require their own premises. Physical realization maps identify records, currents, clocks, stress and scales with laboratory or spacetime objects. Spherical support gives Lorentz symmetry; realized null balance, Ward conservation, the Bianchi identity and an independent scale give the Einstein implication.

Table 2 separates these implications from diagnostics and prospective tests. Companion papers give the longer proofs and executable evidence [1, 5, 6, 7, 8, 11, 10, 2]. Each step is identified as a Lean theorem, exact executable certificate, conditional paper argument, numerical receipt, or explicit unproved premise.

Table 1 gives the seven epistemic claim classes used throughout.

Throughout, "source" means computed from the declared finite source specification, including every explicitly named source law, with no measured physical input. "Source-only" marks a quantity whose entire dependency cone is of that kind. A physical realization map is not source-only merely because the finite branch calculation is target-clean.

The distinction is mathematical: a finite implication and a physical realization of its antecedents are different claims. Machine-readable receipts record that distinction and reject a physical verdict when a required map is absent.

Boundary 1.1. This paper claims no completed derivation of particle masses, no source-only determination of the fine-structure constant, no completed cosmology, and no claim that formal verification establishes physical truth. Quantitative surfaces that consume measured values are labeled diagnostics. The primitive twelve-port propagation branch is a frozen prospective conditional

Table 1: Epistemic claim classes used in the manuscript.

Finite theorem	A deductive result proved from displayed premises on a stated finite domain. An exhaustive computation counts as part of the proof only with a specified domain and a soundness argument.
Certified finite computation	An executable witness whose inputs, controls, and output are fixed by a reproducible certificate; execution alone does not promote its interpretation to a theorem.
Reconstruction implication	An exact implication whose antecedents include displayed geometric or physical premises.
Unsupplied realization map	A required map from a finite object to a physical event, current, field, clock, scale, or continuum structure for which no construction is asserted.
Diagnostic	A certified computation compared against measured values that it consumes; never counted as a prediction.
Prospectively fixed conditional test	A diagnostic with a custody-bound target definition, kill band, precision floor, and decision rule fixed before comparison.
Frozen prospective branch prediction	A source calculation and decision rule fixed before any eligible comparison, conditional on a named physical branch. Failure rejects that branch; its scope extends to the whole framework only if the branch is proved forced and exclusive.

prediction. Section 13 states its decision rule and the physical premises that set the scope of any verdict.

Definition 1.2 (Physical realization map)**.** A physical realization map assigns finite public records, event data, currents, modular generators, and scales to physical records, spacetime events, laboratory currents, clocks, and units. It must preserve the algebraic operations, overlap restrictions, causal order, normalizations, and refinement relations consumed by the conclusion in question. A resemblance of spectra, dimensions, or symmetry names is insufficient.

Throughout this paper, a *contract* is a named, typed theorem interface. It records the inputs, domains, maps, normalization conventions, premises, outputs, and realization obligations for one implication. The term keeps the full dependency surface visible and introduces no additional physical law. Thus a transaction contract, matter contract, or Einstein contract names the specific interface consumed by that result.

1.1 Principal contributions

Finite records connect to geometry, matter and dynamics. Table 2 states hypotheses and conclusions. The normal-form layer gives a canonical partial normalizer, sharp stability and refinement bounds, and complexity barriers. A transactional local diamond yields schedule-independent public records; on the adaptive branch, finite stabilization is source-derived, while fairness, completeness and confluence supply normality and uniqueness. Bounded waste controls attempt cost, and certified read-from provenance supplies a finite causal order. The carrier layer returns an intrinsic rank-three position readback and forces the gauge Lie type, with a conditional $\mathbb{Z}_6$ matter image and exhaustive exterior selection. The quantum-record layer derives Born weights from finite effect valuations and carries Lüders conditioning, Tsirelson's bound and the copying obstruction on a declared algebra-state pair.

The dynamical layer includes a controlled nonlinear continuum limit of a supplied charged-scalar/Maxwell action. Its real, zero-current sector converges at first order in the energy norm on a uniform conforming spatial refinement, under smooth-solution and initial-projection hypotheses. The fixed-mesh interacting kinetic metric is complete; its declared Laplace–Beltrami Hamiltonian has a unique self-adjoint closure and a nonzero neutral Hilbert space. Bounded self-reading software patches execute the same charged model, and a configuration-based duration recovers its action time on nonturning solutions. These are mathematical and computational realizations on supplied geometry and couplings, with physical identification stated separately. The geometry layer constructs a $(1+3)$ target from an independent real axis and the rank-three carrier, then isolates the common physical refinement, stress, entropy and scale premises under which null balance yields the Einstein field equation.

Operational quantum reconstructions [19, 20, 21], relational and modular accounts of physical facts and time [47, 48], thermodynamic routes to gravitational dynamics [58, 59], and compact-group reconstruction [66, 67] are the principal comparison points. Born and Lüders rules, Tsirelson's inequality, conformal–Lorentz isomorphisms, Tannaka reconstruction, and the familiar exterior representation of one generation are used as established ingredients. The originality asserted here is limited to the observer-fiber normalizer with sharp refinement control, the transaction contract yielding the local diamond, the A1–A2 forcing theorem with its fixed-space exclusion, the conditional four-band witness and exhaustive exterior selection, the nine-direction null design with exact determinant and decoder constants, the frozen coefficient-linked primitive-port prediction and the carrier-class band that contains it, the finite effect-valuation representation with its sharp- and unsharp-web countermodels and exact phase-lift boundary, the converse ladder characterizing the generator of a supplied continuous flow, the kernel-and-reference-fixed log-transition action with its real-Legendre non-identifiability theorem, the two independent declared-family selections of the internal-energy slope at nonzero ledger, with the hopping-charge gauge and conservation equivalence, the oriented-face compact-family discriminator with its complete invariant-metric phase diagram, the F/G invariant-form dimension-drop theorem, the exact local finite face-curvature action with its scalar Coulomb sector, the slot-interface Tsirelson saturation with its record-diagonal boundary, the unique deep-regime enclosed-mass law with its self-contained Cauchy lemma, the capacity-to-equation-of-state map with its displayed scoring asymmetry, and the identification of the continuum Yang–Mills gap with the limiting repair gap under named reconstruction certificates.

1.2 Reproduced structures

Table 2 lists the structures of observed physics that the reconstruction returns, with the class of each result in the vocabulary of Table 1 and the section that proves or states it. The table carries no claim beyond the class in its third column: a finite theorem is a theorem about the declared finite objects, a reconstruction implication carries the physical antecedents displayed with it, and a diagnostic consumes measured input and predicts nothing.

Table 2: Reproduced structures of observed physics, the result that returns each one, its claim class, and its location.

Structure	What the reconstruction returns	Class	Where
Probability weights on outcomes	Every nonnegative normalized valuation additive on coexisting effects is the trace against a unique density operator, in every finite dimension, with no continuity axiom	finite theorem	§4

Table 2 continued

Structure	**What the reconstruction returns**	**Class**	**Where**
State update after an outcome	A declared Lüders instrument has the exact channel and conditioning identities; a swap-twisted instrument has the same effects, so the effect table does not select the update; invariance on the outcome-certain states does	finite theorem	§4
The correlation ceiling $2\sqrt{2}$	Tsirelson bound on commuting record subalgebras, attained exactly on a declared slot-split entangled witness, with record-diagonal states confined to 2	finite theorem	§4
Impossibility of copying an unknown state	One ancilla-free isometry copying two sharp states from a common blank forces their overlap to 0 or 1	finite theorem	§4
A three-dimensional position carrier	The normalized twelve-port repair response has an intrinsic rank-three Gram quotient; the signed record module $\mathbb{Z}^6$ and the seam current module D_6 are dense in one abstract three-dimensional Euclidean carrier	finite theorem	§2
Lorentz kinematics and the velocity space	$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ with H^3 the space of future unit timelike directions, on the supplied support-visible cap–BW branch	conditional scaling theorem	§6
A source-derived $1+3$ causal/Lorentz precursor	The independent real axis and exact rank-three source Gram quotient define a four-dimensional ambient target carrier with Lorentz inertia $(1,3)$ and the exact S^2/future-null-ray boundary. The authenticated parent order canonically supplies the exact longest authenticated-parent-chain height, used only as the temporal coordinate of an event placement after multiplication by a positive `timeScale`. This is not an intrinsic dimension estimator for the finite event poset. A physical manifold on the direct event-order route requires the separately stated source-causal continuum certificate	finite theorem plus reconstruction implication	§6
The relativistic mass shell	$E^2 = m^2 + \lvert p\rvert^2$ for the declared-mass momentum, rest characterization $E = m$ exactly at vanishing spatial coordinates, invariance in every oriented chart, and the massless branch	finite theorem	§9
Internal energy in inertia	For a nonzero ledger, frame covariance selects slope one in the declared momentum family; independently, refinement removes the per-step term and the declared proper-time principle selects slope one in the length-action family, giving inertial coefficient $m + E$; the ledger is additive over product references with an exact binding defect	finite theorem	§9

Table 2 continued

Structure	**What the reconstruction returns**	**Class**	**Where**
A nonlinear matter continuum	First-order trajectory convergence in $H^1 \times L^2$ for the real quartic-wave sector of the supplied charged action, on uniform conforming refinements with smooth Neumann data and Ritz initialization	conditional analytic theorem	§9.6
The arrow of time	Every finite stochastic repair kernel preserving a supplied faithful reference contracts relative entropy; the deterministic strict-descent normalizer need not, as an exact counterexample shows	conditional finite theorem	§11
The four laws of thermodynamics	One exact conditional package with five typed source and physical receipts, including the transition-side reading of the third axiom; it is not a source-realized theorem of the three axioms	conditional finite theorem package	§11
Linear transport coefficients	Symmetric positive-semidefinite finite Green–Kubo matrix with an exact cutoff remainder	finite theorem	§11
Entropy that scales with a boundary	Invariant collar decomposition with the one-sided edge term $\sum_\alpha p_\alpha \log d_\alpha$	finite theorem	§5
The inverse-square law	Spherically symmetric flux at constant shell charge falls as r^{-2}, the exponent being the carrier dimension minus one, with the dimension supplied by the completion theorem	finite theorem	§7
Coulomb and Maxwell structure	Symmetric rational Green matrix with a unique minimal-energy neutral solution, and a local twenty-face curvature action gauge invariant exactly for conserved seam currents	finite theorem	§9
The Einstein field equation with Λ	Tensor completion of a realized null balance under Ward conservation, the Bianchi identity, and one scale identification	reconstruction implication	§7
The Standard Model gauge Lie type	Complete A1 response and endogenous A2 transport force $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ on twelve ports, with no ambient gauge group among the premises	finite theorem	§8
Common kernel and maximal faithful image on the declared matter module	Common $\mathbb{Z}_6$ kernel on the declared matter tensors, so their maximal faithful image is $S(U(3) \times U(2))$; this does not select the physical global form	conditional finite algebraic theorem	§8
One anomaly-free generation of fifteen states	Exhaustive scan of the declared exterior menu leaves exactly one charge-conjugate pair of rank-fifteen chiral anomaly-free masks	finite theorem	§8
Absence of the minimal grand-unified exchange channel	The product adjoint $(8,1,0) \oplus (1,3,0) \oplus (1,1,0)$ contains no X/Y generator	finite theorem	§8

Table 2 continued

Structure	What the reconstruction returns	Class	Where
Massless photon, gluon, and graviton	Three conditional structural statements on their declared branches, each with a zero hard mass parameter: two photon modes, two transverse modes per Yang–Mills generator (sixteen on $SU(3)$), and two transverse–traceless gravitational modes	reconstruction implication	§8.4
A positive Yang–Mills mass gap	The continuum Hamiltonian gap identified with the limiting repair gap only on the support-visible compact-gauge branch carrying the finite-rate, transfer, and reconstruction certificates	reconstruction implication	§8.5
The Koide relation among charged leptons	Positive-chamber circulant identity with $Q = 2/3$ exactly at $\rho/a = 1/\sqrt{2}$, plus a custody-bound tau window	finite theorem; fixed test	§10
Charged-lepton mass magnitudes	Outward-rounded enclosures of logarithmic half-width 1.732% containing the measured triple	diagnostic	§10
The fine-structure constant	Interval-certified root of the screen-grain closure equation returning 137.035660 against the measured 137.035999	diagnostic	§10.1
The cosmological constant	Capacity closure candidates 3.2921×10^{122} and 3.3001×10^{122} against the coordinate 3.3129×10^{122} read back from the weighted Planck chain	diagnostic	§10.1
Flat rotation curves and the baryonic Tully–Fisher relation	Given declared deep-regime scale covariance and quadrature source composition, the unique law is $M_A(r) = r\sqrt{M_b a_0/G}$ and gives $v^4 = GM_b a_0$; neither premise nor a_0 is source-derived	conditional characterization theorem	§11.2
A gravitating non-luminous sector	Modular-charge sourcing classifies any geometric excess over the luminous term, with the anomalous collar energy as a candidate carrier	reconstruction implication	§11.2
The dark-energy equation of state	Under the declared inverse-density reading $\rho_{\mathrm{DE}} = \kappa/N$, fixed record capacity gives the exact point $(w_0, w_a) = (-1, 0)$, and monotone nondecreasing capacity gives $w \geq -1$ at every epoch	reconstruction implication	§11.2
The de Sitter transfer sign	Exact capacity identity with $\Delta S_{\mathrm{gen}}^{\max} = \log(1-f) < 0$ and a one-sided boundary maximum	finite theorem	§10
Isotropy of propagation to angular rank five	The first symmetry-allowed anisotropic rank of the carrier class is six, with two frozen rays inside one exact band	finite theorem; frozen prediction	§10.6
An interacting quantum state space	Complete reduced kinetic metric on $\mathbb{R}^{30} \times \mathbb{C}^{13}$, unique self-adjoint closure for the declared Laplace–Beltrami Hamiltonian, and invariant neutral states	conditional analytic theorem	§9.6

Every row carries the premises stated at its own theorem, and the physical realization maps that separate a finite implication from a claim about nature are collected in Section 11.

1.3 The standing unsupplied maps

One physical realization must carry compatible events, fields, action, states and readouts across refinement. Its five obligations are:

(M1) Position and scale. An identification of the intrinsic carrier completion with physical position, together with a physical length, cofinal overlap gluing between carriers, and a refinement limit.

(M2) Clock and energy calibration. An identification of the internal step with a laboratory clock and of the modular ledger with a laboratory energy, including conversion into measured units. Configuration-derived model duration does not select laboratory time or its units.

(M3) Laboratory current. An identification of the finite port response with a physical gauge current, together with a physical matter action and a continuum operator limit.

(M4) One common physical tower. A single physical refinement family realizing the event, modular, stress, entropy, continuum, vacuum, and scale premises together, rather than several independently declared islands.

(M5) A source-selected action. An action produced by the source rather than declared, with its field sector, coupled interaction, relative normalizations, and operational readout.

A finite theorem that consumes none of these stands on its own domain. A statement about nature consumes some of them, and each boundary block below names which. No result in this paper discharges any of the five.

1.4 The three axioms

The model uses the following three mathematical postulates.

A1 (carrier federation and spherical support). In plain language, the model assigns local state, readback, records, repair moves, and checkpoints to a federation of observer-patch carriers. Each local carrier has an oriented twelve-port boundary. Typed overlaps form the federation nerve, which maps to a separate oriented spherical support. Concisely, for each regulator r there is a typed object

$$\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$$

whose carrier boundary $K_{r,i} = (P_{r,i}, E_{r,i}, F_{r,i}, o_{r,i})$ has $|P| = 12$, $|E| = 30$, $|F| = 20$, and the oriented icosahedral incidence relations. The bridge $b_r : N_r \to S_r$ carries a designated oriented two-cycle to the fundamental class of the spherical support, and the typed maps commute with refinement. For each complete carrier, put $V_{r,i} = \mathbb{R}^{P_{r,i}}$. A1 supplies a finite-dimensional unitary space $H_{r,i}$ and an injective real-linear response derivative

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i})$$

whose image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators. The primitive port probes span $V_{r,i}$, their ordered compositions generate every accepted infinitesimal reversible response, and no public response direction is omitted. The pairing

$$\langle v, w \rangle_D = -\operatorname{Tr}(D_{r,i}(v) D_{r,i}(w))$$

is positive definite. The response data and their completeness commute with refinement, and

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0$$

is the connected response group. A1 constrains the finite carrier, federation, operational interfaces, spherical-support tower, and complete public reversible-response tangent. It does not imply semantic agreement, repair termination, confluence, a particular matrix response, an inverse-port law, a compact Lie type, a global group, or a laboratory identification [4].

A2 (observer agreement). In plain language, observers assign the same operational meaning to accepted data on their shared boundary. Every proper recharting of a complete carrier is implemented by their own reversible overlap response. Formally, the interpretation functor

$$\mathcal{J}_r : \mathsf{Data}_r \longrightarrow \mathsf{Meaning}_r$$

is natural under every visible restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map. On an overlap O,

$$\mathcal{J}_O(\mathrm{res}_{P\to O}\, d_P) = \mathcal{J}_O(\tau_{Q\to P}\, \mathrm{res}_{Q\to O}\, d_Q).$$

The accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, let $\mathrm{Hol}_r(o)$ be its closed overlap paths. A2 requires the port projection

$$\Pi_{r,i} : \mathrm{Hol}_r(o) \longrightarrow \mathrm{Aut}^+(K_{r,i})$$

to be surjective. For every proper carrier automorphism $a \in \mathrm{Aut}^+(K_{r,i})$, A2 supplies a closed path γ_a and a projective unitary implementer $[U_a]$ satisfying

$$\Pi_{r,i}(\gamma_a) = a, \qquad \mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a\cdot v).$$

The implementer is endogenous to the same response:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \qquad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

The centralizer factor acts trivially on $\mathfrak{g}_{r,i}$. A2 constrains operational meaning on accepted shared data and makes proper carrier rechartings inner on the complete port-response algebra. It does not imply global state extension, termination, confluence, unique normal forms, durable records, a global compact group, or a laboratory current [4].

A3 (maximum randomness under agreement constraints). In plain language, the selected state is least informative relative to the declared reference after every observer-visible constraint has been imposed. At finite regulator r, let $\mathcal{K}_r$ be the nonempty convex set of compatible local state families and let

$$\mathcal{D}_r(\rho\|\tau_r) = \sum_{P\in\mathcal{G}_r} w_{r,P} D(\rho_{r,P}\|\tau_{r,P}), \qquad w_{r,P} > 0.$$

The observer cover is state-determining on $\mathcal{K}_r$, and

$$\rho_r = \arg\min_{\rho\in\mathcal{K}_r} \mathcal{D}_r(\rho\|\tau_r).$$

A3 constrains state selection inside one A1-fixed feasible space. It does not select a field list, repair law, response map, particle multiplicity, or continuum limit [4].

None of the axioms contains a gauge group, a particle list, a spacetime dimension, or a recovery law. A1 does fix a two-dimensional spherical support and hence the Lorentz kinematic type used below. The results below state which structures follow from these axioms, which require added premises, and which realization maps are consumed as premises.

1.5 The self-referential closure hypothesis

The closure hypothesis is the overarching structural premise. The universe is modeled as a timeless, closed, self-referential mathematical structure in which the simulating description and the simulated system are one object, and the physical configuration is a fixed point $T(\mathfrak{U}) = \mathfrak{U}$ of a universe-level closure operator. No primitive or preferred global clock is assumed or derived. Repair order is a partial order on commits, an observer history is an ordered chain of completed records, and that order is invariant under arbitrary strictly increasing regradings, so affine time and proper time require separate calibration and event-geometry receipts.

A structure closed under its own description contains subsystems that read, record, compare, and reconstruct it. Within their emergent subjective time, observers assemble a consistent account of the structure, recover its simulator architecture, and construct that recovered specification, whose constructed hardware and protocol generate the same class of observer records from which it was inferred. Globally there is no earlier reconstruction and later construction; that ordering belongs to the observers' internal records. The closure condition requires the recovered specification, its constructed realization, and the inhabited structure to return the same invariant quantities, and it supplies no finite repair law, current derivative, physical kinetic operator, or existence and uniqueness proof for the cosmic fixed point.

No temporal paradox arises. The equation $T(\mathfrak{U}) = \mathfrak{U}$ relates parts of one timeless object, with no global time coordinate along which information travels backward. Repair descent is a well-founded order on authorized state updates used to prove termination and compare schedules, and it is neither an observer's experienced time nor a universe-level clock, so observers can recover the specification and later construct it in their own record order while both stages belong to the same global fixed point. On this reading A1–A3 provide the finite consistency conditions under which the loop closes, and the A2 endogeneity clause is the hypothesis's local expression inside the axioms: every proper recharting is implemented from inside the response. The necessity claims proved here concern the quotient-visible specification recovered inside the inhabited fixed point, within the stated premise sets.

Two finite results give the hypothesis exact content on declared branches. The packet-quotient closure theorem proves, on the declared finite consensus branch, that the closure map is an affine idempotent whose fixed points are exactly the packets supported on consensus normal forms, with no habitat-level extension [7]. The universe-level record-closure equation $N = \log M_0(\mathfrak{U}_N)$ is posed exactly, and the all-rung arithmetic proves nonidentifiability for the bounded completion class defined by base agreement, positivity, and the carrier bound; universal all-rung membership in a complete A1–A3 capacity-source contract is unproved, so direct N is not evaluable on that incomplete antecedent [9]. A positive result requires a complete antecedent and a proof of one physical zero.

The hypothesis also fixes how the declared carrier is read. The twelve-port carrier belongs to the fixed-point solution inhabited by the observers, and A1 is their recovered description of its quotient-visible specification. The hardware present in the structure, the specification inferred from public records, and the realization constructed later in observer record order must return the same invariant quantities under closure. This internal fidelity is the hypothesis's necessity claim. The mathematical theorems consume the specification as a premise; universe-level closure remains the separately stated hypothesis. Its quantitative candidates, the screen-grain equation and the capacity coordinate, are developed in Section 10.1, where existence, uniqueness, and stability of the physical fixed point are separate determinacy tests.

1.6 How the constraints work together

Physical structure is selected at the intersections of constraints. The carrier incidence fixes a finite response space, its symmetry action, and its invariant subspaces, and that geometry by itself admits many equivariant response laws. A1 restricts the menu to faithful, complete, compact, commutator-closed reversible responses, and A2 requires every proper recharting visible through overlap agreement to act internally on that same response. For the twelve-port carrier the resulting one-dimensional fixed space excludes the sole centreless compact alternative and fixes the local Standard Model gauge algebra, so geometry, reversible dynamics, and agreement are all used in that conclusion.

Repair dynamics occupy a distinct part of the architecture. A named repair law produces a schedule-independent public record only when its termination and local-diamond premises hold. The gauge theorem imports no such law: it consumes the complete response clause of A1 and the endogenous transport clause of A2 directly, and a simulator that claims to realize those clauses owes an implementation-level check that adds no premise to the implication. The same discipline applies to measures. Carrier counting and algebraic traces supply finite candidate measures, agreement and repair determine whether they descend to the public quotient, and a physical realization map must then preserve them as spacetime or laboratory measures. This dependency discipline separates forced structure from formal resemblance.

1.7 The theorem chain

The results form branches with different premises:

$$\begin{array}{rcl}
\text{finite carrier and convergent repair} & \longrightarrow & \text{consensus normal form},\\
\begin{array}{r}\text{signed port records}\\ \text{normalized repair response}\end{array} & \longrightarrow & \begin{array}{c}\text{abstract continuous local}\\ \text{three-dimensional Euclidean carrier}\end{array},\\
\text{algebra-state reading of the completed record} & \longrightarrow & \text{finite quantum record identities},\\
\text{A1 response and A2 transport} & \longrightarrow & \text{Standard Model local Lie type},\\
\text{displayed matter representation} & \longrightarrow & \text{conditional } \mathbb{Z}_6 \text{ matter image},\\
\text{oriented spherical support} & \longrightarrow & \text{Lorentz observer-velocity kinematics},\\
\text{authenticated read-after-write provenance} & \longrightarrow & \begin{array}{c}\text{finite informational causal poset}\\ \text{and canonical source height}\end{array},\\
\begin{array}{r}\text{source causal poset, rank-three carrier, real axis,}\\ \text{celestial } S^2\text{, and finite placement contract}\end{array} & \longrightarrow & \begin{array}{c}\text{populated finite (1+3) Lorentzian precursor}\\ \text{with exact order–cone embedding}\end{array},\\
\begin{array}{r}\text{physical event/link attachment and compatible refinement,}\\ \text{calibrated count–volume density, independent dimension,}\\ \text{manifoldlikeness, topology, and uniqueness}\end{array} & \Longrightarrow & \begin{array}{c}\text{effective four-dimensional time-oriented}\\ \text{Lorentzian event manifold}\end{array},\\
\begin{array}{r}\text{same-family tensor-curvature convergence or}\\ \text{continuum small-ball/null balance, with}\\ \text{stress, Ward/Bianchi, coupling, and scale}\end{array} & \Longrightarrow & G_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab},\\
\begin{array}{r}\text{deep-regime scale covariance}\\ \text{and quadrature composition}\end{array} & \Longrightarrow & v^4 = G M_b a_0,\\
\begin{array}{r}\text{repair relaxation and}\\ \text{reconstruction certificates}\end{array} & \Longrightarrow & \text{positive Yang–Mills mass gap.}
\end{array}$$

The normal-form, carrier, quantum-record, gauge-Lie-type, Lorentz-kinematic, finite-poset, and finite-precursor rows are finite or exact results under their displayed premises; the matter-image row is a conditional finite algebraic theorem. The effective-manifold, Einstein, deep-gravity, and mass-gap rows are reconstruction implications. The spacetime branch is exact through the populated finite $(1+3)$ precursor and its finitely checkable order-embedding constructor, and conditional at the physical continuum and curvature promotions. A common modular–stress–entropy tower is the proposed route to the null-balance antecedent of the Einstein row; no physical realization of that tower is supplied. Sections 2 through 10 develop these branches. Section 11 records the physical interpretation maps. Section 12 describes the verification stack. Section 13 states the falsification architecture and the empirical decision rule. Section 14 compares the program with neighboring approaches. The conclusion states the program's full ambition, a derivation of the constants and structural properties of nature from consistency requirements alone.

2 The primitive finite architecture

Definition 2.1 (Observer patch)**.** An observer patch is a finite object with local state, an observable boundary map, durable records, readback, and a set of authorized repair moves. A patch sees a fragment of the world: its own state and the boundary data of its authorized overlaps.

Definition 2.2 (Overlap and repair)**.** For patches x_i, x_j with an authorized overlap $e = (i, j)$, both induced boundary records must agree for the pair to hold public data. A repair move changes local state while preserving protected readout. A configuration is a normal form when no authorized repair applies.

The carrier realization used throughout is the twelve-port icosahedral architecture of A1: each carrier's boundary packet is combinatorially the oriented icosahedron boundary (P, E, F, o) with $|P| = 12$, $|E| = 30$, $|F| = 20$; carriers federate through typed seam algebras with coherent triple-overlap cocycles into a nerve carrying a degree-one bridge to the oriented spherical support. Under the closure hypothesis, this is the quotient-visible source architecture recovered by inhabitants of the fixed point. Every result below displays its dependence on that architecture.

Two exact features of this architecture enter the later constructions. First, the proper rotation group of the icosahedral packet is the alternating group A_5, with sixty rotations acting on the twelve ports; the real port-coefficient space has the character decomposition

$$P_{12} \;\cong_{A_5}\; \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3'} \oplus \mathbf{5}, \tag{1}$$

an exact multiplicity-free decomposition obtained by standard finite-group character theory [53] and checked by certified projectors. The two triplet bands are machine-checked at the level of the projectors themselves. On the six-dimensional golden sector the spectral projectors have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$ with eigenvalues $2\varphi^2$ and $2/\varphi^2$, each is the entrywise Galois conjugate of the other, their characters take the values $3, -1, 0$ off the order-five classes and φ or $1 - \varphi$ on the twenty-four order-five elements, and a Burnside span certificate makes each piece an irreducible real representation of the listed automorphism group. Irreducibility survives extension of scalars: the nine listed automorphisms span the complex endomorphism algebra with the same inverse certificates, the two complexified pieces are inequivalent because the traces on the order-five row are $1 - \varphi$ and φ, and the real endomorphism algebra of each piece is the real scalars, so both are of real type (`Lean/Screen/GoldenSectorComplexIrreducibility.lean`). Second, record keeping is integer valued: a write appends $+1$, a retraction appends -1, and readback sums atomic events per port; a conservative repair transfers one unit across a seam whenever the oriented mismatch has magnitude at least two. With $V(N) = \sum_i N_i^2$, each such repair strictly decreases V by $2(d - 1)$ for mismatch $d \geq 2$, so repair terminates by a finite theorem. Divisibility of total load by twelve is a necessary consensus condition and is not sufficient by itself. For the certificate's source-generated full-pile packet, an explicit eighteen-move schedule reaches consensus.

The same incidence fixes the dimension of the carrier's own position readback, which is the reason the number three occurs in this framework without being declared.

Theorem 2.3 (Intrinsic three-dimensional carrier completion)**.** *Let A be the twelve-port adjacency, let $T = I - (5I - A)/60$ be the uniform scalar repair mean, and take the complete equally weighted centered probe census. The normalized infinite-response kernel converges to the intrinsic Gram $G = 4P_{\text{slow}}$, of rank three. The twelve integer port loads have antipodal-odd quotient $\mathbb{Z}^{12}/\{x : x_p = x_{-p}\} \cong \mathbb{Z}^6$, and the thirty-seam boundary has image exactly the even-sum sublattice D_6, with one residual parity coset. Under the pullback of the response-selected Gram metric, both integer modules embed densely in one abstract continuous three-dimensional Euclidean carrier. The sixty proper carrier maps act faithfully and isometrically on the rank-three range and extend to that completion, and cumulative records act by exact internal translations.*

Boundary 2.4. The completion is taken after slow-response normalization and the infinite-step limit; completing at finite step count gives a different object, since the kernel has rank eleven on the centered space and rank six on the signed antipodal sector. The metric here is the response-selected Gram pullback and is distinct from the usual six-dimensional lattice metric on D_6. The nonlinear repair kernel does not descend through the signed quotient even though its conditional mean does, so working state and ordered history form a separate fiber. Thirteen finite covariance and isometry table identities used by the proper-action packet are discharged in Lean with `native_decide`; those subreceipts therefore trust the native compiler and runtime in addition to Lean's kernel and are not kernel-only proofs. Reading the completion as physical carrier position consumes the source readback and distinguishability premises; scale refinement, cofinal overlap gluing, physical space, physical scale, and field attachment are not constructed [1, 8].

2.1 Hardware and protocol: one substrate, typed dynamics, several readouts

The finite substrate and the operations acting on it are distinct parts of the model. Every physical conclusion requires a specified operation and its own realization map.

Under the closure hypothesis, the substrate is also the specification recovered by observers inside the model and constructed by them within their emergent subjective future. Its ports, seams, readback, records, and repair interfaces form the hardware side of the strange loop. The observations used to infer the specification and the constructed simulator that realizes it belong to the same globally closed structure.

The *substrate and reversible-response tangent* are the content of A1: a federation of finite carriers, each with the oriented twelve-port icosahedral boundary packet, joined through typed seam algebras with coherent triple overlaps into a nerve that bridges to the oriented spherical support, together with the complete reversible response space of each carrier. The companion microphysics paper specifies this substrate at the level of patch hardware, records, and synchronization interfaces [8]. This layer fixes what exists at each finite regulator: ports, seams, records, and response directions. It does not select a repair schedule, convergent repair law, or physical propagation operator.

The *agreement and state-selection protocol* combines A2 and A3 with named branch dynamics. Patches write and retract integer record events, read back port sums, compare induced records on shared boundaries, and interpret accepted data through the natural functor of A2. A named repair law moves oriented mismatches under quadratic descent; its termination and local-diamond premises are proved separately. A3 selects the residual state by constrained information projection after the agreement constraints are fixed. The companion consensus paper develops this layer as an asynchronous agreement protocol with explicit safety and liveness assumptions [7]. The protocol fixes what happens: which configurations count as repaired, which records become public, and which transports implement recharting.

The claim structure of the paper is that distinct readouts of this one protocol-on-hardware system yield the distinct pillars of observed physics. Repair to a fixed point under quadratic descent and the transactional diamond gives the schedule-independent public record of Section 3, and the algebra-state reading of that record surface gives the quantum identities of Section 4. Overlap transport around closed paths gives the central defects and edge-center entropy of Section 5. Authenticated semantic commits give the informational event carrier and order. Independently, the standard real axis, source rank three, and celestial/Lorentz cone data give its finite (1+3) ambient model; canonical source height enters only through event placement. A physically faithful manifoldlike refinement limit gives the conditional effective spacetime of Section 6, and modular flow with entropy stationarity on that same limit gives the conditional field equation of Section 7. Endogenous recharting of the complete port response gives the gauge type and conditional matter image of Section 8. Record readback, the modular ledger, and the realized path law give the position, energy, and action of Section 9. Capacity accounting and closure readback give the quantitative surfaces of Section 10, and homogeneous propagation on a selected carrier action gives the coefficient rays of Section 10.6. Table 2 indexes the resulting structures by physical name.

One feature of this pipeline carries the unification claim: the branches consume one substrate under typed operations, so the same completed records that define measurement feed the event geometry, and the same overlap transport that defines agreement feeds the gauge forcing. The universe claim of the program is the conjunction. A physical carrier that realizes the hardware and runs the protocol produces public records whose readouts include quantum probability, Lorentzian causal structure, Einstein dynamics, and the Standard Model gauge and matter skeleton, with the maps of Section 1.3 as additional premises.

3 Observable normal forms and finite consensus

A completed finite configuration x is a fixed point of the named repair-and-agreement map:

$$\mathcal{R}_{\mathrm{cons}}(x) = x, \tag{2}$$

where $\mathcal{R}_{\mathrm{cons}}$ is the composite finite dynamics. This map is distinct from the universe-level closure operator T of Section 1.5. Fixed-point existence, observable determination, and schedule independence are separate questions. The following framework isolates them [5].

3.1 Exact observable fibers

Definition 3.1 (Observable quotient system)**.** An observable quotient system is a tuple

$$\mathfrak{S} = (Q, C, \mathcal{B}, B),$$

where Q is a set of configurations, $C \subseteq Q$ is the consistent subset, $\mathcal{B}$ is a set of protected records, and $B : Q \to \mathcal{B}$ is the record map. For $b \in \mathcal{B}$, write $C_b = C \cap B^{-1}(b)$.

The fiber C_b is unrealizable, reconstructing, or ambiguous according as its cardinality is zero, one, or greater than one. This trichotomy avoids assigning a state to a record that has either no consistent extension or several observationally indistinguishable extensions.

Theorem 3.2 (Canonical observable normalizer)**.** *The following statements are equivalent:*

(i) $B|_C$ *is injective;*

(ii) there is a unique map $N : Q \to C \sqcup \{\perp\}$ that preserves B on its C-valued outputs, fixes every point of C, is constant on each B-fiber, and returns $\perp$ exactly on fibers with no consistent extension;

(iii) $B|_C : C \to B(C)$ is a bijection.

If $B(Q) \subseteq B(C)$, the map is a total idempotent retraction $Q \to C$.

Proof. Injectivity makes every nonempty C_b a singleton, which defines N and forces all its properties fiber by fiber. Conversely, if $c, c' \in C$ have $B(c) = B(c')$, the fixed-point and fiber clauses give $c = N(c) = N(c') = c'$. The third statement is the restriction of the first to the image $B(C)$. $\square$

This normalizer is defined by observable fibers, independent of a chosen repair schedule. A repair relation implements it only when the relation preserves B, reaches the consistent set, and satisfies an appropriate liveness condition. Confluence from one source does not by itself prove determination across distinct sources with the same protected record.

For a separately declared finite scheduler, protected observations admit four nested first-hit layers: nonempty consistent fiber, positive reach from some active source, almost-sure reach from every active source, and a single first-hit endpoint class modulo the declared silent equivalence. Their complementary cuts partition the empty-fiber, inaccessible, liveness, and selection obstructions, the layers instantiate the behavior-cut interface of this section exactly under explicit support–rewrite laws, and an exact finite-state morphism transports the whole profile. The four cuts compare coordinatewise, a product preorder rather than a scalar score. The definitions, machine-checked theorems, native fixtures, and countermodels are in the observable-normal-forms component paper [5]; the statement is conditional on the declared scheduler and morphism packet and supplies neither hitting-time rates nor an implementation-refinement result.

For the repository's canonical single-site repair, the attempt semantics is sharper. Every adaptive run is eventually constant because each nonstuttering attempt is an accepted repair and strictly lowers the mismatch count. Pathwise weak fairness makes the stable state a normal form, while work conservation gives a normal horizon no larger than the initial mismatch count. Under completeness and confluence, every such weak-fair run reaches the canonical raw repair endpoint and hence the same public fixed-point object. These results select no scheduler and provide no stochastic hitting law, rate, physical clock, or refinement theorem.

The attempt count has its own sharp capacity theorem. Every chosen-site invocation costs one unit, including equality stutter. At fixed initial mismatch one, a scheduler can delay the first genuine repair for any prescribed finite number of attempts and then normalize on the next attempt, so the mismatch rank alone gives no finite attempt horizon. If every reducible record instead sees a genuine change among the next $q + 1$ attempts, the bounded-waste upper horizon is $(q+1)$ times the initial mismatch count. An independent-defect family attains the work-conserving case exactly at $q = 0$, and the different-cardinality TwoCell source gives a second sharp instance. With a sufficient budget and the existing completeness and confluence premises, the same bound reaches the canonical raw and public fixed-point endpoint before budget exhaustion [5, 7]. This mathematical capacity is not a physical clock, rate, energy, bandwidth, fee, or hardware quota.

The precedence layer beneath these results is itself source-derived. Every quotient-visible register version carries the identifier of its last semantic writer, each commit certifies the register versions its acceptance genuinely depends on, and the informational order is the transitive closure of authenticated read-from parenthood. The `causalSupp` field is supplied semantic data: Lean requires it to lie inside the read set, while its soundness and completeness as the set of genuine dependencies are producer and independent-verifier obligations rather than kernel-derived dependence analysis. A direct edge requires the child to certify the resource, the parent to write it, the child's pre-commit snapshot to name that parent as writer, and the pre-commit value to equal the parent's post-commit value. A raw writer label alone is a negative control. The result is a strict order under a rank increasing along direct edges and is least among strict transitive extensions of those edges. The executed-history model derives such a rank from execution position, and any exact precedence adapter carries the generated order verbatim. Static mismatch scores are writer-blind, so a standing residual determines no causal arrow. Under fresh-ID, duplicate-free, and visible-support hypotheses, a hidden commit changes no visible mismatch and supplies no outgoing direct raw or authenticated parent edge to a visibly supported child; incoming edges into the hidden commit are not excluded. For one adjacent independent swap, the authenticated relation and its transitive closure are preserved under fresh-ID and duplicate-free hypotheses on both executions. The broader equivalence-generated swap chain preserves only the raw writer-citation relation; it does not prove arbitrary labeled-payload invariance. Executor stutters are removed before commits are formed. The generated relation is an authenticated read-from dependence suborder and analogue of happened-before in distributed computing [43]. Record or program order contributes only when a later commit genuinely consumes the earlier record.

Lean packages reflexive generated ancestry as a partial order and proves that every interval is finite at each finite cutoff. Thus, as an abstract finite poset, this object satisfies the order-theoretic causal-set axioms. Standard causal set theory starts from a locally finite order whose relation is interpreted as proto-causality [30, 31]; finiteness gives local finiteness here, but read-from dependence is informational rather than physical. A standard faithful-embedding interpretation additionally requires an order embedding, manifoldlikeness and dimension, and the approximately Poisson count-density condition that completes a causal-set faithful embedding [32, 33, 34]. On the continuum side, causal order determines conformal geometry only under the relevant distinguishing or causality hypotheses [41, 42]; event number or an equivalent volume datum is needed for the conformal factor. The standard Lorentz-invariant continuum approximation uses Poisson sprinkling rather than a regular lattice [40]. The OPH commit log supplies no calibrated map from event counts to spacetime volume. No faithful embedding, four-dimensional manifoldlikeness, volume law, continuum existence, or Hauptvermutung uniqueness theorem follows from the finite provenance order. This exact statement is exposed as `generatedBeforeEq_isPartialOrder` and `finiteCausalSetAxioms`; it is deliberately an informational theorem.

Theorem 3.3 (Finite causal-order compiler)**.** *Let E be finite and let $R \subset E \times E$ be a decidable irreflexive transitive relation. There is an abstract authenticated semantic log L_R on E whose direct semantic-parent relation and generated strict precedence are both exactly R:*

$$L_R.\mathsf{ParentEdge}(e, f) \iff R(e, f), \qquad L_R.\mathsf{GeneratedBefore}(e, f) \iff R(e, f).$$

No rank is supplied: the compiler uses the number of strict predecessors, which increases strictly along R. In this generic construction every supplied strict predecessor is an authenticated semantic parent: it represents R itself rather than its Hasse reduction alone.

Proof. Give each event its own register. Event e writes that register, and event f certifies it exactly when $R(e, f)$; unit values make value continuity automatic while the writer stamp authenticates the event identity. Irreflexivity and transitivity make predecessor sets grow strictly along R, so predecessor cardinality supplies the rank. Authenticated parenthood is R by construction, and transitivity makes its generated closure R. $\square$

This theorem shows that chainlike captured histories are not forced by the authenticated-provenance grammar. Every finite causal set can be represented exactly as an abstract semantic log, and Theorem 6.2 then gives it a separated one-way realization in the exact source (1+3) carrier. The relation R is supplied to the compiler. Its per-event snapshots are not proved to arise from one threaded executor, and no OPH dynamics, physical event identification, order embedding with a calibrated density law (and hence no causal-set faithful embedding), or manifold limit is selected by this expressivity result (`Lean/ObserverPatchHolography/Provenance/FiniteCausetCompiler.lean`).

The finite OPH construction does not leave the ambient carrier's dimension and signature to a chart receipt. Authenticated semantic provenance fixes the event carrier and generated order. From its parent relation, Lean computes root height zero and one plus the maximum direct-parent height otherwise, proving strict increase on parents and generated ancestry. Independently, the exact source Gram quotient is a positive rank-three carrier. Its direct sum with a real axis,

$$W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}, \qquad Q(t,x) = t^2 - g_{\mathrm{src}}(x,x),$$

has real dimension four and Lorentz inertia $(1,3)$. Source-unit directions map exactly to future-null rays, so the source S^2, the celestial S^2, and the null boundary are the same algebraic direction type. This removes the free event order, supplied ordinal placement coordinate, rank-four chart, and fitted signature from the finite precursor (`Lean/Geometry/SourceDerivedSpacetimeCarrier.lean`).

Every finite log has a separated one-way realization in this carrier without a placement-existence premise. Choose a finite enumeration $i : E \simeq \mathrm{Fin}(n)$, place $i(e)$ along one fixed source-spatial axis, and take height scale $n+1$. The enumeration separates events, its spatial diameter is smaller than the time increment across every parent edge, and generated ancestry enters the future cone. Distinct events at the same source height are spacelike. This construction is auxiliary and enumeration-dependent; it supplies neither order reflection nor source-selected physical coordinates.

For any supplied spatial readback $x(e)$ and positive height scale τ, the placement $F(e) = (\tau h_L(e), x(e))$ sends every parent edge, and hence every generated ancestry, into the future cone when the source-metric edge speed bound holds. If the converse cone/order implication holds, exact order–cone equivalence follows. Antisymmetry of authenticated ancestry then forces F to be injective, so the placement is an exact finite order embedding and every source-poset interval is exactly the corresponding cone interval on the placed event image. Composing this placement with the proved source-carrier equivalence constructs a separated one-chart finite causal-chart interface. The all-pairs converse need not be checked as a black box. Lean derives it from two finite geometric conditions: the spatial readback separates events within every equal-height layer, and every increasing-height pair unsupported by ancestry is strictly spacelike. These conditions remain supplied for a physical candidate; provenance does not select the readback. That global finite chart is not an open manifold atlas. The formal diamond witness is non-chain, its branches are spacelike, and every parent edge is null. Optional affine frame transports supply covariance after placement; the ambient target-carrier dimension comes from the direct sum, while causal order comes from the semantic log. Neither fixes the intrinsic dimension of the finite poset. The canonical height is ordinal rather than an operational clock, and the spatial readback is not derived generically.

Exact-to-conditional composition. The strict causal-set route requires one source-selected family. It must identify physical events and links, make order and cone agree exactly in both directions, and cover the source S^2 densely and isotropically. It must also calibrate $\#I/\rho \to \mathrm{Vol}(I)$, refine compatibly, pass independent dimension and manifoldlikeness tests, recover stable thickened-antichain topology or homology, and converge to a unique distinguishing Lorentzian limit. Causal order then fixes the conformal geometry and the calibrated measure fixes scale; neither conclusion follows from finite cardinality alone.

On the same finite event type, a minimal source-direction Einstein-shape theorem asks for nine fixed algebraic source-direction balances, supplied symmetric geometry and stress fields, four discrete step maps, Ward/Bianchi identities, and step connectedness. Exact null tomography derives balance on every null vector, the pointwise metric ambiguity, its constancy, and

$$\mathsf{geometry}_{ab} = \kappa\, \mathsf{stress}_{ab} + \Lambda \eta_{ab}.$$

The minimal theorem contains no normalization, vacuum calibration, Newton constant, or universal source field. No provenance edge selects a tensor field, step, or balance. A physical Einstein reading further requires either same-family discrete tensor-curvature convergence or the separately premised continuum small-ball/null-balance identification, together with same-source stress, Ward/Bianchi data, coupling and scale calibration, and controlled remainders on the same physical continuum family. The Benincasa–Dowker and dimension-dependent causal-set operators give concrete curvature tests for that limit [38, 39].

A declared tensor scalar action on prepared golden addresses supports classical waves, Fock states and bounded smeared detectors. Its field edges fit inside the causal read radius. A Gram-corrected isometry and resolvent estimates control the full frequency residual, including off-subspace leakage. Fixed-band errors vanish as source gaps shrink.

A compact coherent preparation in $y_1 \le 0.45$ and a compact detector in $y_1 \ge 0.55$ resolve a response on the actual golden addresses. For $q = 233$ and model time $0.95 \le s \le 1$, outward-rational certificates give

$$|P_{q,c}(s) - P_c(s)| < 0.050055, \qquad |P_{q,c}(s) - \tfrac{1}{2}| > 0.137754.$$

The total error includes both localization tails and is smaller than the induced signal throughout the window. The spacetime paper proves the comparison with the massive Dirichlet continuum field; independent arithmetic replays the certificate. Axes, boundary, protected addresses, action, quantization and model time are declared. Authentication of the field evolution and its identification with the count clock remain separate.

The input-independent compiler separates immutable inputs from its generated federation. Generated formula nodes form one federation independent of input, while the initial record carries immutable input ports. Every genuine canonical node repair lowers a dependency-order rank, every attempt run stabilizes, and pathwise weak fairness makes the stable record a consensus with the correct output. Same-input normal endpoints agree on that output through the observation-determined normal-form interface. The historical weak patch-frame relation is not strengthened: even its one-member execution can remain fairly stuttering forever.

The execution classification sharpens this mathematical layer. Pathwise weak fairness on one fixed attempt run is equivalent to eventual stable consensus; the reverse is vacuous on the stable consensus tail and gives no recurrence. Member and register-site recurrence coincide because generated output registers are unique, but recurrence is strictly stronger than tail fairness. A state-blind round-robin function is recurrent and bounded-waste and reaches the correct output. Its first n attempts are exactly the compiler's emission-order sweep, the order in which it emits nodes, so it reaches stable consensus and the correct output in a linear emitted-node horizon. The actual compiler's single-consumer structure bounds all genuine accepted paths by $n(n+1)/2$, while an emitted nested-NAND family has a consensus path satisfying $8m + 1 = n^2 + 8n$. This is an order-sharp $\Theta(n^2)$ result in the same node-count metric, not equality of finite constants. Well-formedness alone admits fewer than 2^n accepted steps and an explicit fanout chain attains $2^n - 1$, so the single-consumer structure is load-bearing for the quadratic bound. Weak fairness alone has no uniform attempt horizon; bounded-waste q gives the conditional horizon $(q+1)n(n+1)/2$, with round robin inhabiting $q = n-1$; the direct one-cycle theorem is the sharper bound for that scheduler. A neutral ranked-attempt theorem is reused by both this fixed route and the adaptive route through distinct adapters and actual Tower receivers. None of these mathematical attempts is a physical computer, scheduler source, clock, rate, energy, bandwidth, fee, hardware resource, full raw-state uniqueness, or finite `PublicWorld` [5, 7].

3.2 Approximate outputs, refinement, and cost

Three further results control the normalizer where exactness fails, and their proofs are in the component paper [5]. Let (Q, d_Q) be finite, $(\mathcal{B}, d_{\mathcal{B}})$ metric, and $C = \Phi^{-1}(0)$ nonempty for a nonnegative consistency residual Φ, with $\eta_\Phi(t) = \max\{\mathrm{dist}_Q(x, C) : \Phi(x) \le t\}$ turning residual error into distance from consistency and $\omega_B(r) = \max\{d_Q(c, c') : c, c' \in C,\ d_{\mathcal{B}}(Bc, Bc') \le r\}$ measuring how well protected observations distinguish consistent outputs. For L_B-Lipschitz B, any two outputs with $\Phi(x) \le \delta_x$, $\Phi(y) \le \delta_y$, and $d_{\mathcal{B}}(Bx, By) \le \varepsilon$ satisfy

$$d_Q(x, y) \le \eta_\Phi(\delta_x) + \eta_\Phi(\delta_y) + \omega_B(\varepsilon + L_B[\eta_\Phi(\delta_x) + \eta_\Phi(\delta_y)])\,, \tag{3}$$

which is exact in the ω_B term at zero residual, with the coefficient one on each residual-distance term optimal in the class of finite metric systems. Along a refinement tower with restriction maps $\rho_{m,n}$ and level normalizers N_n, the naturality defect accumulates as $d_n(\rho_{m,n} N_m q, N_n \rho_{m,n} q) \le \sum_{j=n}^{m-1} K_{j,n} a_j$ with $K_{j,n} = \mathrm{Lip}(\rho_{j,n})$ and a_j the one-step defect, so nonexpansive restrictions with a summable defect sequence give asymptotic naturality of the normalizers. Canonicity carries no promise of cheapness: for succinct Boolean boundary systems, deciding whether a protected record has a consistent extension is NP-complete, deciding whether $B_S|_C$ is injective is coNP-complete even when all but one variable are observed, and deciding whether every protected collar value has some consistent interior extension, equivalently whether an unrestricted total strong repair exists, is Π_2^P-complete.

3.3 Repair realization on the twelve-port carrier

For a prepared batch of support-closed transactions, join τ and σ in the conflict graph when

$$W_\tau \cap (R_\sigma \cup W_\sigma) \neq \varnothing \quad \text{or} \quad W_\sigma \cap (R_\tau \cup W_\tau) \neq \varnothing.$$

The conflict components below are obtained by iterating the following finite closure: compute semantic support to a fixed point inside every aggregate, rebuild the conflict graph on the expanded read and write sets, merge every connected component, and repeat until the partition and all aggregate supports are unchanged.

Proposition 3.4 (Transactional local diamond)**.** *Let an accepted repair be an aggregate transaction with read set R_τ, write set W_τ, a read snapshot, and a snapshot-determined payload. Assume:*

(i) every read set contains the semantic dependency closure of its write set, and every affected acceptance functional is revalidated at commit;

(ii) the terminal closure above exists, and each of its connected conflict components has one coherent canonical aggregate payload;

(iii) a prepared component whose snapshot remains valid survives commits of independent components; and

(iv) accepted commits preserve the protected record and strictly decrease the integer descent functional.

Then every one-step quotient peak

$$t \longleftarrow s \longrightarrow u$$

has a quotient state v with $t \longrightarrow v \longleftarrow u$.

Proof. Two different first steps cannot originate in one conflict component, since that component has a unique aggregate payload, so they originate in distinct components of the terminal closure. Rebuilding the graph after every aggregate support expansion makes their final read and write sets disjoint in the sense $W_\tau \cap (R_\sigma \cup W_\sigma) = W_\sigma \cap (R_\tau \cup W_\tau) = \varnothing$, so neither commit changes the other's read snapshot, and semantic dependency closure ensures neither changes an acceptance functional consumed by the other. Revalidation and the surviving component rule make both second commits legal, and with snapshot-determined payloads and disjoint writes the two application orders agree; protected-record preservation carries the equality to the physical quotient. □

Theorem 3.5 (Consensus normal form)**.** *Suppose the accepted repair relation satisfies Proposition 3.4, the exact quadratic descent of Section 2, and repair completeness, meaning that its normal forms are exactly the consistent states. Every maximal asynchronous repair schedule from one initial state then terminates at the same consistent quotient normal form. The induced global repair map is idempotent, fixes consistent states, and preserves the protected record.*

Proof. Quadratic descent makes the accepted relation terminating. Proposition 3.4 makes it locally confluent, so Newman's lemma [46] makes it confluent. A terminating confluent relation has a unique normal form below each source. Repair completeness puts that normal form in the consistent set. Applying the normalizer a second time does nothing, and protected-record preservation holds along every accepted edge. □

Boundary 3.6. Atomic commits, disjoint write sets, or termination alone do not imply the local diamond. Countermodels omitting semantic dependency closure, canonical aggregation, or the surviving component rule are retained with the finite receipt. The reference engine exhaustively checks its states, peaks, payload hashes, protected records, and descent comparisons. The theorem is finite and asserts no continuum limit.

The declared class also has gauge compatibility (repair commutes with local relabeling), idempotence of the completed dynamics, and refinement compatibility where proved. The public object is the protected quotient record, not an arbitrary intermediate configuration.

The public OPH-FPE run of this law uses the level-six cell rung of the geodesic icosahedral refinement tower: 81,920 exact twelve-port carriers, 983,040 port readings, the thirty seams of every carrier, and one declared gluing of ports to neighbouring cells, 2,580,480 seams in all. The repair primitive is the seam-mean retraction, symmetric between the two sides of a seam and conservative, so uniform scheduling gives the repair mean $T = I - L/60$ of Theorem 2.3 on every carrier. Under the integer law, sixteen shuffled schedules from one seeded load terminate in 51 to 79 sweeps at the balanced class, every unit transfer lowers $V = \sum_i N_i^2$ by exactly $2(d-1)$, no accepted move raises V, and all sixteen return one quotient hash equal to the component multiset fixed by the initial totals. Under the real seam-mean law the isolated carriers terminate in 26 or 27 sweeps at the component means within 10^{-9} with one terminal hash; on the glued federation the component mean is the unique fixed point by linearity and conservation, and a declared budget of 256 sweeps lowers the seam mismatch from 1.5×10^7 to 17. The seam-sum form rises on 23 to 41 percent of the non-wait moves, so the descent functional of Section 2 is V and the seam sum is a termination diagnostic only. At levels zero to three the same receipts hold in exact rational arithmetic, and the per-carrier normalized response kernel read from the run equals $4P_{\mathrm{slow}}$ to 10^{-12} on isolated carriers; under the declared gluing the slow-band share reads 0.91 at thirty steps and the glued-port pattern replaces it at large depth, a property of the gluing convention, which is the open cofinal-gluing item. A standalone verifier that imports no simulator code rebuilds the component multisets from the primitive arrays, replays one schedule from its seed, and recomputes every hash [2].

The self-reading component is a closure loop at carrier scale. From an event log alone, with no access to the generator, a fixed recovery returns the twelve ports, the thirty seams, the inverse-port pairing at distance three, the automorphism group of order 120 with its rotation subgroup of order 60, the seam-mean rule, the uniform schedule, the rank-three Gram, and the Lie split $1 + 3 + 8$; the carrier instantiated from that specification returns the same invariant vector on two further iterations. The tetrahedral and octahedral carriers recover their own specifications, the Gram rank alone being three for all three carriers, and the random endpoint-overwrite law recovers as nonconservative with a schedule-dependent terminal state and no fixed point. Neither receipt supplies a continuum or laboratory realization.

The architecture sits close to distributed agreement in the tradition of Lamport, Shostak, and Pease [44]: patches play the role of protocol nodes, overlap repair of a quorum vote, and the completed fixed point of the decided state. A component paper of the stack proves safety and liveness for a Byzantine-tolerant reading of repair under explicit quorum and partial-synchrony assumptions [7]; the impossibility boundary of Fischer, Lynch, and Paterson [45] is the reason those assumptions are named.

4 Finite record algebra and quantum identities

Once a compare, write, and verify slice has completed, its accessible events are represented in a finite-dimensional $*$-algebra with a normalized state. This algebra-state representation is an explicit input at this stage. The consensus theorem identifies which record is public; it does not derive the classification of finite-dimensional C^*-algebras or the trace pairing.

Theorem 4.1 (Finite public-record identities)**.** *Let P_E be the projector of a completed public event E and ρ the normalized state of the completed record surface. The event weight and nonzero-weight update*

$$\Pr(E) = \mathrm{Tr}(\rho P_E), \qquad \rho \,|\, E = \frac{P_E\, \rho\, P_E}{\mathrm{Tr}(\rho P_E)} \tag{4}$$

define a normalized probability measure on every orthogonal event partition and a normalized positive state on the post-event corner. For self-adjoint dichotomic observables A_0, A_1 and B_0, B_1 in commuting record subalgebras, with $A_i^2 = B_j^2 = I$, define

$$S_{\mathrm{CHSH}} = \mathrm{Tr}[\rho\{A_0(B_0 + B_1) + A_1(B_0 - B_1)\}]\,.$$

Then

$$|S_{\mathrm{CHSH}}| \le 2\sqrt{2}. \tag{5}$$

Proof. Positivity of ρ and $0 \le P_E \le I$ give $0 \le \mathrm{Tr}(\rho P_E) \le 1$. For a complete orthogonal partition (P_k), linearity and $\sum_k P_k = I$ give $\sum_k \mathrm{Tr}(\rho P_k) = 1$. If the event weight is nonzero, $P_E \rho P_E / \mathrm{Tr}(\rho P_E)$ is positive, normalized, and supported on $P_E \mathcal{A} P_E$.

For $\mathcal{B} = A_0(B_0 + B_1) + A_1(B_0 - B_1)$, the commuting-subalgebra calculation gives

$$\mathcal{B}^2 = 4I - [A_0, A_1][B_0, B_1].$$

The two commutator norms are at most two, so $\|\mathcal{B}^2\| \le 8$ and $\|\mathcal{B}\| \le 2\sqrt{2}$. Taking its expectation in ρ proves the stated bound. $\square$

The carrier supplies a candidate that reaches past the classical bound without reaching the quantum one. On the declared binary-icosahedral spinor branch the defining two-dimensional representation has a unique invariant line in its tensor square, and for the incidence-defined 120-row setting family carried by the twelve ports the invariant singlet gives $|S_{\mathrm{CHSH}}| = 1 + 3/\sqrt{5} = 2.3416407865\ldots$, every declared setting row having correlations $(-1/\sqrt{5}, -1/\sqrt{5}, -1/\sqrt{5}, +1)$. An independent exact verifier reconstructs the invariant multiplicity, all 720 covariance identities, the joint probabilities, the no-signalling marginals, and the complete 12^4 setting census, which contains 960 maximizers. The displayed family forms two proper rotation orbits and is selected by no carrier property, and the finite source packet supplies neither a completed two-wing record instrument nor source-selected settings, so this is an exact projective candidate rather than a physical Bell prediction.

Theorem 4.2 (Exact Tsirelson saturation and the diagonal boundary)**.** *The bound of Theorem 4.1 is attained exactly on the committed slot-split interface: the normalized maximally entangled state on the four-dimensional product carrier, with the two Pauli projections on the left slot and the two quarter-rotated projections on the right slot, gives $S_{\mathrm{CHSH}} = 2\sqrt{2}$, and the four events inhabit the declared finite subsystem split with the identity Kraus family. Conversely, every state diagonal in the record basis, paired with any four diagonal observables valued in* $[-1, 1]$*, obeys the classical bound* $|S_{\mathrm{CHSH}}| \le 2$*; in particular the counted correlation state of the committed source pair obeys it.*

The two halves bracket one committed record surface: the algebra-state representation attains the full quantum value on a declared entangled witness, while one fixed record-diagonal state with four jointly diagonal bounded readouts stays inside the classical bound. The Bell state and all four settings are real matrices, so the exact gap here is entanglement, non-diagonal coherence, and noncommutativity rather than the complex phase direction. The result covers no arbitrary context-dependent classical operational model, and the saturating state and events are declared library data rather than source-produced preparations.

The first identity is the Born rule on this finite surface [22]; the second is the declared Lüders conditioning instrument [23]. The projectors alone do not select that instrument: an explicit swap-twisted instrument has the same effects and different updates. Kraus form with induced effect P together with invariance on the states certain of P is equivalent to being the Lüders map, so certain-state invariance is what selects the instrument the effect table leaves open; the swap-conjugation map on two dimensions has Kraus form with induced effect the identity and fails that invariance, and the committed repeatability clause is strictly weaker and does not select (`Lean/EventAlgebra/InstrumentSelectionByCertainStates.lean`). Events, preparation, readback, and a source implementation are unbuilt. The third is the Tsirelson bound [25] for the Clauser–Horne–Shimony–Holt combination [24] at fixed cutoff. Compatibility structure for commuting events follows the standard finite operational pattern within the same library.

Two maps built from a partition of pairwise orthogonal projectors (P_i) with $\sum_i P_i = I$ organize conditioning on the same surface. The pinching $\mathcal{P}(X) = \sum_i P_i X P_i$ and the average $\mathcal{A}(X) = \sum_i \frac{\mathrm{Tr}(XP_i)}{\mathrm{Tr}(P_i)} P_i$ are positive, unital, trace-preserving idempotents, with a zero projector contributing zero. The exact range of $\mathcal{P}$ is the commutant of the partition and the exact range of $\mathcal{A}$ is the commutative span of the P_i, so $\mathcal{AP} = \mathcal{A} = \mathcal{PA}$; the average preserves every partition-event weight, and on a member of nonzero weight it commutes with Lüders conditioning, both orders returning the normalized projector. Each statement follows from orthogonality, the identity $\mathrm{Tr}(\mathcal{A}(X)P_i) = \mathrm{Tr}(XP_i)$, and the two range descriptions.

These are standard finite-matrix identities related to trace-preserving conditional expectations [26, 27]. The accompanying formal development contributes a checked arbitrary-partition interface, exact range and uniqueness statements, and adapters from projection events to the state-level Clauser–Horne–Shimony–Holt bound [6]. No priority claim is made for Born probability, Lüders conditioning, pinching, or Tsirelson's inequality as identities of the represented state.

The same surface carries the copying obstruction. After zero projectors are removed, trace coordinates and projector synthesis are mutual inverses, so the public record algebra is star-equivalent to complex functions on the active labels. Independently, if one ancilla-free linear isometry of the tensor square copies two sharp states from the same normalized blank, preservation of inner products forces their overlap z to satisfy $z^2 = z$, so z is zero or one and the overlap-one branch forces equality. Two distinct alternatives copied by one device are therefore orthogonal, which is the finite form of no-cloning on the completed record. The mixed-state no-broadcasting implication is exposed as an explicit adapter premise [6].

At the level of weights the Born form is derived rather than assumed. Every valuation on the effects of the finite surface that is nonnegative, normalized, and additive on coexisting effects equals $E \mapsto \mathrm{Tr}(\rho E)$ for a unique density operator ρ, in every finite dimension including two, where the projector-only Gleason theorem fails, and the derivation consumes no continuity axiom.

Finite webs do not by themselves supply its hypothesis, and the gap is located exactly. Sharp two-outcome webs admit the non-Born valuation $F_z(\mathbf{n}) = (1 + n_z^3)/2$; the exhibited unsharp trine excludes that response, yet every axis in the finite battery has $n_y = 0$, so the planar response $F_y(\mathbf{n}) = (1+n_y^3)/2$ agrees with the maximally mixed Born weight on the whole battery while staying non-affine off it. The source-attached real S_3 algebraic web is blind to the Pauli-Y direction in the same way, and its missing coordinate is exact: for its noncommuting projector candidates P, Q the commutator phase lift $I/2 - (2\sqrt{3}/3)i(QP - PQ)$ is the $+Y$ projector, and $P, Q, +Y$ separate every fixed-trace 2×2 matrix. Closure under real coarse graining and real Kraus pullbacks stays Y-blind, so a complex phase-sensitive effect is load-bearing for this tomography. What the missing step needs is a source-produced complex-tomographically-complete effect and instrument web together with an operational theorem giving full coexistent-effect additivity [6]. Outcome frequencies can validate such an instrument and cannot by themselves create the universal valuation law, and the algebraic phase lift is neither a source instrument nor an outcome receipt.

Boundary 4.3. This is a finite operational record interface. It does not reconstruct an arbitrary quantum state space or an interacting continuum quantum field theory. Its role is narrower: the declared algebra-state representation of the completed consensus record carries the finite probability, conditioning, expectation, and correlation operations consumed by later branches.

5 Overlap defects and edge-center entropy

Overlap agreement has a second layer beyond equality of scalar records. Transport maps can compose only up to a central multiplier. The resulting defect is a finite gluing invariant; removing triangle defects and obtaining endpoint-only transport are distinct operations.

Theorem 5.1 (Central defect strictification and residual holonomy)**.** *Let N_Σ be the nerve of a finite overlap cover. Suppose $U_{ji} = U_{ij}^{-1}$ and*

$$U_{ij}U_{jk} = z_{ijk}U_{ik}, \qquad z_{ijk} \in Z_\Sigma,$$

where the identified coefficient group Z_Σ is abelian and overlap transport acts trivially on it. Then:

(i) z is a Čech 2-cocycle,

$$z_{jkl}z_{ikl}^{-1}z_{ijl}z_{ijk}^{-1} = 1;$$

(ii) its class $[z] \in \check{H}^2(N_\Sigma, Z_\Sigma)$ is invariant under changes of local frame;

(iii) a central edge rephasing removes every triangle multiplier exactly when $[z] = 0$;

(iv) after such a strictification, endpoint-only transport on a charge block $\mathcal{C}_\alpha$ holds exactly when the represented holonomy of every closed loop is the identity on $\mathcal{C}_\alpha$.

Proof. Associativity compares $(U_{ij}U_{jk})U_{kl}$ with $U_{ij}(U_{jk}U_{kl})$ and gives the cocycle identity. A local frame change conjugates the multiplier, hence leaves a central element unchanged. An edge rephasing by a 1-cochain changes z by its Čech coboundary, proving the third statement. Once the edge maps form a strict 1-cocycle, the discrepancy between two paths with common endpoints is the holonomy of the closed loop obtained by composing one path with the inverse of the other. □

The theorem gives a precise finite meaning to gluing curvature. A vanishing triangle class does not imply trivial transport around noncontractible loops. If the center varies as a local coefficient system, the cocycle equation must use the transported, twisted Čech differential; that case lies outside the untwisted statement above.

5.1 Finite edge-center decomposition

Let a regulated collar be cut into left and right halves with common interface Σ. Suppose finite-dimensional Hilbert spaces $\widetilde{H}_L, \widetilde{H}_R$ carry diagonal unitary actions of a finite compact group G_Σ. Write

$$\widetilde{H}_L = \bigoplus_\alpha V_\alpha \otimes H_{L,\alpha}, \qquad \widetilde{H}_R = \bigoplus_\beta V_\beta^* \otimes H_{R,\beta}.$$

Here the V_α are pairwise inequivalent irreducible unitary representations of G_Σ and $d_\alpha = \dim V_\alpha$.

Theorem 5.2 (Invariant collar decomposition and one-sided edge entropy)**.** *The invariant collar space and its sector-preserving algebra have the exact forms*

$$H_C = (\widetilde{H}_L \otimes \widetilde{H}_R)^{G_\Sigma} \cong \bigoplus_\alpha H_{L,\alpha} \otimes H_{R,\alpha},$$

$$\mathcal{A}_C = \bigoplus_\alpha \mathcal{B}(H_{L,\alpha}) \otimes \mathcal{B}(H_{R,\alpha}), \qquad Z(\mathcal{A}_C) = \bigoplus_\alpha \mathbb{C}\, P_\alpha^C.$$

Let $\Omega_\alpha \in V_\alpha \otimes V_\alpha^*$ be the normalized invariant vector corresponding to $d_\alpha^{-1/2} I_{V_\alpha}$. A sector-preserving collar state has the form

$$\rho_C = \bigoplus_\alpha p_\alpha \rho_\alpha, \qquad \rho_\alpha \in \mathcal{B}(H_{L,\alpha} \otimes H_{R,\alpha}),$$

where $p_\alpha \geq 0$, $\sum_\alpha p_\alpha = 1$, and each block with $p_\alpha > 0$ satisfies $\rho_\alpha \geq 0$ and $\operatorname{Tr} \rho_\alpha = 1$. Embed it in $\widetilde{H}_L \otimes \widetilde{H}_R$ through Ω_α and trace out $\widetilde{H}_R$. If $\rho_{L,\alpha} = \operatorname{Tr}_{H_{R,\alpha}} \rho_\alpha$, the resulting left state is

$$\rho_L = \bigoplus_\alpha p_\alpha \frac{I_{V_\alpha}}{d_\alpha} \otimes \rho_{L,\alpha},$$

and its entropy splits as

$$S(\rho_L) = H(p) + \sum_\alpha p_\alpha S(\rho_{L,\alpha}) + \sum_\alpha p_\alpha \log d_\alpha.$$

The edge term is the expectation in ρ_L of the sector observable

$$Z_L = \sum_\alpha (\log d_\alpha) P^L_\alpha,$$

where P^L_α projects onto $V_\alpha \otimes H_{L,\alpha} \subset \widetilde{H}_L$.

Proof. Schur's lemma on $V_\alpha \otimes V^*_\beta$ gives a one-dimensional invariant space at $\alpha = \beta$ and zero otherwise, which yields the invariant direct sum and the center of its sector-preserving algebra. The partial trace of $|\Omega_\alpha\rangle\langle\Omega_\alpha|$ over V^*_α is I_{V_α}/d_α, giving the displayed ρ_L. Entropy of a block-diagonal state is the Shannon entropy of the block weights plus the mean block entropy, tensor-product additivity contributes $\log d_\alpha$ from each maximally mixed factor, and $\mathrm{Tr}(\rho_L Z_L) = \sum_\alpha p_\alpha \log d_\alpha$. $\square$

The theorem distinguishes the gauge-invariant collar state from its one-sided reduced state; the representation-dimension term belongs to the latter. Reading Z_L as an area observable requires a physical realization map. Exact quantum Markovity alone also does not identify an arbitrary state-dependent Markov decomposition with these preselected edge-center factors; alignment of the state with the collar decomposition is an additional premise. Related edge-mode and algebra-center decompositions occur in lattice gauge theory [63, 64].

6 Source-derived causal spacetime and the Einstein limit

6.1 How the finite results assemble

The slow-response band supplies an abstract continuous three-dimensional Euclidean carrier under the premises of Theorem 2.3. Its unit directions identify with the celestial S^2 and future-null rays; adjoining a real axis gives the $(1+3)$ Lorentz carrier. Authenticated records separately supply events, generated order and longest-parent-chain height. The finite placement theorems join these structures: the edge-speed bound preserves order, and converse support gives order reflection, separation and interval preservation. The local three-dimensional result therefore enters the spacetime composition. Direct causal-set reconstruction requires events filling this carrier with compatible causal and volume scaling. Effective field reconstruction instead controls local readouts and interventions on one source-selected family.

One common physical refinement family promotes this finite precursor to an effective spacetime when it supplies physical event and link identification, source-selected order/cone embeddings, calibrated count density, independent dimension and manifoldlikeness tests, stable topology, and a unique distinguishing Lorentzian limit. On that same family, either convergent tensor-curvature reconstruction or the independent continuum small-ball/null-balance identification, together with physical stress, Ward/Bianchi conservation, universal coupling, and scale calibration, yields the Einstein equation. Thus the populated finite causal precursor is an exact result, while the smooth curved $(3+1)$-dimensional manifold and its Einstein dynamics are the stated continuum composition theorem.

6.2 Exact kinematics of the spherical support

The oriented conformal spherical support carries

$$\mathrm{Conf}^+(S^2) \;\cong\; \mathrm{PSL}(2,\mathbb{C}) \;\cong\; \mathrm{SO}^+(3,1), \qquad H^3 \;\cong\; \frac{\mathrm{SO}^+(3,1)}{\mathrm{SO}(3)}, \qquad \dim H^3 = 3. \tag{6}$$

These are classical isomorphisms of conformal geometry, consumed as exact mathematics [52]. They identify the three-dimensional hyperbolic homogeneous space of future unit timelike directions. Interpreting those directions as physical observer velocities requires realized events, clocks, and local frames.

Boundary 6.1. Observer-velocity geometry and physical event localization are distinct objects. The finite route does not begin with a supplied event atlas, a freely declared rank-four chart, or a fitted signature. Authenticated semantic commits supply a finite event carrier and generated order; their exact longest authenticated-parent-chain height supplies one ordinal coordinate only when events are placed. Independently, a real axis and the source Gram quotient define the ambient carrier with three positive spatial directions. The scaled height coordinate is not an operational clock, and the source quotient does not define physical position. The velocity space H^3 and event localization remain typed as separate constructions.

Theorem 6.2 (Source-derived finite (1+3) causal precursor)**.** *Let L be an authenticated finite semantic-event log with event type E and generated strict order $\prec_L$. Let $h_L : E \to \mathbb{N}$ be its canonical source height: roots have height zero and every nonroot has one plus the maximum height of its authenticated direct parents. Then h_L strictly increases on every parent edge and on $\prec_L$. For every event, it is exactly the maximum length of an authenticated direct-parent chain ending at that event, and a chain attaining the maximum exists. Let V_{src} be the source Gram quotient with positive form g_{src} and real rank three. Define*

$$W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}, \qquad Q(t,x) = t^2 - g_{\mathrm{src}}(x,x).$$

Then:

(F1) W_{src} has real dimension $1+3=4$, and Q has one positive and three negative directions;

(F2) every source-unit direction u gives the future-null vector $(1,u)$, and the exact equivalences $\mathsf{SourceUnitDirection} \simeq S^2 \simeq \mathsf{FutureNullRay}$ *identify the boundary of this cone;*

(F3) for any spatial readback $x : E \to V_{\mathrm{src}}$ and scale $\tau > 0$, the placement $F(e) = (\tau h_L(e), x(e))$ sends every authenticated parent edge into the future cone whenever

$$g_{\mathrm{src}}(x(f) - x(e), x(f) - x(e)) \le \tau^2 \big(h_L(f) - h_L(e)\big)^2$$

on each parent edge $e \to f$; cone transitivity then sends every generated precedence $e \prec_L f$ into the future cone;

(F4) every finite log admits an explicit auxiliary placement of this form: for a finite enumeration $i : E \simeq \mathrm{Fin}(n)$, put the spatial coordinate $i(e)$ along one fixed source axis and take $\tau = n+1$. This placement is injective, sends reflexive generated ancestry into the future cone, and makes distinct same-height events spacelike; and

(F5) for a placement satisfying the edge-speed condition of (F3), the abstract converse-support condition follows from two concrete finite geometric checks: the spatial readback separates distinct events within each equal-height layer, and whenever $h_L(e) < h_L(f)$ but $e \not\prec_L f$,

$$\tau^2 \big(h_L(f) - h_L(e)\big)^2 < g_{\mathrm{src}}(x(f) - x(e), x(f) - x(e)).$$

Consequently every future-causal comparison between placed events is supported by reflexive authenticated ancestry, so $e \preceq_L f$ holds exactly when $F(f) - F(e)$ is future causal. This equivalence and antisymmetry force F to be injective, so it is an exact finite order embedding of the generated poset and preserves every causal interval exactly on its image. Composed with the source-carrier equivalence, it supplies a separated one-chart finite causal-chart interface.

The first two clauses construct the finite (1+3) carrier and signature from the independent real axis and rank-three source carrier; they do not assume a rank-four event chart. Canonical source height appears only in the placements of clauses (F3)–(F4). Clause (F4) removes existence of some separated forward-causal placement as an assumption, while its enumeration is not a source-selected spatial readback. The physical spatial placement, edge-speed condition, converse cone/order equivalence, the two concrete geometric checks, physical interpretation, and clock calibration remain independently checkable data.

Proof. Dimension is additive on the direct sum, and positivity of g_{src} gives the displayed inertia. The source-unit/null-ray statement is the exact algebraic equivalence. Induction over authenticated parents bounds every ending chain by source height and constructs an attaining chain, while the source-height theorem makes the height-coordinate increment positive. The edge inequality makes its Lorentz square nonnegative, hence each parent displacement is future causal; transitivity of the cone and of authenticated ancestry gives the closure statement. For the enumerated placement, spatial separation follows from the enumeration, its diameter is less than $n + 1$, and each parent height increment is at least one. Equal-height event differences are nonzero and purely spatial. For the concrete order-faithful-placement constructor, diagonal pairs follow from reflexivity, decreasing-height pairs are excluded by future orientation, generated pairs are forward causal by the preceding result, and the two geometric checks rule out every remaining equal- or increasing-height unsupported cone comparison. Thus converse support follows. If two event images coincide, cone reflexivity gives comparisons in both directions; converse support and antisymmetry of reflexive ancestry identify the events. Interval preservation follows by applying the two-way equivalence at both endpoints. These finite statements are formalized in the geometry proof stack; no limiting topology is used. □

Definition 6.3 (Source-causal continuum certificate)**.** A cofinal family of finite precursors from Theorem 6.2 carries a source-causal continuum certificate when one common family supplies all of the following:

(C1) a physical-event interpretation for the retained commits and an adequacy argument that authenticated support captures the relevant physical signal dependencies;

(C2) source-selected placements satisfying the edge-speed condition and either converse support directly or its equal-height-separation and incomparable-spacelikeness criterion, so that their induced cones agree with the generated order in both directions and event separation follows; local link directions that become dense and isotropic on the source S^2, with ordinal regradings kept separate from operational-clock calibration;

(C3) refinement maps preserving events, generated order, source directions, and placements, with a specified notion of convergence rather than a post-hoc subsequence;

(C4) a locally calibrated density law $\#I_n/\rho_n \to \mathrm{Vol}(I)$ on causal intervals, with public capacity N excluded unless a theorem identifies it with event number;

(C5) agreement across independent dimension and manifoldlikeness tests; interval-abundance and height/count scaling; and stable thickened-antichain topology or homology, compared with Poisson-sprinkled 3+1-dimensional Minkowski and appropriate curved controls [34, 35, 36, 37]; and

(C6) convergence of the local cone/overlap data to a distinguishing, time-oriented Lorentzian geometry, together with the stated uniqueness or Hauptvermutung-strength control for the admitted limit.

This is one certificate with independently falsifiable rows, not a relabeling of a supplied atlas as a derivation.

Theorem 6.4 (Effective manifold and smooth-curvature promotion)**.** *For a source-causal continuum certificate, causal order fixes the conformal Lorentzian geometry of the admitted distinguishing limit and the calibrated count measure fixes its conformal scale [30, 31, 41, 42]. The local model is (1+3), so a limit passing the independent dimension and topology rows is an effective four-dimensional time-oriented Lorentzian event manifold. Scalar-curvature convergence alone does not identify the Ricci or Einstein tensor. Suppose, on that same family, a causal-set d'Alembertian/scalar-curvature estimator converges as an independent curvature diagnostic and either an explicit discrete tensor-curvature reconstruction converges to the smooth Einstein tensor or the separately stated continuum small-ball/null-balance route identifies the limiting finite* `geometry` *tensor with G_{ab}. If the stress field and Ward/Bianchi data also converge and the finite Einstein-form identity has controlled remainder and calibrated coupling, its smooth limit is*

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}.$$

Dimension-dependent causal-set estimators test the scalar-curvature and d'Alembertian channels [38, 39]; they do not supply the required tensor identification, which is not inferred from a finite matrix named `geometry`*.*

The dimensional statements fit together on one carrier: the selected rank-three response quotient supplies the spatial metric, its unit directions are the source S^2, and adjoining a scalar axis gives the $(1+3)$ Lorentz module. The normalized seam average is also selected under the naturality and unique-minimizer hypotheses of Proposition 10.5. These determine spatial geometry and an internal spatial operator. The following construction realizes a causal order and count measure on source records; the subsequent covariance criterion restricts which operational orders can have that geometry.

Theorem 6.5 (A source-record causal and count limit)**.** *In the rank-three conservative repair carrier, let $\varphi = (1+\sqrt{5})/2$, $L = 2/\sqrt{\varphi+2}$ and $q = F_n$, $n \geq 4$. The population*

$$S_q = L\{b\varphi - \lfloor b\varphi \rfloor : 0 \leq b < q\}^3 \subset [0, L]^3$$

consists of q^3 distinct conservative source-record images. At each layer retain this population and let every next-layer event read all preceding-layer sites within $a_q = L/\sqrt{q}$, including itself. Assign model duration $\Delta_q = a_q/c$, $c > 0$. On bounded model-time windows, the posets have width q^3; each event has edge-count height equal to its layer index. As $q \to \infty$, their order converges away from null separation to the speed-c causal order. With $\rho_q = q^3/(L^3\Delta_q)$, normalized counts converge to $dt\, d^3x$ volume. For fixed interior timelike diamonds,

$$\frac{|I_q|}{\rho_q} \longrightarrow \mathrm{Vol}(I), \qquad \frac{2C_q}{|I_q|(|I_q|-1)} \longrightarrow \frac{1}{10},$$

where C_q counts strict causally ordered pairs. For vertical duration T, $\mathrm{Vol}(I) = \pi c^3 T^4/24$.

Proof. The exact integer section realizes each site by a conservative seam word. Consecutive Fibonacci approximants permute a uniform reference partition, with equal cell volume L^3/q^3 and site-assignment error $H_q = 2\sqrt{3}L/q$. For any h-covering population in a convex window, segment subdivision constructs a k-read path whenever $d \leq k(a-2h)$; every such path satisfies $d \leq ka$. The finite path construction, the layered order with its graph-distance characterization, width and height, and the flat constants $1/10$, $8/35$ and $1/2$ are formalized in Lean. Here $h_q \le H_q$ and $H_q/a_q \to 0$, so the two cones coalesce. The product cell assignment proves count convergence; null boundaries have zero volume. The continuum strict-pair integral is $\mathrm{Vol}(I)^2/20$, giving the ordering fraction. The spacetime derivation supplies the source records, explicit volume error bounds, and the full argument [10]. $\square$

Boundary 6.6. This construction supplies a flat effective order and volume for an explicit source-record population, neighbour read law and model clock. At finite regulator the family is produced by the carriers themselves: each carrier holds its record as port loads, reads its position through the rank-three response with the source metric recovered at scale exactly one, decides its neighbours from that readback, and the precedence generated from the carriers' authenticated reads equals the layered order at every level through $q = 34$. On the interior diamond the inverted Myrheim–Meyer dimension reads 4.15, 3.61, 4.10 and 4.09 at $q = 13, 21, 34, 55$ against 3.0 and 2.0 for the two- and one-dimensional control populations, the ordering fraction reads 0.093 at $q = 55$ against the flat $1/10$, the moving-tip diamond reads 0.106, and the count clock reads 1.91 against the model-time ratio 2; the spacetime derivation carries the full account [10]. Counts exclude auxiliary read instructions and seam operations. It does not select that law from the observer axioms, calibrate a physical clock, or identify its cone with a matter action. These are distinct requirements of a physical source-causal continuum certificate. Curved limits, topology stability and the common physical field realization require their stated additional hypotheses.

Theorem 6.7 (Operational selection of the Lorentz cone)**.** *Let E be the source Gram space and $C \subset \mathbb{R} \oplus E$ a nonzero closed convex pointed displacement cone. Suppose the irreducible proper icosahedral rotations preserve C, and boosts along one unit axis preserve it for every rapidity. Then C is the future or past cone of $t^2 - \|x\|^2$. A time orientation selects the future cone.*

Proof. The rotation orbit of the axis spans E. Conjugating its boost and taking finite-matrix product limits gives every boost. The finite rotation average sends (t, x) to $(t, 0)$. A spacelike vector could be boosted to both signs of time, contradicting pointedness. A nonzero causal vector supplies one time-axis ray; boosts and scaling generate its open timelike cone, and closedness supplies the null boundary. Pointedness excludes the opposite orientation. The spacetime derivation gives the product-limit proof [10]. $\square$

Here covariance must preserve possible influence on actual histories, interventions and clock readouts. An algebraic Lorentz action on source directions alone supplies no such operational transformation. The criterion selects cone shape under that hypothesis; it does not select read weights or a temporal field equation.

Corollary 6.8 (Proper-time ratios from record counts)**.** *In Theorem 6.5, let I_q, J_q approach interior timelike diamonds of proper durations $\tau_I, \tau_J > 0$. Then the observer readout*

$$R_q(I; J) = \left(\frac{|I_q|}{|J_q|}\right)^{1/4} \longrightarrow \frac{\tau_I}{\tau_J}.$$

Within the class of positive volume-only readouts additive on collinear inertial intervals, the fourth root is unique up to units.

Proof. Diamond volume is $\pi c^3 \tau^4/24$; the common count weight and constant cancel in the ratio. For uniqueness, $g(t) = F(\pi c^3 t^4/24)$ is positive and additive, hence increasing and linear by rational approximation. $\square$

The readout uses authenticated ancestry and counts, with a reference interval setting its unit. It reconstructs elapsed duration after the upper endpoint and requires access to the interval records. It supplies a geometric clock for the declared source family, independently of a Hamiltonian or oscillator. Identifying a running matter clock with this readout requires their common physical realization.

7 Einstein reconstruction

Theorem 7.1 (Null tomography and metric ambiguity)**.** *Let V be four-dimensional with Lorentz form η. Suppose a symmetric form X satisfies $X(k,k) = 0$ for every null vector k. Then $X = \phi\eta$ for a scalar ϕ. Null-null values therefore determine a symmetric tensor modulo the metric line. In an η-orthonormal basis with $\eta = \mathrm{diag}(-1, 1, 1, 1)$, set $s = \sqrt{3}/3$ and take the nine null vectors*

$$(1, \pm 1, 0, 0), \quad (1, 0, \pm 1, 0), \quad (1, 0, 0, \pm 1),$$
$$(1, s, s, s), \quad (1, s, s, -s), \quad (1, s, -s, s).$$

On the η-trace-free representative, use coordinates

$$x = (X_{00}, X_{01}, X_{02}, X_{03}, X_{11}, X_{12}, X_{13}, X_{22}, X_{23}), \qquad X_{33} = X_{00} - X_{11} - X_{22}.$$

The corresponding nine-charge design reconstructs the quotient

$$\mathrm{Sym}^2(V^*)/\mathbb{R}\eta.$$

Its determinant is $8192/27$. With the supremum norms on coordinates and charges, its exact decoder obeys

$$\|x\|_\infty \le (2 + \sqrt{3})\|q\|_\infty.$$

Proof. Choose coordinates with $\eta = \mathrm{diag}(-1, 1, 1, 1)$ and write every future null ray as $(1, n)$ with $|n| = 1$. The identity

$$X_{00} + 2X_{0i}n_i + X_{ij}n_i n_j = 0$$

holds on the unit sphere. Its odd part gives $X_{0i} = 0$. The degree-two spherical-harmonic part gives $X_{ij} = \phi\delta_{ij}$, and the constant part gives $X_{00} = -\phi$. Thus $X = \phi\eta$. The quotient therefore has dimension $10 - 1 = 9$. Substitution of the nine displayed algebraic directions into the stated trace-free coordinate basis gives determinant $8192/27$. The explicit decoder has maximum absolute row sum $2 + \sqrt{3}$, which gives the supremum-norm bound. These two constants are exact analytic calculations. The accompanying Lean development proves that the nine directions are null and formalizes the design map, an explicit left-inverse decoder, injectivity, and the metric-line ambiguity [1, 10]. □

The four-dimensional small-ball coefficients of entanglement-equilibrium arguments [59] are isolated from their physical reading by one machine-checked identity: with $G_N > 0$ and $\ell > 0$, scalars satisfying

$$\delta S_{\mathrm{bulk}} = \frac{8\pi^2\ell^4}{15}\,t, \qquad \delta A = -\frac{4\pi\ell^4}{15}\,f, \qquad \delta S_{\mathrm{bulk}} + \frac{\delta A}{4G_N} = 0 \tag{7}$$

obey $f = 8\pi G_N t$ exactly, substitution cancelling the common nonzero prefactor.

The tensor completion begins at an explicitly realized null-balance relation. This is the point at which the physical bridge enters the mathematical implication.

Theorem 7.2 (Tensor completion of a realized null balance)**.** *Let (M, g) be a connected, time-oriented, C^3 Lorentzian four-manifold with Einstein tensor $G^{(g)}_{ab}$. Let T_{ab} be a symmetric C^1 tensor satisfying the independent Ward premise $\nabla^a T_{ab} = 0$. Suppose a common realized tower supplies a constant $\kappa > 0$ such that, at every point and for every null vector k,*

$$G^{(g)}_{ab} k^a k^b = \kappa T_{ab} k^a k^b, \tag{8}$$

and suppose $G_N > 0$ with $\kappa = 8\pi G_N$. Then there is a constant Λ such that

$$G^{(g)}_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab} \tag{9}$$

on M. If a reference event p_0 and scalar Λ_0 satisfy

$$(G^{(g)}_{ab} - 8\pi G_N T_{ab})\big|_{p_0} = -\Lambda_0 g_{ab}\big|_{p_0},$$

then $\Lambda = \Lambda_0$.

Proof. At each point, Theorem 7.1 applied to $G^{(g)} - \kappa T$ gives a scalar ϕ with $G^{(g)}_{ab} - \kappa T_{ab} = \phi g_{ab}$. The contracted Bianchi identity $\nabla^a G^{(g)}_{ab} = 0$, the Ward premise, and metric compatibility give $\nabla_b \phi = 0$. Connectedness makes ϕ constant. Set $\Lambda = -\phi$ and use $\kappa = 8\pi G_N$. Evaluation at p_0 gives $\Lambda = \Lambda_0$ under the final calibration premise. □

For OPH to supply the null-balance premise (8), one common refinement tower must realize all of the following:

(G1) the effective manifold of Theorem 6.4, with sufficient regularity for its Levi–Civita connection and contracted Bianchi identity;

(G2) geometrically normalized cap modular flow and half-sided modular inclusions whose directional charges define one symmetric tensor T_{ab};

(G3) the Ward identity for that tensor, independently of its reconstruction from directional charges;

(G4) the edge-center split of Theorem 5.2, the physical normalization $\delta\langle Z_L \rangle = \delta A/(4G_N)$, the modular first law, and generalized-entropy stationarity;

(G5) one family with $\ell_r > 0$, $\ell_r \to 0$, all named remainders $o(\ell_r^4)$, the continuum diamond-kernel and fixed-volume area formulas used in the small-ball identity (7), and enough local observer directions to establish (8); and

(G6) universal coupling of the stress and entropy branches, a vacuum reference, and a physical scale calibration establishing $\kappa = 8\pi G_N$.

The exact finite layer contains null tomography, the edge-entropy split, and the coefficient identity, and the machine-checked composition proves the tensor step from an explicit null-balance premise while separately checking the small-ball arithmetic. The bridge between them is (M4). One Newtonian ingredient is carried by a machine-checked shell law on the committed carrier: a spherically symmetric flux with constant shell charge falls off exactly as the inverse square, the exponent being the carrier dimension minus one with the dimension three supplied by Theorem 2.3 rather than assumed. Its flux premises are declared, and its join to the composed Einstein branch is unbuilt.

8 Two gauge reconstructions and a finite matter image

8.1 Sector reconstruction

The overlap and edge-center branch gives a structural gauge result before a particular compact group is identified.

Theorem 8.1 (Bosonic sector reconstruction)**.** *Suppose a cofinal refinement tail carries zero-obstruction, trivial-holonomy bosonic edge sectors with compatible fully faithful pullback functors, that their closure* Sect_∞ *is an essentially small, additive, idempotent-complete, semisimple rigid symmetric* C^**-tensor category with simple unit and finite-dimensional Hom spaces, and that it carries a faithful, unitary, strong symmetric monoidal,* $*$*-preserving fiber functor* $\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$*. Then* $G_{\mathrm{Tan}} = \mathrm{Aut}_\otimes^{u,*}(\mathcal{F})$ *is compact in the topology of pointwise operator convergence,* $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G_{\mathrm{Tan}})$ *as symmetric* C^**-tensor categories, and the reconstructed group is unique up to isomorphism for the specified category and fiber functor.*

Proof. Every unitary monoidal natural $*$-automorphism of $\mathcal{F}$ has a component in the compact group $U(\mathcal{F}(X))$ at each object X, and naturality, tensor compatibility, symmetry, and $*$-compatibility are closed equations, so the automorphism group is a closed subgroup of $\prod_X U(\mathcal{F}(X))$ over a small skeleton and is compact in the subspace topology. Doplicher–Roberts/Tannaka reconstruction gives the equivalence [66, 67], and a second compact group compatible with the same fiber functor is identified with the same automorphism group. □

The theorem classifies the group encoded by the realized sector data. A trivial sector category reconstructs a trivial group, so the statement selects the Standard Model group only given a suitable sector witness, and it identifies its group with the finite response group of the next subsection through a separate premise.

8.2 The twelve-port response algebra

Let A be the icosahedral adjacency operator and J the antipodal involution on the twelve ports. Exact incidence gives

$$10J = A^3 - 4A^2 - 5A + 10I. \tag{10}$$

Put $\varphi = (1+\sqrt{5})/2$. Choose from each antipodal axis one of the certificate's exact coordinate vectors $u_i \in \mathbb{Q}(\sqrt{5})^3$, each with $\|u_i\|^2 = 2+\varphi$, and put $U = [u_1 \ \cdots \ u_6]$. For a real port field f, write

$$b_i = \frac{f_i + f_{Ji}}{2}, \qquad d_i = \frac{f_i - f_{Ji}}{2}, \qquad c = \frac{1}{6}\sum_i b_i, \qquad b^0 = b - c\mathbf{1}.$$

The even coordinates split as $\mathbf{1} \oplus \mathbf{5}$. The odd coordinates split as the rank-three frame channel detected by U and its Galois-conjugate companion detected by $\sigma(U)$, where $\sigma(\sqrt{5}) = -\sqrt{5}$, giving the source-module decomposition

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{5} \oplus \mathbf{3} \oplus \mathbf{3}'.$$

Theorem 8.2 (A1–A2 gauge Lie-type forcing)**.** *Let* $D : P_{12} \to \mathfrak{u}(H)$ *be the complete reversible response derivative supplied by A1, and put* $\mathfrak{g} = D(P_{12})$*. Suppose the endogenous transport clause of A2 covers every proper carrier action* $a \in A_5$ *and gives projective implementers satisfying*

$$\mathrm{Ad}_{[U_a]}\, D(v) = D(a\cdot v), \qquad [U_a] = [g_a c_a],$$

where $g_a \in G_D^0$ and c_a centralizes $\mathfrak{g}$ pointwise. Then

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

The center is the unique trivial line. Up to exchanging $\mathbf{3}$ and $\mathbf{3}'$ by the outer automorphism of A_5, the $\mathfrak{su}(2)$ ideal carries $\mathbf{3}$ and the $\mathfrak{su}(3)$ ideal carries $\mathbf{3}' \oplus \mathbf{5}$.

Proof. Injectivity of D gives $\dim \mathfrak{g} = 12$. Its commutator-closed image is a finite-dimensional Lie subalgebra of $\mathfrak{u}(H)$, and the positive trace pairing makes it compact. For every $a \in A_5$, the centralizer factor c_a acts trivially on $\mathfrak{g}$, so $\mathrm{Ad}_{[U_a]}|_{\mathfrak{g}} = \mathrm{Ad}_{g_a}|_{\mathfrak{g}}$ is inner. The chosen projective lifts need not form a homomorphic section. Covariance through the injective map D nevertheless gives the homomorphism

$$a \longmapsto (D(v) \mapsto D(a \cdot v)) \quad \text{from } A_5 \text{ to } \mathrm{Int}(\mathfrak{g}).$$

The transitive proper-carrier action fixes only the uniform port line, hence $\dim \mathfrak{g}^{A_5} = 1$.

Compactness gives the reductive decomposition $\mathfrak{g} = \mathfrak{z} \oplus [\mathfrak{g}, \mathfrak{g}]$. Inner automorphisms fix the center pointwise, so $\dim \mathfrak{z} \leq 1$.

If $\mathfrak{z} = 0$, the low-dimensional classification of compact simple Lie algebras forces a twelve-dimensional compact semisimple algebra to be $\mathfrak{su}(2)^4$ [65]. Inner automorphisms preserve each simple ideal. On each three-dimensional ideal, the restriction of the A_5 action is either trivial or faithful because A_5 is simple. Its fixed-space dimension is therefore three or zero: a faithful A_5 subgroup of $\mathrm{SO}(3)$ cannot fix an axis, since a finite rotation group fixing an axis is cyclic. The fixed-space dimension of $\mathfrak{g}$ would be a multiple of three, contradicting the one-dimensional fixed line. Hence $\dim \mathfrak{z} = 1$.

The semisimple part has dimension eleven. The same low-dimensional classification leaves only $11 = 8 + 3$, realized by $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$. Adding the center proves the Lie-algebra claim. The center must be the unique fixed line. The three-dimensional simple ideal cannot carry the trivial action, since that would add three fixed directions; hence it carries $\mathbf{3}$ or $\mathbf{3}'$, and the eight-dimensional ideal carries the complementary triplet together with $\mathbf{5}$. ☐

Boundary 8.3. No ambient compact group or candidate Lie algebra occurs among the premises of Theorem 8.2. A1 supplies the faithful compact commutator-closed response algebra, A2 supplies its inner proper-carrier action, and the theorem fixes the abstract local Lie-algebra type. The port count twelve equals the dimension of the forced algebra, so A1 fixes that dimension by declaration; the content of the theorem is the exclusion, through compactness and the one-dimensional fixed space, of the only other twelve-dimensional compact type $\mathfrak{su}(2)^4$, together with the forcing of the unique center. The source receipts derive the carrier module, the inverse-port response, and its relative band signs, while the reconstruction of D from ordered reversible histories or same-current holonomy is consumed as a premise, and the branch supplies no physical matrix current, coupling, matter action, global quotient, or laboratory attachment (M3).

Two exact negative controls bound how a source realization could be built. For the declared adjacency recurrence and inverse-port readback the full response-word algebra is exactly

$$\mathrm{span}\{I, A, A^2, A^3\},$$

four-dimensional and commutative, so it carries neither twelve independent generators nor a nonzero bracket nor any nonidentity proper recharting; the target-free diagonal phase lift supplies rank twelve with an abelian bracket, and adjoining the declared connected adjacency tangent generates $\mathfrak{u}(12)$ with derived rank 143. Neither control ranges over order-sensitive port perturbations or excludes other port actions. A source-positive realization requires both composition orders on one carrier, twelve independent first-order port derivatives, exact bracket closure with derived rank eleven and the constant generator spanning the center, and closed words implementing all sixty proper rechartings. On the canonical oriented carrier an exact Reynolds calculation gives $\dim_{\mathbb{Q}} \mathrm{Hom}_{A_5}(\Lambda^2\mathbb{Q}^{12}, \mathbb{Q}^{12}) = 14$, the complete target-free alternating-bracket search space, whose complete Jacobi condition has quadratic coefficient-row rank 38; conditional on three named compact-Lie inputs the real compact locus is classified into P, F, G [2, 1]. No uniqueness claim over carriers is made, and the closure reading of the carrier in Section 1.5 enters no theorem here.

Let

$$\Phi_0(b^0) = \sum_{i=1}^{6} b_i^0 u_i u_i^\mathsf{T}, \qquad \widehat{x}\, y = x \times y.$$

The conditional matrix witness uses the space

$$H = \mathbb{C}_E^3 \oplus \mathbb{C}_W^3$$

and four nonzero rational coefficients $\lambda_1, \lambda_5, \lambda_3, \lambda_{3'}$, one on each source band. These are explicit branch premises. The two three-dimensional response blocks match the two rank-three odd channels; they are not asserted to be a derived physical Hilbert space. Write v_p for the signed coordinate vector at port p. For $g \in A_5$, let $R_g \in \mathrm{SO}(3)$ be the exact rotation satisfying $R_g v_p = v_{g(p)}$ on the oriented port frame, and define

$$\Pi(g) = \mathrm{diag}(R_g, \sigma(R_g)).$$

The target A_5 action on $\mathfrak{u}(H)$ is conjugation by $\Pi(g)$.

Theorem 8.4 (Conditional four-band finite port-response algebra)**.** *On the declared twelve-port carrier and supplied response representation, define*

$$K(f) = \mathrm{diag}\big(\lambda_3 \widehat{Ud} + i\{\lambda_1 c I_3 + \lambda_5 \Phi_0(b^0)\},\; \lambda_{3'} \widehat{\sigma(U)d}\big) \in \mathfrak{u}(H). \tag{11}$$

Then K is injective, its image is commutator-closed, and

$$K(g\cdot f) = \Pi(g) K(f) \Pi(g)^*.$$

The induced A_5 action on $\mathrm{im}\, K$ is inner. Let $K^{-1} : \mathrm{im}\, K \to P_{12}$ denote the inverse of K onto its image. With the bracket pulled back from that image,

$$[x, y]_K = K^{-1}([K(x), K(y)]_{\mathrm{mat}}),$$

one has

$$(P_{12}, [\,\cdot\,,\cdot\,]_K) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2). \tag{12}$$

Its center is the uniform one-dimensional port line, its derived algebra has dimension eleven, and the five-dimensional A_5 band is noncentral. The map has a positive-definite invariant Hilbert–Schmidt pullback.

Proof. Evaluating (10) on the four adjacency eigenspaces gives the antipode relation, and the source protocol solves the common farthest-shell filter to give the signed response $-J$. The constant coordinate maps to $i\mathbb{R}I_3$; the six rank-one axis matrices form a basis of the symmetric 3×3 matrices with $\sum_i u_i u_i^\mathsf{T} = (5 + \sqrt{5})I_3$, so their sum-zero coefficients map isomorphically onto the traceless symmetric matrices under Φ_0. The map U is an isomorphism on the **3** band and annihilates $\mathbf{3'}$, with

$\sigma(U)$ complementary, so K has rank $1+5+3+3=12$ and image exactly $\mathfrak{u}(3)\oplus\mathfrak{so}(3)$, closed under commutators, with derived algebra of dimension 11 and center the uniform port line. Equivariance, positive definiteness of the invariant pairing, and the statement that $\Pi(g)$ is the exponential of an element of $\operatorname{im} K$ are exact calculations in $\mathbb{Q}(\sqrt{5})$ verified over the full sixty-element domain, and by the exhaustive-computation rule of Table 1 they are part of the proof [1]. □

Boundary 8.5. The A_5-module decomposition alone does not determine a Lie bracket: the same module also carries the zero bracket, whose trivial response-generated group fails the A2 endogeneity clause. The independent inputs to Theorem 8.4 are the carrier incidence and orientation, the supplied response representation, and four nonzero band coefficients. The block architecture and coefficients are declared. No measured particle, mass, or coupling value enters the exact verification. The theorem constructs the pulled-back bracket for this response map; it does not select K from source histories or prove a uniqueness claim over all A_5-equivariant brackets or finite carriers. The abstract Lie type is forced independently by Theorem 8.2.

8.3 Global form and anomaly-free exterior module

Theorem 8.6 (Finite matter image and exhaustive exterior-menu selection)**.** *Let*

$$V = \mathbb{C}^3_{-1/3} \oplus \mathbb{C}^2_{1/2}$$

be an additional, explicitly supplied trace-balanced matter representation of the Lie algebra forced in Theorem 8.2. This $3 \oplus 2$ matter space is distinct from the supplied response space $H = \mathbb{C}^3_E \oplus \mathbb{C}^3_W$ and is not derived from it. Let

$$\widetilde{G} = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$$

act on V by

$$(g_3, g_2, z) \cdot (v_3, v_2) = (z^{-2} g_3 v_3, z^3 g_2 v_2).$$

The differential of this action gives the displayed hypercharges. Its exterior module has the exact branching

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L \tag{13}$$

with

component	$\mathrm{SU}(3)\times\mathrm{SU}(2)$ *type*	Y
Q	$(3,2)$	$1/6$
u^c	$(\bar{3},1)$	$-2/3$
e^c	$(1,1)$	1
d^c	$(\bar{3},1)$	$1/3$
L	$(1,2)$	$-1/2$

This is the familiar one-generation $\mathbf{10} \oplus \overline{\mathbf{5}}$ branching associated with $\mathrm{SU}(5)$ *unification [68]. It has total complex dimension fifteen. All perturbative gauge and mixed gravitational anomalies vanish, and the number of weak doublets, counted with color multiplicity, is four. The common kernel of the action of $\widetilde{G}$ on V, and hence on $\Lambda^2 V \oplus \Lambda^4 V$, is $\mathbb{Z}_6$. Hence the maximal faithful matter image is*

$$S(U(3)\times U(2)) \cong \frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}. \tag{14}$$

Define

$$G_{\text{packet}} := S(U(3)\times U(2)),$$

the maximal faithful image of the displayed declared exterior module. This quotient is familiar as a possible global form of the Standard Model gauge group [70]. The scan menu consists of the ten nontrivial irreducible summands of $\Lambda^\bullet V$ after the invariant vacuum $\Lambda^0 V$ and invariant top line $\Lambda^5 V$ have been excluded by declaration. Among all $2^{10} = 1{,}024$ subsets of this menu, imposing nonemptiness, chirality, and vanishing of the mixed gravitational–$\mathrm{U}(1)$, $\mathrm{SU}(3)^2\mathrm{U}(1)$, $\mathrm{SU}(2)^2\mathrm{U}(1)$, and $\mathrm{U}(1)^3$ traces leaves exactly two rank-fifteen masks. They are exchanged by charge conjugation, and the exterior-parity grading is computed as an output.

Proof. The exterior-sum identity $\Lambda^k(A \oplus B) = \bigoplus_{p+q=k} \Lambda^p A \otimes \Lambda^q B$ gives the displayed five components and charges, and the anomaly-cancellation conditions are standard [69]. With the conventional quadratic indices $T(3) = T(2) = 1/2$ the pure color anomaly is $2 - 1 - 1 = 0$, the two color fundamentals sitting in the weak doublet against the antifundamentals u^c and d^c; the local $\mathrm{SU}(2)^3$ anomaly vanishes identically because the doublet is pseudoreal; and the mixed sums are $\frac{1}{6} - \frac{1}{3} + \frac{1}{6} = 0$ and $\frac{1}{4} - \frac{1}{4} = 0$. The abelian cubic and mixed gravitational sums are

$$6\Big(\tfrac{1}{6}\Big)^3 + 3\Big(-\tfrac{2}{3}\Big)^3 + 1 + 3\Big(\tfrac{1}{3}\Big)^3 + 2\Big(-\tfrac{1}{2}\Big)^3 = 0, \qquad 6\Big(\tfrac{1}{6}\Big) + 3\Big(-\tfrac{2}{3}\Big) + 1 + 3\Big(\tfrac{1}{3}\Big) + 2\Big(-\tfrac{1}{2}\Big) = 0,$$

and with three color copies of Q against one L the Witten parity is even [71]. For the displayed cover action the element $(e^{2\pi i/3} I_3, -I_2, e^{i\pi/3})$ acts trivially on every row and generates a cyclic group of order six; direct center-action enumeration excludes a larger common kernel, and the independent integer presentation has Smith invariants $(1, 1, 1, 1, 1, 6)$. The cover action has determinant one on V, so $\Lambda^4 V \cong V^*$ equivariantly and its kernel on the exterior module equals its kernel on V, which gives (14). The finite scan evaluates nonemptiness, chirality, and the four displayed traces on all 1,024 masks of the declared menu; its only survivors are the two exterior-parity masks, exchanged by the explicit conjugation permutation, with parity recorded after selection rather than used as a selection predicate. □

The structure layer composes as one conditional statement with a coarse two-mask capstone. A single theorem on the declared premise bundle proves the forced Lie type, the unique type-matched nearest family at the bundle's balanced point, and the fifteen-state count, anomaly forms, and $\mathbb{Z}_6$ stabilizer directly on the bundled selection mask. Within the declared exterior-table grammar, exactly two masks survive at the selected coarse G label and the mask permutation exchanges them. This does not count bracket points in the continuous G family or select a global form, physical matter carrier, current, or action, so it is not a two-model uniqueness theorem for complete Standard Model structures.

One corollary of the product type concerns a channel that the framework does not contain. For the forced abstract type $\mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2)$, instantiated by the declared charged-double-triplet current witness, the adjoint is the direct sum $(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0)$. It contains no mixed $(3, 2, -5/6)$ generator and no conjugate, so the ordinary minimal simple grand-unified gauge-exchange channel is absent from the twelve-dimensional current algebra. This is an algebraic statement about that adjoint. It establishes no proton lifetime and excludes neither scalar mediators, nor higher-dimensional baryon-number violation, nor other ultraviolet gauge mechanisms.

Boundary 8.7. The finite response group G_{packet} and the sector-reconstructed group G_{Tan} arise from different data, and their equality is a separate commuting-square premise. The theorem classifies a finite declared exterior representation and its maximal faithful image; laboratory current attachment, physical matter identification, exclusion of extra light sectors, scalar attachment, family multiplicity, Yukawa couplings, mixing, masses, and a continuum quantum-field realization are unsupplied, and the exclusion of $\Lambda^0 V$, $\Lambda^5 V$, direct sums, vectorlike additions, and neutral singlets is underived by this scan. Section 9.7 states term by term what the reconstruction supplies for the textbook Lagrangian and what it consumes.

8.4 Massless carriers, conditionally

Three conditional masslessness statements occur on distinct premise sets, each a consistency consequence of its declared branch and none an independent discriminator from the Standard Model or General Relativity. For the photon, a finite one-loop calculation importing the electroweak action and its symmetry-restored vacuum relation certifies a transverse self-energy and a null direction of the neutral mass matrix, so that fixture carries no hard photon mass term; a derivation additionally requires a source-generated action, (M3), and a photon-pole map. For the gluon, the color factor in (12) lies in the unbroken subalgebra of the conditional global form, and under the separately declared scalar completion no scalar direction carries color, so the finite action contains no hard color-vector mass term, with the physical gluon spectrum and confinement outside the finite calculation. For the graviton, linearizing the composed relation of Theorem 7.2 about a suitable vacuum yields a null spin-two kernel with two transverse-traceless modes, conditional on the same premise vector and supplying no source-selected Hilbert space, positive-residue pole, or rest-mass prediction.

8.5 The Yang–Mills mass gap

Confinement sits on the other side of the same gauge sector, and the finite repair dynamics reaches it as a conditional identity rather than as a spectrum calculation. The Clay problem asks for a nontrivial four-dimensional quantum gauge theory with a positive mass gap for every compact simple gauge group [18]. The companion paper states what the architecture supplies and what it consumes [17].

The finite mechanism is exact. On the support-visible compact-gauge branch, local holonomy data and the screen scaling chart give the candidate Euclidean form, and exact local repair acts on it as a positive Euclidean relaxation generator at finite cutoff. Atomic heat-bath collars supply that generator, and a finite source-type table together with a uniform approximate-tensorization receipt gives the explicit rate floor $\delta_\star = c_\star/A_\star$.

Lean checks only a narrower conditional statement. In its legacy finite-gap result, commuting collar projections and strictly positive rates are hypotheses. The separately machine-checked legacy Lemma 7.2 gives scalar relaxation on a homogeneous fiber, but no Lean theorem bridges its matrix and scalar to the collar projections and rates. The minimal fixed-space witness instantiates only abstract finite operator premises; it is not a compact-gauge construction, a continuum certificate, or a physical mass-gap derivation. The noncommuting Dobrushin and fiber-dependent-rate routes remain separate viable routes under their stated receipts.

Boundary 8.8. The four-dimensional conclusion consumes convergence of the Schwinger functions, reflection positivity, Euclidean locality, nontriviality, and convergence of the transfer data. Under those certificates the continuum Yang–Mills Hamiltonian gap is identified with the limiting repair gap, so the statement is a conditional implication whose hard content sits in the continuum construction. The finite repair theorem is separate from the Clay claim and carries no continuum limit.

A finite numerical diagnostic tests the identification on $\mathbb{Z}_2$ gauge systems at $L = 2, 3$. The Wilson constant-rate receipt is analytic in the free limit and fails on the tested diagonal interacting grid, while the Kogut–Susskind Doob transform has exact positive fiber-dependent rates $c_\ell = \lambda(r + 1/r)$ with the analytic floor $c_\ell \geq 2\lambda$. That grid is not a no-go: it leaves anisotropic limits, alternative transfer objects, and the variable-rate route untouched. Quotient-space approximate tensorization and continuum transfer are unconstructed.

9 Position, energy, matter, and the action

A reconstruction that starts from records owes an account of the objects physics starts from. This section states what position, energy, matter, and the action are inside the finite architecture, which of them are derived and which are declared, and how far the route to a Standard Model Lagrangian is carried. The layer boundary is sharp throughout: the finite objects are constructed, and their identification with laboratory position, laboratory energy, physical matter, and a physical action consumes the maps of Section 1.3.

9.1 Position

Position is the carrier's own readback of its accumulated records. Theorem 2.3 makes this exact: the normalized repair response has intrinsic rank three, the signed record module and the seam current module embed densely in one abstract three-dimensional Euclidean carrier under the response-selected Gram metric, and cumulative records act on it by exact internal translations while the sixty proper carrier maps act by isometries. Displacement is therefore a record difference, and the dimension of the carrier is an output of the incidence rather than an input. Physical position, physical length, and cofinal gluing between carriers are unsupplied.

9.2 Energy

Energy is the modular ledger of the record state. For a reference τ the modular Hamiltonian is $K = -\log \tau$, the second law of Section 11 appears as $\Delta S \geq \Delta\langle K\rangle$ on τ-preserving channels, and the exact identity

$$S(p) - S(\tau) = \langle K\rangle_p - \langle K\rangle_\tau - D(p\|\tau)$$

makes the entanglement first law $\delta S = \delta\langle K\rangle$ a first-order corollary. The quantity $\langle K\rangle$ is the internal energy ledger every later statement weighs.

That ledger composes exactly. Against a positive product reference it is additive over parts for every joint state, correlated or otherwise, and the binding defect against a general joint reference is the expectation of $\log(\tau_1\tau_2/\tau_{12})$, which vanishes on the product reference. Against the Gibbs reference of a declared total $H_1 + H_2 + V$ with part references at H_1 and H_2 and one inverse temperature, the defect is $\beta\langle V\rangle + \log(Z_{12}/Z_1Z_2)$, whose constant vanishes at $V = 0$. For a declared attractive $V \leq 0$ at $\beta \geq 0$ the state part is nonpositive and the constant is nonnegative, so attraction alone leaves the sign of the defect unfixed; the two-point uniform state has the exact positive defect $-1/4 + \log((3 + e)/4)$.

The mass–energy identity has its two halves here. The kinematic half is a normalization: for a declared positive mass parameter along a future unit timelike frame direction of the Hermitian Lorentz module, the Lorentz square of the scaled frame vector is m^2, the scalar coordinate satisfies $E^2 = m^2 + |p|^2$ and equals m exactly at vanishing spatial coordinates, every oriented Lorentz map transports the construction with the same Lorentz square, and a future null vector satisfies the zero-mass identity. The declared plane wave of that momentum, read along its own worldline, advances in phase by exactly m per unit proper time, with period set the integer multiples of $2\pi/m$.

The dynamical half is the slope with which $\langle K\rangle$ enters inertia, and two independently declared selection routes constrain it. On the momentum map, in the family $P_\lambda = (m + \lambda E)\,u + (1 - \lambda)E\,e_0$ every member has rest energy $m + E$, yet P_λ is covariant under oriented Lorentz maps and sits on the shell of its rest energy at every frame exactly when $\lambda = 1$ or $E = 0$, with transport defect $(1-\lambda)E\,(e_0 - Le_0)$ and Lorentz-square defect $2(1-\lambda)E(m+\lambda E)(u^0-1)$, the converse witnessed by an explicit boost. On the action, midpoint refinement of a worldline leaves the proper length invariant and doubles the step count, so, for $E \neq 0$, refinement invariance of the internal action $E(aL + bM)$ forces $b = 0$ and excludes the per-step additive ledger. Separately, for $E \neq 0$, the declared proper-time principle selects coefficient one in the family $E\lambda L$; the declared internal clock is the proper-length member, and the length action $(m+E)L$ has equation of motion $(m+E)\,\Delta T =$ impulse on timelike windows, the unit-tangent difference carrying the inertial coefficient $m + E$. Under the same declared shape the inertial coefficient of a composite is $m_1 + m_2 + E_1 + E_2$ plus the binding defect, which is the shape a mass defect takes.

The stability window of Section 9.5 carries a dimensionless carrier-coordinate causal bound once the seam-step worldline unit is identified with the scaled field step. A crossing step is timelike exactly when $4 < h^2$, and no step lies both inside the window and above that value, so in the window every per-step crossing is spacelike, the ratio of the two thresholds being $2\varphi^2 = 3 + \sqrt{5}$. A block of k rests and one crossing is timelike exactly when $2 < (k+1)h$, which inside the window forces $\sqrt{2}\,\varphi < k+1$ and therefore at least two rests per crossing, and its speed $2/((k+1)h)$ is below one exactly when the block is timelike, is never one on a timelike block, and has supremum one over the window, approached and never attained. The identification is declared, and without it the two thresholds are independent, so no signal speed is claimed.

Boundary 9.1. The slope family, the covariance requirement, the refinement, and the proper-time principle are declared, and every shape is a declared enrichment under the non-identifiability theorem of Section 9.4. Nothing here derives a mass from repair data. Reading $\langle K\rangle$ against a laboratory energy runs through the declared calibration row at one declared inverse temperature, with an affine law, slope and reference uniqueness, and tick conversion; no theorem establishes that the record reference is a Gibbs state, and the calibration carries one declared tick, with two distinct ticks giving two distinct laboratory frequencies.

9.3 Matter

Matter enters as charge carried by the record structure, in two registers that the framework keeps distinct. The gauge register is the exterior module of Theorem 8.6: a declared trace-balanced representation whose exterior algebra branches into the fifteen-state chiral generation, with the exhaustive scan selecting exactly one charge-conjugate pair inside the declared menu and the common $\mathbb{Z}_6$ kernel fixing the maximal faithful image. The transport register is integer port load. A write appends $+1$, a retraction appends -1, and, at a declared nonzero step h, a declared charge hopping along the twelve ports has conserved total charge, satisfies the committed continuity equation identically, and makes the source pairing gauge invariant under endpoint-vanishing gauge functions exactly when continuity holds on the interior steps. Its endpoint step difference is the polarization load of the committed coupling at the charge fixing $\kappa = -qh/(12 - 4\varphi)$. That charge carries an exact discrete balance. Under closed two-step variations of the transported action, exchanging a forward crossing of a seam with an adjacent rest moves the action by $qhE(e)$ at the delayed step with zero clock cost, so, for nonzero charge q and nonzero step h, the exchange is stationary exactly when the scaled electric seam field on that seam vanishes there; the block momentum is unchanged by every in-block move. Along the respective scaled Ampère evolutions sourced by the original and delayed hopping currents, and with the original field static across the crossing step, the delay difference equals h times the field-energy transfer at the hop. The worldline class and the potentials are declared. The hopping sources are consistent with the field sector: the hopping load and current satisfy the discrete continuity equation, the Gauss constraint propagates under the sourced

Ampère step exactly when the sources are conserved, and a Gauss-consistent history exists if and only if the initial load is neutral and the sources are conserved, so on the closed carrier one hopping charge alone is admissible only at zero charge and a join carries a neutral partner or a neutralising background (`Lean/Screen/SeamChargeContinuity.lean`).

The neutral-partner branch is realized by one finite action on this carrier, not by adding independently prescribed background sources after variation. Two declared seam-step worldlines carry charges q and $-q$, and the action is the source-free Maxwell window functional plus both minimal-coupling terms and both clock terms. For $h \neq 0$ it is invariant under gauge functions that vanish at the two window endpoints. Its load is neutral at every step and its summed load and current obey the continuity equation identically. Variation of this same action in the two field slots, with the seam-potential endpoints fixed, is equivalent to the sourced scaled Ampère equations at $m < N$ and Gauss equations at $n < N+1$ for those two worldlines. A closed two-step replacement of either path, with the other path and the fields fixed, gives an exact off-shell clock-plus-interaction action difference; this identity is not by itself a path equation of motion. At $q = 1$, $h = 1/2$, the opposite crossings of seams 0 and 29 recover the explicit nonzero-current Coulomb-started history and make its field-sector stationarity non-vacuous. The same packet fails the declared closed-two-step exchange-stationarity condition in both path slots: for every $N \geq 1$, delaying either the positive seam-0 crossing or the negative seam-29 crossing while fixing the fields and the other path changes the action by $5/12$ for every pair of clock-unit values within the fixed quadratic clock-action form. The two results are charge conjugates, and applying both canonical exchanges simultaneously changes the action by $5/6$. These are only the displayed discrete exchanges, not continuum first-variation claims or a no-go for other paths, fields, or action shapes. That packet does not exhaust the same action. On the window $N = 1$, a second exact rational temporal-gauge history at the same $q = 1$, $h = 1/2$, and the same two crossing paths satisfies the sourced Ampère and Gauss equations. Every admissible closed two-step replacement of either path has one of four intermediate ports. The two endpoint routes have zero action difference, while each triangular detour has clock difference -4 and interaction difference $+4$. The same finite action is therefore stationary in both field slots and unchanged under every in-window closed two-step replacement in either path slot. Its sources are nonzero and opposite on seams 0 and 29, and at the separately declared clock unit $\tau = 3$ both crossings are timelike in the existing Lorentz-module metric (`Lean/Screen/NeutralPairCoupledAction.lean`, `Lean/Screen/NeutralPairJointStationaryWitness.lean`). This supplies an exact same-instance stationary inhabitant rather than a source selection. The pair, charge, step, field history, action terms, clock unit, and variation class remain declared. No minimum, stability, or dynamic-selection theorem follows. In particular $\tau = 3$ is independent of $h = 1/2$; setting $\tau = h$ would make a crossing spacelike. No laboratory-time calibration or continuum limit follows.

On a declared flat Spin chart, the registered representations compose local Yang–Mills, anticommuting Weyl, Higgs and Yukawa terms with derivative-sensitive gauge identities. Their physical connection to transported port loads, selected coefficients and matter clock remains unsupplied.

9.4 The action, and why it is stationary

Given a pointwise-continuous one-parameter group of record-preserving star-automorphisms on the finite private algebra, every such group is blockwise unitary conjugation generated by one time-independent self-adjoint Hamiltonian per Wedderburn block, unique up to a real scalar [9, 1]. The Schrödinger form is a theorem about that supplied flow; no theorem selects the flow parameter as physical time or selects one flow from the record alone.

Once the kernel and reference are supplied, the finite path action is fixed rather than independently posited. For any strictly positive row-stochastic kernel and strictly positive normalized initial law, the Markov path law is the exponential tilt of the step-uniform reference by the log-transition action at multiplier one, and an action-multiplier pair reproduces the law exactly when the multiplier-weighted action equals the log-transition action plus a constant. The action is therefore unique up to an additive constant and a multiplier rescaling, the bare convention choosing multiplier one. The step-uniform reference is itself the unique reference invariant under independent target scrambling at fixed source. The declared object is the reference measure; given it, the dynamics selects its own action, and the committed source chain instantiates both statements through kernel-decided integer receipts.

Single-site minimality of that action is then a statement about probability. Single-site minimizers of the log-transition local action coincide with most probable paths of the realized chain through an exact corner identity, so single-site least-action minimization and local maximum likelihood are two readouts of one functional. Stationarity alone is weaker: an explicit concave stationary history is nonminimal and not a positive-Gibbs mode. A finite Legendre bridge joins the two faces: the discrete Euler–Lagrange condition at a junction holds exactly when one step of the discrete Hamilton flow carries the incoming junction state to the outgoing one, for the quadratic class and for strictly convex Lagrangians with a solver section, and the constant Noether current of the chain face equals the quadratic Legendre momentum, conserved with the energy along the free Hamilton orbit that reproduces the committed witness path. One machine-checked theorem composes them: a fixed-endpoint single-site-minimizing embedded history realizes the derived action of the same kernel whose exponential tilt is the path law, is an interior most-probable update of that law, and satisfies the discrete Hamilton equations.

Boundary 9.2. The reverse direction is an exact non-identifiability theorem. The bilinear real extension of the source corner table is velocity-affine with no global Legendre solver, while every $L_a = L_0 + \frac{a}{2}y(y-1)$ with $a > 0$ agrees on every realized history yet is strictly convex with an explicit Hamiltonian, so no theorem produces a unique real Hamiltonian continuation from this finite history law without an additional curvature receipt. Inside the declared one-parameter family, requiring the constant-one transition-weight maximizer to be stationary under every fixed-endpoint single-site variation forces the member uniquely at $a_\star = 2\log(W_{11}^2/(W_{10}W_{01}))$, twice the log of the interior mode-dominance ratio, while a two-parameter corner-invisible counterfamily selects no unique enrichment beyond that ansatz. The selection is grammar-bound: for every $0 < \lambda < a_\star$ the velocity-only quartic continuation $L_{a_\star} + \lambda y^2(y-1)^2$ shares corner invisibility, junction stationarity, the single-site minimum, and momentum invertibility, and a fourth clause of second-order boost null covariance pins the committed point inside the full polynomial grammar only, every clause proved load-bearing.

9.5 The field sector on the screen

The same carrier supports an exact static field theory. The seam Laplacian has an explicit symmetric rational Green matrix whose kernel is exactly the constants, every neutral load has a canonical Coulomb solution that is the unique cycle-orthogonal and unique minimal-energy solution of the discrete Gauss problem, and the committed uniform seam repair is one minus one sixtieth of that same Laplacian, so the repair operator and the field operator are the same object up to normalization. The $U(1)$-valued seam connections modulo port gauge transformations are classified exactly by their nineteen chord holonomies, the first cohomology of the thirty-seam graph with circle coefficients being $U(1)^{19}$, matching the source-free Gauss cycle rank. The twenty oriented faces give a genuinely local curvature CA, with C of rank nineteen, kernel exactly the port gradients, and five nonzero entries per seam row in $C^{\mathsf{T}}C$. The local quadratic action is gauge invariant exactly for conserved seam currents, its stationary points exist exactly for those currents, and they are global minima unique modulo gauge.

The complete seam-current symbol also carries a quantitative continuum field theorem. On a supplied common Euclidean domain, set $\omega_a(k) = \sqrt{\Lambda_a(k)}$, and use the opposite-sign curl pair $\mathcal{G}_a(E, B) = (D_a B, -D_a E)$ with $D_a = i\omega_a(k)|k|^{-1}k\times$ at nonzero momentum and zero at the origin. The exact source second and fourth moments imply the global bounds

$$0 \le |k|^2 - \Lambda_a(k) \le \frac{a^2|k|^4}{20}, \qquad 0 \le |k| - \omega_a(k) \le \frac{a^2|k|^3}{20}.$$

Its Fourier propagator assembles to a unitary group on real $L^2(\mathbb{R}^3; \mathbb{R}^6)$: opposite momenta obey the conjugate reality condition, and longitudinal components are fixed. For the local Maxwell group U_0, Plancherel gives

$$\sup_{|t|\le T} \|(U_a(t) - U_0(t))F\|_2 \le \frac{Ta^2}{20} \, \||D|^3 F\|_2.$$

For arbitrary finite-energy data the convergence is strong and uniform on bounded time intervals. Duhamel's formula extends it to the same prescribed current on every scale; charge continuity propagates the Gauss constraint. The global scalar estimates are kernel-checked in `Lean/Screen/SeamMaxwellContinuum.lean`; the real-field, strong-limit, and forcing arguments are proved in the detailed microphysics paper [8]. This is continuum Maxwell dynamics for a declared field family. The Euclidean domain, reversible evolution, common time and physical current identification are supplied, and the finite-scale curl is generally nonlocal. A map from authenticated source histories and decoded observer fields into this family is not constructed.

Boundary 9.3. A finite packet adds a typed continuity equivalence conditional on the declared unit-step Ampère recurrence, whose Faraday identity, gauge invariance, static Coulomb join, and staggered quadratic-form balance are exact. For nonzero h, every zero-current datum has uniformly bounded electric seam energy exactly when $h^2(3+\sqrt{5}) < 4$. Inside that strict window the magnetic face energy is also uniformly bounded; at equality and above, an explicit zero-current solution has unbounded electric seam energy. This does not bound the gauge potential A, which can shear or grow in the kernel sector. The committed unit step lies above the window. Inside it, for every eigenvector of the local operator with eigenvalue λ and a step h with $0 < h^2\lambda < 4$, the zero-current temporal-gauge evolution on the amplitude-velocity plane is a determinant-one symplectic map conjugate to the rotation by $\arccos(1 - h^2\lambda/2)$, it embeds in a continuous one-parameter group whose h-step it is, and that group preserves the committed staggered energy; on the kernel, which is exactly the gradient space, the evolution is the h-step of the shear flow. The assembled theorem acts on coefficient states and generates field histories; because it permits zero listed vectors, a faithful flow on the actual field span additionally needs nonzero linear independence or a quotient of redundant coefficients. The staggered energy supplies the inner product rather than leaving it declared: the per-mode form is positive definite exactly on the window, its polarization is an inner product for which the committed step and the continuous flow are isometries, and at nonzero step the step-invariant symmetric forms are exactly its real multiples, with the committed energy fixing the multiple. Each mode is then a complex line carrying the phase $\exp(i\theta t/h)$ with the generator exhibited rather than obtained from Stone's theorem. The energy-derived reading covers the whole curl sector: nineteen listed seam vectors over $\mathbb{Z}[\varphi]$, at eigenvalues two, three, five, and the golden pair, are kernel-checked pairwise orthogonal eigenvectors spanning the nineteen-dimensional range of C^T. Inside the window their weighted direct sum is a positive definite real inner product with a preserved Hermitian form, explicit phase generator, energy diagonal, and coefficient recovery. On the complementary gradient modes, precisely the static gradient-amplitude directions with zero electric component form the radical; gradient velocity/electric directions are non-null. Every zero-current temporal-gauge solution is the assembled history of a coefficient state (`Lean/Screen/FieldSectorEnergyInnerProduct.lean`, `Lean/Screen/CurlSectorEigenbasis.lean`). A selected principal-angle interpolation also places the energy-weighted coefficient vector on a separately constructed $M_{19}(\mathbb{C})$ block, with an exact self-adjoint diagonal Hamiltonian and a rank-one outer-product intertwiner whose trace equals the field energy (`Lean/Screen/CurlStoneClockBridge.lean`). The vector retains absolute phase; the outer product loses it. This representation is not an algebra isomorphism with the two-dimensional private carrier, and its arbitrary input is not identified with the joined record's field history. The branch and clock matching are declared; mode-resolved channels and larger carriers are not excluded. Reading the group parameter as physical time remains the declared clock and duration premise. The listed rows are orthogonal rather than normalised, but normalisation is not a premise of the weighted energy form. These modules therefore identify no source-selected public representation or Hamiltonian, physical photon Hilbert space, laboratory Schrödinger evolution, physical field, source-produced charge or current, calibrated clock, or massless physical particle.

9.6 Coupled matter, continuum trajectories and quantum states

One supplied charged-scalar/Maxwell action on a solid twenty-tetrahedron cone supports both a nonlinear continuum estimate and a finite-mesh interacting Hilbert space [9]. Its real, zero-current sector is invariant under the full coupled equations, including terms from the gauge-dependent scalar interpolation. With zero electromagnetic fields it reduces to $u_{tt} - \Delta u + m^2 u + gu^3 = 0$. For $m^2 > 0$, $g \geq 0$, a smooth $C_t^2 H_x^2$ Neumann solution and Ritz-projected initial data, a nested conforming refinement, $n = 2^k$, with $20n^3$ cells and uniform shape bounds gives

$$\sup_{0 \leq t \leq T} \left(\|u_n - u\|_{H^1} + \|\dot{u}_n - \dot{u}\|_{L^2} \right) \leq C_T / n.$$

For a smooth magnetic-Neumann reference and magnetic Ritz data, complex matter has the same rate in a prescribed uniform magnetic field. Self-consistent Maxwell backreaction requires a separate estimate.

On the fixed mesh with unwrapped edge coordinates and $e \neq 0$, eliminating the twelve mean-zero gauge directions using the full kinetic form gives a complete metric γ on $\mathcal{Q} = \mathbb{R}^{30} \times \mathbb{C}^{13}$. For supplied $\hbar > 0$ and Laplace–Beltrami quantization,

$$\widehat{H} = -\frac{\hbar^2}{2} \Delta_\gamma + V \quad \text{on} \quad L^2(\mathcal{Q}, d\mathrm{vol}_\gamma)$$

is essentially self-adjoint on compactly supported smooth functions. The nonzero subspace invariant under the remaining constant matter phase reduces its closure and carries unitary evolution. Gauge-invariant scalar-density and magnetic smearings are the same configuration functions classically and quantum mechanically: their maximal multiplication operators are self-adjoint on the neutral sector and have a joint spectral probability law. In the real sector, bounded Lipschitz density readouts inherit the $O(1/n)$ trajectory rate. Quantum continuum convergence and selection of quantization are separate requirements.

Five self-reading patches evolve field records through feedback. Interval integration and authenticated replay bound 81 decoded position/velocity checkpoints within 1.0001×10^{-10} of the charged solution on $[0, 2]$, whose lift satisfies 68 Euler–Lagrange and 13 Gauss equations. For regular paths in $E > V$, the temporal-gauge duration $d\tau = \sqrt{G_q(dq, dq)/(2(E - V(q)))}$ is reparameterization invariant; on nonturning solutions of energy E it recovers action time. Geometry, couplings, preparation, quantization and laboratory identification are inputs [9, 1].

9.7 Standard Model action structures

Table 3 distinguishes gauge and fermion forms constrained by the finite representation, scalar and Yukawa interactions supplied as inputs, and parameters not determined by the construction. It records action shapes and relative structures; numerical coupling values are separate inputs.

Table 3: The canonical Standard Model term-and-sector inventory, classified by what the reconstruction supplies.

Term or sector	**Classification**
SU(3), SU(2), and U(1) gauge kinetic terms	partial
Nonabelian gauge self-interactions	partial
Quark doublet, up-singlet, down-singlet, lepton doublet, and electron-singlet kinetic and covariant terms	partial
Right-handed neutrino stance	derived conditional
Higgs kinetic and covariant term	registered premise
Higgs potential	registered premise
Up-type, down-type, and lepton Yukawa terms	registered premise
Theta-QCD term	absent
Generation triplication	partial
Gauge couplings g_1, g_2, g_3 as numerical parameters	absent
Quark and lepton mixing structure	partial

The four classifications mean different things. The single derived conditional row is the right-handed neutrino stance: within the declared grammar the exhaustive scan admits exactly one unordered charge-conjugate pair of chiral anomaly-free packages, each with fifteen Weyl states and no right-handed neutrino, so minimal content is a theorem of that grammar, with neutral anomaly-free singlets outside the scanned space. The partial rows carry an attained shape without a source-produced action: ad-invariance over each certified compact bracket forces a block-diagonal kinetic form with zero cross terms and one free positive coefficient per simple factor, which is the Yang–Mills shape, and the registered exterior grammar fixes the covariant-derivative coupling pattern of each fermion line through its hypercharge. The registered-premise rows are consumed as declared: no theorem produces the scalar doublet, the quadratic-plus-quartic potential, either of its coefficients, the vacuum, the electroweak scale, or any Yukawa matrix, and an exact non-definability result leaves a free positive rescaling family of admissible up-type spectra. The two absent rows have no structure in the framework.

Two exact discriminators constrain the gauge sector without selecting it. A declared equal-weight cyclic rule on the twenty oriented faces gives the A_5-equivariant bracket $B_{\text{face}} = 60R_{13}$, which fails Jacobi in exactly 240 of 2640 independent coordinates, split evenly between $+1$ and -1; exact primal–dual certificates compare it with the classified compact locus under three coordinate edit norms,

$$\begin{array}{c|ccc} & G & F & P \\ \hline \ell_1 & 30(\sqrt{5}-1) & 60\ (\text{infimum}) & 60 \\ \ell_2^2 & (615-123\sqrt{5})/22 & (615+123\sqrt{5})/22 & 45 \\ \ell_\infty & (5-\sqrt{5})/10 & \sqrt{5}/5 & 1/2, \end{array}$$

so the family G wins all three. Independently, ad-invariance reduces the F/G three-weight space to dimension two through the exact conjugate relations $F : w_{3+} = \sqrt{5}\, w_5$ and $G : w_{3-} = \sqrt{5}\, w_5$, while the P control stays two-to-two, and a symbolic two-factor theorem makes the relative factor coefficient identifiable from a constructed Gibbs kernel without selecting its value.

Boundary 9.4. These are representation-level statements over committed finite packets. The face rule, metric, compact-locus projection, reference measure, step lattice, quadratic cost, and Gibbs kernel are declared or constructed objects, and no source selection, physical unit, clock, complex amplitude, continuum limit, or laboratory field is claimed. Relative coefficients between simple factors, the abelian kinetic ray, and the matter, scalar, and hypercharge sectors of any composed effective action require the prospectively fixed source-selection protocol; mismatch against its fixed Standard Model target rejects that branch.

10 Exact quantitative closures and physical tests

The finite architecture supports a small set of quantitative surfaces sharp enough to be tested numerically. Each surface below is classified by Table 1: the theorems are finite theorems, the numerical enclosures are certified finite computations, and every empirical surface is typed as a diagnostic, a conditional empirical test, or a frozen prospective branch prediction. The closure hypothesis of Section 1.5 contributes its quantitative candidates here, in Section 10.1, under the same typing.

10.1 Closure candidates: the screen grain and the capacity coordinate

Two declared map candidates motivated by the closure hypothesis are examined, one for the screen grain and one for the record capacity, and both carry interval-certified roots. Neither clears the source-only selection and same-quantity gates. The certified gauge-width grain map has its fixed point at $P = 1.6309682414$ and returns the inverse electromagnetic coupling 137.0356601, a relative distance of 2.5×10^{-6} from the 2022 CODATA value 137.035999177(21) [79]. The capacity branch gives the candidates 3.2921×10^{122} and 3.3001×10^{122} against the coordinate 3.3129×10^{122} read back from the weighted Planck base-ΛCDM chain, residuals of -0.63 and -0.39 percent. Both comparisons are diagnostics of declared source maps in the vocabulary of Table 1, and both were exposed before the branch choice, so neither carries predictive weight. What the two rows record is the size of the residual for frozen formulas that do not tune a parameter against the displayed target; the choice of map is itself declared. For the coupling, the absolute residual 3.3904×10^{-4} exceeds the experimental standard uncertainty 2.1×10^{-8}; numerical proximity is not agreement at that precision. A physical endpoint for either one additionally requires target-independent selection of the map, a typed bridge identifying the two sides as readings of one quantity, and, for the coupling, same-scheme hadronic spectral transport to the Thomson endpoint.

The capacity coordinate carries the cosmological constant itself through one dictionary. Reading the record capacity as the de Sitter horizon entropy in nats gives

$$N = \frac{3\pi}{\Lambda\, \ell_P^2}, \qquad \text{equivalently} \qquad \Lambda = \frac{3\pi}{N\, \ell_P^2}, \tag{15}$$

with $\ell_P^2 = \hbar G/c^3$. The two capacity candidates then read $\Lambda = 1.0959 \times 10^{-52}\,\mathrm{m}^{-2}$ and $1.0933 \times 10^{-52}\,\mathrm{m}^{-2}$, against $\Lambda = 3\Omega_\Lambda H_0^2/c^2 = 1.0891 \times 10^{-52}\,\mathrm{m}^{-2}$ at the Planck base-ΛCDM centrals, distances of $+0.62$ and $+0.38$ percent. Equivalently the vacuum energy densities are 5.277×10^{-10} and $5.265 \times 10^{-10}\,\mathrm{J\,m}^{-3}$ against 5.245×10^{-10}. Equation (15) is the same dictionary that produces the comparison coordinate, so this is the capacity diagnostic expressed in the constant rather than a second and independent comparison, and it consumes the measured $\hbar$, G, and c through the calibration import (M2). Large capacity explains the small dimensional constant, not its agreement with the comparison: a relative capacity residual δ gives a relative constant residual $-\delta/(1+\delta)$.

The official DESI DR2 release supplies four base-ΛCDM chains for BAO+CMB. Each was authenticated against its manifest. Transforming every weighted sample gives a covariance-preserving retrospective coordinate,

$$\Lambda \ell_P^2 = (2.96770 \pm 0.03978) \times 10^{-122},$$

with weighted 2.5–97.5 percent interval $[2.88997, 3.04530] \times 10^{-122}$ [77, 1]. The transform retains the sampled H_0–Ω_Λ covariance rather than inserting a separate Gaussian covariance. It is a target-informed ΛCDM posterior display, not an OPH likelihood, theory uncertainty, or predictive pull.

In the displayed proposals, $A_T(\cdot)$ is the inverse electromagnetic coupling the declared Thomson-limit source map returns at a trial screen grain, $\alpha_U(\cdot)$ the coupling the same declared map returns at its unification point, $\mathfrak{U}_N$ the regulated universe-level observer system at capacity budget N, and M_0 the correctable-code capacity of its terminal public record; the grain scalar of these equations is distinct from the port set and the event projectors used elsewhere in this paper. Certified roots of both displayed maps are compared against the 2022 CODATA inverse coupling [79]. The symbol $\varphi = (1+\sqrt{5})/2$ is the golden ratio of Section 8.

The closure principle. The axioms describe the observer screen. The global closure principle states that the simulating and the simulated description are one system. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity. Constructing that bridge and the return map is part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling returned by the declared source map, has one interval-certified fixed point on its declared analytic domain. Interpreting that root as the laboratory coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at $P = 1.6309720959$ and returns $A_T = 136.9948352$; the certified gauge-width variant's fixed point sits at $P = 1.6309682414$ and returns 137.0356601. Through the same outer equation the measured 2022 CODATA coupling $\alpha^{-1} = 137.035999177(21)$ reads back the grain value $P = 1.6309682094$. The comparisons stand at 3.0×10^{-4} relative for the displayed map and at 2.5×10^{-6} for the gauge-width variant, as diagnostics of the declared source maps. The latter absolute residual is 3.3904×10^{-4}, compared with experimental standard uncertainty 2.1×10^{-8}; this is not agreement at measured precision.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. The incomplete antecedent determines no direct N, and no theorem extends the bounded-class conclusion to the complete source class. A positive result must complete the source antecedent and prove one physical zero, either within the three-axiom source contract or through a separately named stronger source law.

A separate common-load branch starts from

$$N_0 = \pi \exp\left(\frac{6\pi}{P\,\alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation $P/4$ and six-class equidistribution give presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\text{pres}} = N_0\left(1 - \frac{P}{24}\right), \qquad \ln\frac{N_{\text{pres}}}{\pi} = \frac{6\pi}{P\,\alpha_U(P)} + \ln\left(1 - \frac{P}{24}\right).$$

The exponential alternative $N_{\text{Pois}} = N_0 e^{-P/24}$ requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The source class determines no named-law value, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical $\mathbb{Z}_6$ seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is 3.2921×10^{122} and the exponential candidate is 3.3001×10^{122}. The full weighted Planck base-ΛCDM chain gives the comparison coordinate 3.3129×10^{122}. The respective residuals are -0.63 percent and -0.39 percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

10.2 The positive-chamber Koide identity

The icosahedral carrier's face structure supplies order-three cyclic response operators. The following exact identity governs every Hermitian three-cycle response [13].

Theorem 10.1 (Positive-chamber Koide identity)**.** *Let $R^3 = I$ generate a cyclic action and let*

$$C = aI + bR + \bar{b}R^2, \qquad a > 0, \quad b = \rho e^{i\delta} \in \mathbb{C},$$

be the associated Hermitian circulant, with eigenvalues

$$\lambda_k = a + 2\rho\cos\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2.$$

Suppose all three eigenvalues are nonnegative and, for one scale $s > 0$, $\sqrt{m_k} = \sqrt{s}\,\lambda_k$. Then

$$Q := \frac{\sum_k m_k}{\left(\sum_k \sqrt{m_k}\right)^2} = \frac{1}{3} + \frac{2}{3}\left(\frac{\rho}{a}\right)^2, \qquad \textit{so} \qquad Q = \frac{2}{3} \iff \frac{\rho}{a} = \frac{1}{\sqrt{2}}. \tag{16}$$

Proof. The three cosines at angles separated by $2\pi/3$ sum to zero and their squares sum to $3/2$, so $\sum_k \lambda_k = 3a$ and $\sum_k \lambda_k^2 = 3a^2+6\rho^2$. With $m_k = s\lambda_k^2$ the ratio is $Q = (3a^2+6\rho^2)/(3a)^2 = \frac{1}{3}+\frac{2}{3}(\rho/a)^2$, and solving $Q = 2/3$ on $a > 0$, $\rho \geq 0$ gives the equivalence. □

The shape depends on ρ/a and δ. Fixing $Q = 2/3$ fixes ρ/a and leaves one phase: the two mass ratios vary together. Equal rank-two event blocks in the finite tracial Gelfand–Naimark–Segal packet supply the balance $\rho/a = 1/\sqrt{2}$ exactly under the declared tracial premises, and the Lean development checks both the circulant identity and the equivalence [13].

Under the balance and mass-ordering premises the measured electron and muon masses [80] determine the tau mass through one quadratic, governed by a prospectively fixed conditional test whose target definition, precision floor of 0.045 MeV, and decision rule were fixed before comparison with any post-registration measurement [3]. The outward-rounded enclosure is [1776.968991, 1776.969063] MeV, a 72-eV window whose center sits 0.43σ from the measured 1776.93 ± 0.09 MeV. The premise ancestry is declared, the balance having been abstracted from the measured triple, so this is a target-informed conditional postdiction whose evidential weight lies in the kill direction: a world-average or dedicated central value more than three standard uncertainties from 1776.969027 MeV refutes the balanced-circulant premise, and with it the equal-rank-two-block tracial reading of the face-circulant sector, the only registered exact balanced-circulant family mechanism. Alternate family-sector carriers, potentials, dynamics, and source attachments are not excluded. The surface supplies no physical family attachment or phase selection.

10.3 The charged-lepton interval diagnostic

A finite eight-path carrier with an empirical electromagnetic transport packet yields outward-rounded enclosures for the three charged leptons of logarithmic half-width 1.732%, one-sided widths -1.72% and $+1.75\%$, under the declared payload-coherent anchor-gap premise, against 6.554% for the premise-free independent-gap enclosures; the measured triple lies inside all three coherent enclosures [12]. Measured mass ratios and transport anchors enter the branch, so this is a diagnostic of internal closure giving a sharp surface that a source-emitted transport bridge must hit, and a physical charged determinant line, an absolute clock, and that bridge are the maps an endpoint prediction requires.

10.4 The electroweak chart and strict pole-consumer checks

On the target-free but declared and incomplete running/map branch, evaluating the supplied formulas gives the electroweak chart coordinates 80.330 and 91.119 GeV for the W and Z carriers, with no measured mass entering the arithmetic. The source does not select the map, so these are not source-emitted coordinates [12]. Writing $S = gv_F/2$, $t = g'/g$, $w = S^2$, and $z = S^2(1+t^2)$, the strict one-loop consumer factorizes on the declared domain as $s_W/s_Z = (1 + d_W)/((1 + t^2)(1 + d_Z))$ with d_W and d_Z the normalized self-energy factors, whose strict expansion is $\frac{1}{1+t^2}(1+d_W-d_Z)+O(\text{loop}^2)$. The common scale cancels under passive rescaling at fixed t, d_W, d_Z, while those three inputs stay unselected. Directed complex-interval receipts exclude zeros of a scalar inverse-propagator entry on declared principal-sheet boxes, and separate receipts isolate, for each of W and Z, one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart [1]. The receipts identify neither chart with the physical resonance sheet and prove no unique continuation, sign bridge, full-matrix Laurent residue, or current amplitude, and their external fixture is uncomposed with the declared-map coordinates, so no chart-to-pole map, physical pole, or mass comparison follows.

10.5 Finite capacity and the de Sitter transfer sign

The screen's finite capacity supports an exact entropy accounting with one conditional gravitational reading [14].

Proposition 10.2 (Capacity transfer sign)**.** *For finite sector dimensions d_i with total $M = \sum_i d_i$ and sector probabilities p_i, the generalized entropy obeys the exact identity*

$$S_{\mathrm{gen}}(p,d) = -\sum_i p_i \log p_i + \sum_i p_i \log d_i = \log M - D\Big(p \,\Big\|\, \frac{d}{M}\Big),$$

where D is relative entropy; its exact maximum is $S_{\mathrm{gen}}^{\max} = \log M$ at $p_i = d_i/M$. If an observer receives a fraction $f \in (0,1)$ of a fixed total capacity and the sectors deplete uniformly, every admissible transfer changes the maximum by

$$\Delta S_{\mathrm{gen}}^{\max} = \log(1-f) < 0,$$

and the positive-real interpolation in f is strictly decreasing and concave, so the zero-transfer point is a one-sided boundary maximum.

Proof. Expanding $D(p\|d/M)$ gives the identity, and nonnegativity of relative entropy with equality at $p = d/M$ gives the maximum. Uniform depletion replaces M by $(1-f)M$, so the maximum changes by $\log(1-f)$, whose monotonicity and concavity are elementary. □

Identifying capacity with horizon area, the transfer coordinate with observer mass, and the associated finite port operator with the gravitational shock supplies a conditional de Sitter time-advance mechanism with a definite sign: under those dictionaries an observer absorbing a capacity fraction f strictly lowers the maximal generalized horizon entropy by $\log(1-f)$, and the one-sided boundary maximum carries the definite sign of the associated time shift. The finite identities make none of those physical identifications; no horizon or observer dictionary is supplied, and the finite calculation supplies no positive cyclic static-patch trace.

10.6 The unique rank-six invariant of the primitive carrier

In this subsection, "spin six" means spherical-harmonic angular rank $j = 6$. It does not denote a particle with spin six. The mathematical engine is the invariant count for the sixty-element proper icosahedral rotation group. The dimension of the invariant subspace in angular rank L is

$$m_L = \frac{1}{60}\Big[(2L+1) + 15\,U_{2L}(0) + 20\,U_{2L}(\tfrac{1}{2}) + 12\,U_{2L}(\tfrac{\varphi}{2}) + 12\,U_{2L}(\tfrac{\varphi-1}{2})\Big],$$

where U denotes the Chebyshev polynomials of the second kind. The exact table gives $m_1 = \cdots = m_5 = 0$ and $m_6 = 1$. Thus no nonconstant invariant occurs below angular rank six, and the rank-six invariant line is one-dimensional, in agreement with the classical icosahedral-harmonic classification [54].

Theorem 10.3 (Icosahedral angular-rank universality)**.** *Let the carrier kinetics be any operator $\lambda_a(k) = a^{-2}\sum_d w(d)\,[1-\cos(a\,k\cdot d)]$ with a finite direction multiset and weights invariant under the proper icosahedral rotation group. Then the dispersion is exactly isotropic through angular rank five at every order in a; every directional term below angular rank ten is one multiple of the unique normalized invariant $\mathcal{I}_6$, whose stationary structure is the exact 62-direction census (12 and 20 extrema of opposite index, 30 saddles); and whenever the weighted sixth-moment $\mathcal{I}_6$ coefficient is nonzero, a condition certified for the equal-weight stencil and for each fundamental orbit, the first directional artifact appears at order a^4k^6 and one binary refinement step suppresses it by exactly $1/16$ at that order. A vanishing sixth moment removes the displayed a^4k^6 residue. Any surviving directional term occurs at higher order. For a single-radius member whose weighted sixth-moment* *coefficient vanishes, the rank-six component vanishes at every order, so any surviving anisotropy has a higher even invariant rank. Rank six is the least symmetry-allowed nonzero anisotropic rank in the class.*

Proof. The order-$a^{2m-2}k^{2m}$ angular content is an invariant polynomial of degree $2m$, with harmonic components at even spins $L \leq 2m$. The invariant table forces zero at $L = 2, 4, 8$ and a one-dimensional space at $L = 6$, so every anisotropic term below angular rank ten is proportional to $\mathcal{I}_6$, and the possible artifact ranks are exactly the even invariant levels $6, 10, 12, 16, \dots$ The scaling in a is fixed by the expansion order, and the census is the exact critical-point theorem of the fingerprint certificate. The invariant table, the census, the constructive orbit verification, the nonvanishing sixth-moment coefficients, and an exact positive-weight tuned member whose spin-six content cancels are certified in exact arithmetic [1]. □

The universality theorem concerns a class of finite operators, and a physical prediction requires a narrower branch whose operator and transfer are fixed. For the declared equal-weight twelve-port member the stationary structure is exact at every scale: the 62 critical directions of $\mathcal{I}_6$ define 31 projective axes, and three exact projective charts with a 2,624-leaf interval cover prove that the full cosine kernel has exactly those directions for every $0 < |ak| \leq 1$, the twelve port directions remaining maxima, the twenty face-center directions minima, and the thirty edge-center directions saddles [1].

Let $u_1, \dots, u_{12}$ be the unit vectors through the twelve carrier ports. The *primitive twelve-port propagation branch* declares that the intrinsic spatial kinetic symbol lies in the cosine class of Theorem 10.3 with scale-independent coefficients; that the complete primitive orbit is its sole hop support, with no independent isotropic or directional term through order k^6; that the carrier scale a is finite and positive with the continuum quadratic term normalized to k^2; that one carrier rest frame is transported coherently over the experiment; and that the tested scalar or polarization-independent sector realizes the operator, a photon realization additionally requiring equal action on both transverse polarizations. Transitivity of the proper carrier group forces equal weights once the single orbit is selected. The carrier is realized as a quasiperiodic structure or as a graph, since no periodic three-dimensional lattice carries this point group, so the branch presupposes no crystal.

The finite repair packet leaves this branch avoidable. Its certified operator acts on thirty internal seam readings, a spatial translation stencil acts on a field at distinct sites, and a physical-sector readout is a third typed object; an exhaustive classification of the declared serialized artifacts finds no packet joining a complete twelve-port translation operator to a physical readout of that same operator. The equal-weight stencils on the vertex, face, and edge orbits have pairwise distinct rank-six rays, and proper-carrier transitivity selects no orbit as the physical hop support.

Proposition 10.4 (Primitive twelve-port branch prediction)**.** *Under the primitive-port propagation premises, the physical symbol is*

$$\omega^2(k, \widehat{k}) = \frac{1}{2a^2} \sum_{i=1}^{12} \left[1 - \cos\Big(ak\, u_i \cdot \widehat{k}\Big)\right]. \tag{17}$$

With

$$\mathcal{I}_6(n) = \frac{25}{132} \sum_{i=1}^{12} P_6(u_i \cdot n), \qquad \mathcal{I}_6(u_i) = 1,$$

where P_6 is the degree-six Legendre polynomial. The small-ak expansion is

$$\omega^2 = k^2 - \frac{a^2}{20} k^4 + \frac{a^4}{840} k^6 + \frac{2a^4}{7875} k^6 \mathcal{I}_6(\widehat{k}) + O(a^6 k^8). \tag{18}$$

Writing the same expansion in a transported frame as

$$\omega^2 = k^2 + C_4 k^4 + B_0 k^6 + B_6 k^6 \mathcal{I}_6(R^{-1}\widehat{k}) + O(k^8), \qquad R \in \mathrm{SO}(3)/A_5,$$

the branch predicts

$$C_4 = -\frac{a^2}{20}, \qquad B_0 = \frac{a^4}{840}, \qquad B_6 = \frac{2a^4}{7875}, \tag{19}$$

and hence the scale-free relations

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \qquad \frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{B_0} = \frac{16}{75}. \tag{20}$$

All intrinsic anisotropic coefficients at $1 \le j \le 5$ vanish. Once a negative C_4 is resolved on this branch, no scale or anisotropic amplitude is free; only one three-parameter orientation class R in $\mathrm{SO}(3)/A_5$ is profiled. Under binary refinement, $C_4(a/2) = C_4(a)/4$ and $B_0(a/2) = B_0(a)/16$, $B_6(a/2) = B_6(a)/16$ at fixed physical momentum.

Proof. The exact carrier moments are

$$\sum_i (u_i \cdot n)^2 = 4, \qquad \sum_i (u_i \cdot n)^4 = \frac{12}{5}, \qquad \sum_i (u_i \cdot n)^6 = \frac{12}{7} + \frac{64}{175}\mathcal{I}_6(n).$$

Substitution into the cosine series in (17) gives (18). Eliminating a gives (20). The lower-rank nulls and uniqueness of the rank-six line follow from Theorem 10.3. □

Proposition 10.5 (Source-seam edge ray and conditional propagation branch)**.** *The boundary current of the complete thirty-seam source packet maps through the antipodal-odd load readout onto the even-sum module D_6 of Theorem 2.3, with Smith invariants $(1,1,1,1,1,2)$, and every directed seam defines one translation of the internal record carrier whose complete normalized direction multiset is the thirty-direction edge orbit. If the feasible move laws and their objective are natural under the proper carrier action and A3 supplies a unique normalized minimizer, transitivity forces weight $1/60$ on every directed seam, and the selected internal operator is the source-counting homogeneous convolution.*

Suppose these record translations are physical displacements under one homogeneous position action, that the complete edge orbit is the sole direct support through the displayed order, that proper-carrier covariance fixes one common weight, that the same operator acts on one scalar or polarization-independent physical sector with quadratic term normalized to k^2, and that the action carries the stated gluing, scale, frame, readout, nuisance, and exclusivity data. Its spatial kinetic symbol is then

$$\Lambda_a(k, \widehat{k}) = \frac{1}{5a^2} \sum_{j=1}^{30} \left[1 - \cos\!\left(ak\, w_j \cdot \widehat{k}\right)\right], \tag{21}$$

with

$$C_4 = -\frac{a^2}{20}, \qquad B_0 = \frac{a^4}{840}, \qquad B_6 = -\frac{a^4}{12600}, \tag{22}$$

and hence

$$\frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{C_4^2} = -\frac{2}{63}, \qquad \frac{B_6}{B_0} = -\frac{1}{15}. \tag{23}$$

The intrinsic anisotropic coefficients at angular ranks one through five vanish, and the rank-six vector lies on the rotated $\mathcal{I}_6$ orbit.

Proof. Exact integer incidence gives the displayed image and Smith invariants, and the response-selected Gram pullback gives the dense three-dimensional completion of Theorem 2.3. The sixty directed labels map two-to-one onto the thirty signed edge directions, whose unit-normalized moments are

$$\sum_j (w_j \cdot n)^2 = 10, \qquad \sum_j (w_j \cdot n)^4 = 6, \qquad \sum_j (w_j \cdot n)^6 = \frac{30}{7} - \frac{2}{7}\mathcal{I}_6(n).$$

Substitution in (21) gives (22), and elimination of a gives (23). The source incidence, quotient, orbit binding, and moment identities are machine-checked in exact arithmetic [1]. □

The proposition contains an exact finite source theorem and a conditional physical branch. The naturality and minimizer clause supplies a homogeneous internal action whose generator satisfies the positive maximum principle and an exact Dirichlet identity, and whose plane-wave spectrum is the symbol above; Section 9 develops the field structure that generator carries.

The declared spatial symbol admits an exact remainder certificate with no physical premise: writing $q = ak$, the kernel-checked eighth moment, the exact range $-5/9 \le \mathcal{I}_6 \le 1$, and alternating cosine bounds give $|\widehat{\Lambda} - P_6| \le \frac{7}{388800} q^8$ and $\frac{19}{20} q^2 \le \widehat{\Lambda} \le q^2$ uniformly on $0 \le q \le 1$, where $P_6 = q^2 - q^4/20 + (1/840 - \mathcal{I}_6/12600) q^6$. An exhaustive target-free enumeration of all 2^{12} sign-error vectors in one fixed twelve-row design detects the leading coefficient in every replica at noise scale 1/200, while the linked isotropic and rank-six higher-order pair is detected in 209 of 4096, so that stress law leaves the pair unresolved. The certificate concerns the mathematical operator and turns a into no SI length.

If Λ_a is separately identified with a physical frequency squared, the positive leading branch gives exact formal-series coefficients for ω, its radial group velocity, and its transverse gradient, the last proportional to $a^4 k^4 \nabla_{S^2} \mathcal{I}_6$. The spatial-symbol bound supplies no analytic remainder for that physical branch, and physical position, field sector, clock, cofinal gluing, physical scale, frame and boost law, wave-packet dynamics, laboratory readout, and nuisance model are all required before a time-of-flight comparison exists. The edge branch and the primitive-port branch have opposite rank-six signs, and their coefficient rays, custody packages, exposure exclusions, and comparison budgets are separate.

One exact theorem binds the two frozen rays and every member of the declared positive-weight scalar cosine class into a single test surface.

Theorem 10.6 (Carrier-class dispersion band)**.** *Consider the declared positive-weight scalar cosine class of spatial symbols*

$$\Lambda(k, \widehat{k}) = \sum_s w_s \sum_{u \in O_s} \left[1 - \cos(ak\, r_s\, u \cdot \widehat{k})\right],$$

where $a > 0$, each O_s is one orbit of the sixty proper carrier rotations on unit directions, the finitely many shells carry radii $r_s > 0$ and weights $w_s > 0$, the displayed cosine sum is the full spatial symbol and hence complete through order k^8, and the continuum normalization fixes the k^2 coefficient to one. Write the normalized expansion as

$$\Lambda_{\text{norm}} = k^2 + C_4 k^4 + (B_0 + B_6 \mathcal{I}_6(\widehat{k})) k^6 + (D_0 + D_6 \mathcal{I}_6(\widehat{k})) k^8 + O(k^{10})$$

and set $\mu_m = \sum_s w_s |O_s| r_s^m$. Then five exact statements hold.

1. *Sign law: $C_4 = -(a^2/20)\, \mu_4/\mu_2 < 0$ for every member.*
2. *Isotropic floor: $B_0/C_4^2 = (10/21)\, \mu_2 \mu_6/\mu_4^2 \ge 10/21$, with equality exactly on the single-radius members.*

3. *Rank purity: the intrinsic anisotropic ranks one through five vanish and the rank-six residue is one multiple of the rotated* $\mathcal{I}_6$*.*

4. *Rank-six band: every member has* $B_6/B_0 = (16/75)\,\langle \mathcal{I}_6(\widehat{u}_s)\rangle$*, the mean taken with the positive weights* $w_s|O_s|r_s^6$*, and this ratio lies in* $[-16/135,\,16/75]$ *for the declared class. The pure face and vertex orbits attain the endpoints, the pure edge orbit sits at* $-1/15$*, and the vertex-face mixture at per-direction weight ratio* 25:27 *attains zero.*

5. *Eighth-order confinement and common-radius lock: every member has*

$$D_0 = -\frac{a^6}{60480}\frac{\mu_8}{\mu_2}, \qquad \frac{D_6}{D_0} = \frac{64}{125}\,\langle \mathcal{I}_6(\widehat{u}_s)\rangle_{w_s|O_s|r_s^8}\,.$$

No independent angular shape occurs at order k^8*. Every member whose active shells share one radius obeys the division-free identity* $5D_6B_0 = 12B_6D_0$*. This includes the* 25:27 *zero-anisotropy mixture, where both sides vanish.*

Proof. The group-summed sixth-power kernel factors on the invariant line: summing over the sixty proper rotations,

$$\sum_g \big((gu)\cdot\widehat{k}\big)^6 = \frac{60}{7} + \frac{64}{35}\,\mathcal{I}_6(\widehat{u})\,\mathcal{I}_6(\widehat{k})$$

on unit vectors. The kernel is invariant in the seed, the invariant multiplicities $(1,0,0,0,0,0,1)$ through rank six pin the degree-six invariant space to the span of the radial power and $\mathcal{I}_6$, and exact residue-zero verification at seeds with distinct $\mathcal{I}_6$ values determines the identity there. The moment ratios give C_4 and B_0, and the floor is the Lagrange identity $\mu_2\mu_6 - \mu_4^2 = \sum_{i<j} W_iW_jr_i^2r_j^2(r_i^2 - r_j^2)^2 \geq 0$ with $W_s = w_s|O_s|$, vanishing exactly at one radius. The band follows from the same factorization, since the rank-six numerator is the positive-weighted sum of seed $\mathcal{I}_6$ values against the matching isotropic denominator, so every member averages inside the range $[-5/9, 1]$ fixed by the 62-direction census. At degree eight the invariant multiplicities $(1,0,0,1,0)$ and the exact kernel identity $\sum_g((gu)\cdot\widehat{k})^8 = \frac{20}{3} + \frac{256}{75}\mathcal{I}_6(\widehat{u})\mathcal{I}_6(\widehat{k})$ leave the radial term and the rotated $\mathcal{I}_6$ as the complete through-eighth-order basis, and normalizing by the quadratic term gives D_0 and the weighted mean. At one common radius the sixth- and eighth-order means agree and $(64/125)/(16/75) = 12/5$, which proves the lock without dividing by either anisotropy. A second series route and an independent high-precision fit reproduce the table, and one-sign nonnegativity is load-bearing, a signed-weight control violating the floor [1]. ☐

The theorem converts the two frozen branches into two points of one exact map. The declared class shares a support-independent negative quartic sign and the floor $B_0/C_4^2 \geq 10/21$, whose excess is a normalized weighted radial-moment gap vanishing exactly on common-radius support, while the signed rank-six ratio constrains the weighted angular moment and distinguishes the three pure fundamental orbit rays. The frozen vertex branch sits at the endpoint $16/75$ of the band and the frozen edge branch at $-1/15$. The class-level kill surface is correspondingly wider than either branch band: under the same physical-sector premises, a resolved intrinsic dispersion with B_0/C_4^2 below $10/21$, or B_6/B_0 outside $[-16/135, 16/75]$, or a residue at ranks one through five, or a rank-six residue off the rotated $\mathcal{I}_6$ template, excludes every member of the declared class at once. Exact saturation of the floor certifies common-radius support, and a generic rank-six anisotropic model carries independent k^6 and k^8 amplitudes whose relative amplitude the common-radius stratum fixes. The certificate reads no comparison data and changes no frozen bytes [1].

The same class carries an exact auxiliary-frequency contraction bound.

Proposition 10.7 (Global unit Lipschitz bound on the auxiliary frequency)**.** *Let a finite strictly positive weighted direction family satisfy the Euclidean tight-frame identity* $\sum_i \rho_i(v_i{\cdot}x)^2 = t|x|^2$ *with* $t > 0$*, and for nonzero finite* a *put*

$$\Lambda_a(k) = \frac{2}{ta^2}\sum_i \rho_i\big[1 - \cos(a\,v_i{\cdot}k)\big].$$

The sine-feature map $\Phi_i(k) = \sqrt{4\rho_i/(a^2t)}\,\sin(a\,v_i\cdot k/2)$ *satisfies* $\Lambda_a(k) = \|\Phi_a(k)\|^2$ *and* $\|\Phi_a(k) - \Phi_a(p)\| \leq |k-p|$*, so the nonnegative auxiliary frequency* $\Omega_a = \|\Phi_a\|$ *obeys*

$$|\Omega_a(k) - \Omega_a(p)| \leq |k-p|$$

at every pair of momenta. The complete vertex support instantiates $t = 4$ *with prefactor* $1/(2a^2)$*, and the complete edge support instantiates* $t = 10$ *with prefactor* $1/(5a^2)$*.*

Boundary 10.8. The constant one is a certified upper bound in the selected Euclidean carrier chart rather than an optimality result, and Ω_a is an algebraic spectral root. The proposition supplies no physical position, frequency, clock, field sector, wave packet, signal front, frame, boost law, physical scale, or detector readout, so it is not a statement about a laboratory signal speed [1].

The edge ray and its complete decision rule are frozen under public commit custody before any eligible edge-branch comparison payload [3], with the template class, every primitive-port comparison input, and every datum inspected before that freeze excluded. A null carries no verdict unless a pre-exposure source theorem for the same action and sector supplies $a_{\min} > 0$ and a preregistered power and remainder contract makes the entire admitted manifold excludable.

Exposed source-seam diagnostics. Under the external subluminal quadratic convention $a = \sqrt{20}/E_{\mathrm{QG},2}$, published time-of-flight bounds give conditional photon/clock-sector translations. Separately, the Auger air-shower translation reaches $a < 0.546\,\ell_P$ only with photon-only Lorentz violation, Lorentz-invariant leptons, and the paper's alternative UHECR scenario containing a subdominant proton component above 10^{19} eV [75, 76, 74]. These diagnostics derive no scale and carry no evidence weight.

The Lorentz-invariant-lepton premise in the air-shower translation is load-bearing. If instead the hard photon, electron, and positron are assigned the same leading coefficient $\delta = -d = -a^2/20$, additive head-on collinear kinematics gives, at $u = x(1-x) \in (0, 1/4]$,

$$\epsilon_{\mathrm{req}}(k,u) = \frac{m_e^2}{4ku} + \frac{3dk^3}{4}u.$$

The reciprocal-linear envelope and endpoint transition are machine-checked. At and above the transition, Lean also proves that the equality witness is realized by an open physical share, so it is the constrained share minimum rather than merely a lower bound. Explicitly,

$$\epsilon_{\min}(k) = \begin{cases} m_e^2/k + 3dk^3/16, & 3dk^4 \leq 16m_e^2,\\ (\sqrt{3}/2)m_e k\sqrt{d}, & 3dk^4 \geq 16m_e^2.\end{cases}$$

The branches agree at equality. The unconstrained minimizer is $u_* = m_e/(\sqrt{3d}\,k^2)$; it enters the physical interval at $k_t = 2\sqrt{m_e}(3d)^{-1/4}$. At $a = \ell_P$, representative single-energy CMB targets give finite leading kinematic windows $[4.12\times10^{14}, 7.82\times10^{19}]$ eV for $\epsilon = 6.34\times10^{-4}$ eV and $[8.70\times10^{13}, 3.70\times10^{20}]$ eV for $\epsilon = 3.0\times10^{-3}$ eV. These are roots of a conditional leading equation, not optical depths or fluxes. They neither transfer the published photon-only Auger exclusion to the common-coefficient branch nor supply an interaction vertex, radiation spectrum, source population, cascade, shower, or detector response [1].

Primitive-port prediction and decision rule. Proposition 10.4 is a frozen prospective conditional physical-branch prediction [3]. The minimal locally Lorentz-invariant Standard Model plus General Relativity in local vacuum supplies the baseline $C_4 = B_0 = B_6 = 0$ for intrinsic propagation, and nonminimal effective operators, media, curvature, or another icosahedral system can imitate some or all of the pattern, so a match distinguishes the branch from that baseline without identifying OPH uniquely. In the Standard-Model Extension coefficient language [72, 73] the branch populates only the thirteen-component nonbirefringent rank-six multiplet at operator dimension eight, collapsed to one amplitude and one frame orientation, with every anisotropic coefficient at $1 \leq j \leq 5$ exactly zero in the carrier rest frame and frame boosts admixing neighboring ranks at first order in the frame velocity.

The exposure class is fixed. The internal-linear-combination map of the dated template search, its likelihood class, and every data product examined there are ineligible for this prediction; that search included the icosahedral rank-six and higher templates and returned a family-wide p-value of 0.64. The target statement, exact coefficient snapshot, registration record, public commit custody, and detached calendar attestations fix the prospective content and its decision rule [3].

No empirical verdict is licensed without a dataset-specific registration fixing one post-freeze release, a joint likelihood or full covariance for the same-sector coefficient vector, the carrier and observer frames, the $\mathrm{SO}(3)/A_5$ orientation profile, the boost law, the environmental and instrumental nuisance model, trials accounting, sensitivity floor, and calibrated joint threshold. The decision rule scores the complete branch manifold

$$C_4 < 0, \qquad B_0 = \frac{10}{21} C_4^2, \qquad B_6 = \frac{32}{315} C_4^2,$$

with every intrinsic coefficient at $1 \leq j \leq 5$ zero and the $j = 6$ vector on the rotated $\mathcal{I}_6$ orbit. The branch fails at five standard deviations or more if an isolated intrinsic C_4 is positive, an isolated lower-rank coefficient is nonzero, the linked sixth-order terms are excluded at adequate sensitivity, or the calibrated profile likelihood excludes the complete manifold. Support requires exclusion of the zero-coefficient baseline at five standard deviations or more, agreement with the linked manifold within two, rejection of named systematic alternatives, and an independent replication. A null is inconclusive, since the branch supplies no positive lower bound on a, as are incomplete covariance, insufficient sixth-order sensitivity after a negative C_4, and failure to isolate the carrier contribution. A failed verdict rejects the primitive twelve-port physical propagation branch, and applies to the framework as a whole only after a stronger source law makes that branch forced and exclusive.

Boundary 10.9. No statement in this section is a source-only determination of a physical constant or mass. The Koide identity and capacity identities are finite theorems; the tau window is a prospectively fixed conditional test, postdictive in premise ancestry; the lepton enclosures and the screen-grain separation are diagnostics that consume measured inputs; the electroweak chart and external scalar-pole fixture are uncomposed diagnostics and define no W/Z pole comparison; the de Sitter reading is conditional on unconstructed dictionaries. The angular-rank theorem is a finite theorem. The primitive-port and source-seam coefficient relations form separate prospective predictions on their named physical branches, with empirical scoring sealed until the required branch-specific contracts exist.

11 Physical interpretation maps and their consequences

The preceding theorems concern finite records, explicit response maps, and reconstruction implications. They become claims about nature only when a physical carrier realizes the operations and identifications in Definition 1.2. Table 4 records the resulting implications and the premise that can fail, keyed to the standing maps of Section 1.3.

Table 4: Physical questions, reconstruction consequences, and the standing maps each one consumes.

Question	Consequence of the stated realization	Maps
When is a measurement public?	A completed record is a schedule-independent normal form, and on the declared algebra-state surface its event projectors obey Theorem 4.1. A laboratory system must instantiate durable records, readback, protected boundaries, and the algebra-state map.	(M2)
When is event geometry Lorentzian?	The independent real axis and rank-three Gram quotient construct the finite (1+3) carrier and cone; canonical source height enters only through event placement. A common source-selected family satisfying exact order embedding with calibrated density, compatible refinement, manifoldlikeness, topology, and convergence gives the effective Lorentzian four-manifold of Theorem 6.4; the finite capture does not satisfy those physical hypotheses.	(M1), (M4)
What governs the gravitational response?	A realized null balance, Ward conservation, the Bianchi identity, and an independent scale identification give the field equation of Theorem 7.2.	(M2), (M4)
How do the gravitational and gauge branches share a source?	One carrier can feed the event and modular branch and the twelve-port response branch, with equality of G_{packet} and G_{Tan} a separate identification premise.	(M3), (M4)
Why this gauge type and matter image?	Complete A1 response and endogenous A2 transport force the compact local Lie type, and the separately supplied matter representation gives the conditional $\mathbb{Z}_6$ quotient and anomaly-free rank-fifteen module. Laboratory currents, quantum fields, physical matter poles, family multiplicity, and masses require further maps.	(M3), (M5)
What are position, energy, and the action?	Record readback gives the three-dimensional carrier, the modular ledger gives the internal energy with its exact composition law, and the realized path law gives the action with single-site minima equivalent to locally most-probable updates; stationarity alone is not equivalent to maximum likelihood.	(M1), (M2), (M5)
What gravitates without shining, and what does the vacuum weigh?	Modular charge with no electromagnetic readout classifies any geometric excess, the deep law carries one constant with $v^4 = GM_b a_0$, and fixed record capacity, under the declared inverse-density reading, gives $(w_0, w_a) = (-1, 0)$.	(M2), (M4)
Why is the gauge spectrum gapped?	Exact local repair is a positive Euclidean relaxation generator with an explicit rate floor, and its limit is identified with the continuum Hamiltonian gap under the reconstruction certificates.	(M5)
What propagation signature separates the primitive-port branch from the minimal baseline?	Proposition 10.4 fixes B_0 and B_6 from C_4 and fixes the complete rank-six shape up to one carrier orientation. The internal seam-repair operator is no spatial hop operator, and sector bridge, coherent frame transport, carrier-term isolation, and exclusivity are branch premises.	(M1), (M5)

11.1 The gravitational chain and its break point

The route from patches to gravity is one chain: consensus supplies the authenticated finite event poset and exact longest-parent-chain source height; independently, the real axis, rank-three source carrier, and celestial/null data construct the local (1+3) ambient precursor, while height enters its event placement; a physically faithful, count-calibrated, manifoldlike refinement family supplies the effective Lorentzian manifold; modular flow and half-sided inclusions supply the local time generator and null translations on that same family; and the entropy split with its stationarity conditions would establish the null-balance premise that Theorem 7.2 completes. The break point is visible. The normal-form, entropy, and tomography statements have exact finite layers, and the passage to gravity starts when one physical tower realizes (M4). Read through that map, the gravitational response is the large-scale compatibility condition on repaired records: the metric describes the stable causal relation among them, modular flow describes how a patch reads change, and entropy stationarity fixes the response that keeps local accounts compatible.

11.2 The dark sector: non-luminous charge, the deep galaxy law, and the vacuum coordinate

The two dark components can enter the physical Einstein branch only through separate typed adapters. A dark-sector adapter identifies non-luminous modular charge, its stress and the absence of an electromagnetic readout; an independent horizon–record adapter identifies a selected finite capacity with de Sitter entropy and the vacuum metric coordinate. Neither identification is part of the minimal finite Einstein-shape theorem. The companion papers carry the full conditional statements, premises, and comparisons for both [15, 14, 16].

Under the supplied universal-coupling, physical-stress and source-localization premises, sourcing gravity by modular charge gives a conditional classification identity: the rest-frame Einstein relation responds to total modular charge, the luminous-only relation holds exactly when the non-luminous charge vanishes, and a geometric excess over the luminous term is classified as non-luminous modular charge. This does not derive that a nonzero excess exists, its abundance, or a physical carrier attachment. A separate collar-recovery and modular-stress adapter supplies one candidate: anomalous modular energy on overlap collars, bounded by the collar recovery defect, absent at full recovery, and diluting as a^{-3} when the adapter's conservation hypothesis holds. For common envelope parameters $\kappa \geq 0$, $B \geq 1$, $r_0 \geq 0$, $\zeta > 0$, fixed positive collar radius and bounded stress constant, anomalous stress tends to zero as density grows at fixed positive depth, or depth at fixed positive density. This is asymptotic suppression; exact finite-source vanishing, source existence, abundance and galactic localization remain separate premises.

Conditional on two named premises, the deep-regime profile is characterized rather than supplied as an additional functional form. An enclosed-mass law that is degree-one scale covariant in the radius and composes independent sources in quadrature, because variances of independent collar defects add, carries a unique constant $a_0 > 0$ with $M_A(r) = r\sqrt{M_b a_0/G}$; this forces the deep anomalous acceleration $a_A = \sqrt{a_b a_0}$, and hence the asymptotic total-acceleration relation, flat rotation curves, and the baryonic Tully–Fisher relation $v^4 = GM_b a_0$, with each premise proved load-bearing by an explicit counterexample law. A per-cut model reduces quadrature composition to variance additivity on nr nested cuts and gives the exact dictionary $a_0 = Gn^2c$, with only the product n^2c identifiable. The corrected SPARC diagnostic compares the total observed acceleration with $a_b + \sqrt{a_b a_0}$, not with the anomalous term alone. An older comparison against a finite-radius total-speed proxy produced displaced bootstrap intervals. A matched seen-data diagnostic instead compares the baryon-subtracted anomalous speed at the outermost retained point, applies the same deep cut, and excludes bulge galaxies because the snapshot lacks the disk/bulge luminosity split required by its mass denominator. On 97 common galaxies its paired log-ratio interval $[-0.117, 0.026]$ dex contains zero. Because several observable, sample, and combination choices change together, this supports but does not causally identify a mixed-proxy explanation.

A separate source-bound replay calibrates the published empirical full-RAR interpolation rather than the OPH deep law. It authenticates the committed CDS/VizieR SPARC snapshot, reproduces the published 153-galaxy parent selection, and retains 2696 points in 147 contributing galaxies after the strict fractional-velocity-error cut, three points more than the published 2693-point census. At the standard fixed stellar mass-to-light values it gives $a_0 = (1.1613 \pm 0.0802) \times 10^{-10}\,\mathrm{m\,s^{-2}}$, where the displayed width is the galaxy-bootstrap standard deviation, with 0.1327 dex residual scatter. Equal parent-galaxy weighting shifts the fitted scale by -9.58%, while inverse fractional-velocity-variance weighting alone shifts it by $+15.49\%$. These alternatives are estimator-sensitivity controls, not likelihoods. The interpolation and acceleration scale were known before the replay, no source law fixes a_0, and the nuisance model is incomplete, so the result strengthens data custody and arithmetic rather than supplying OPH-specific evidence [15, 1].

A per-point Gaussian data term with gridded nuisance penalties and an equal-correlation covariance scan has also been computed. It is a penalized profile objective rather than a maximum-likelihood analysis: its delta-objective sublevel sets are uncalibrated sensitivity contours, some disconnected, and the data quadratic-form ratios sit far above one under both nominal denominator conventions, so it can reject no future source value, and a prospectively frozen covariance, intrinsic-scatter, and coverage calibration is unbuilt. An isotropic ambient density has zero trace-free quadrupole, and its traceful isotropic tidal scale is four orders of magnitude below the uncertainty of the dimensionally comparable Cassini quadrupole coefficient, although Cassini bounds that isotropic channel only indirectly. The value of a_0, the physical attachment of the envelope, a full relativistic stress and lensing law, and the abundance are among the named open constructions [15].

The vacuum side of the sector is the capacity coordinate of Section 10.1 read as a density. Identify $\rho_{\mathrm{DE}}(a) = \kappa/N(a)$ at constant κ and take the continuity definition $w(a) = -1 - \frac{1}{3}\,\mathrm{d}\ln\rho/\mathrm{d}\ln a$. The two definitions give

$$w(a) = -1 + \frac{1}{3}\frac{\mathrm{d}\ln N}{\mathrm{d}\ln a}, \tag{24}$$

so a fixed record capacity implies $w = -1$ and hence the exact Chevallier–Polarski–Linder point $(w_0, w_a) = (-1, 0)$ with no continuous freedom inside that branch, a monotone nondecreasing capacity implies $w \geq -1$ at every epoch, and $w < -1$ implies local capacity loss. The power-law family has constant $w = -1 + \epsilon/3$ and $w_a = 0$. Equation (24) is an algebraic consequence of the two definitions and derives neither fixed capacity, the density identification, nor a closed dark sector. The scoring asymmetry is displayed with it: exclusion of $(-1, 0)$ in any direction can fail the fixed branch, while compatibility with that point gives no support, since it is also the ΛCDM null. A $w_0 w_a$CDM posterior is a model-dependent and likelihood-dependent projection rather than a direct capacity measurement, so a comparison requires a post-freeze dataset combination, a scored range, a statistic, complete thresholds, fallbacks, and a direction-neutral nuisance policy fixed in advance [3]. Existing survey posteriors and the display-bias mock are seen diagnostics.

A hash-checked retrospective postprocessing of the four official DESI DR2 Cobaya chains for each default CMB combination finds posterior-weight fractions in the full CPL monotone-capacity subset on $0 \leq z \leq 2$ of 0.00705% for BAO+CMB, 0.1125% with Pantheon+, 0.01206% with Union3, and 0.02054% with DESY5 [77, 1]. The respective raw tail counts are only 6, 126, 15, and 11, and at least one of the four chains has no tail row in two combinations. The fractions are therefore prior- and likelihood-dependent retrospective diagnostics, with explicit rare-tail resolution warnings; they are not branch probabilities, a frequentist exclusion, or a frozen score. Gaussian moment distances of the fixed point are likewise recorded only as diagnostics, not as official profile-likelihood or evidence results.

There is also a conditional local clock bridge, but it needs three explicit physical premises beyond the displayed capacity formulas: (B1) the electroweak closure capacity is the same physical N; (B2) the finite closure relation is an epochwise physical law; and (B3) physical evolution selects the committed root branch and co-variation convention while the local clock reads the same homogeneous cosmological α with the required time map. An outward-rounded implicit-function certificate proves the selected declared branch continuously differentiable without finite differences on $|\Delta \ln \alpha| \leq 10^{-5}$ and gives

$$\frac{\mathrm{d}\ln N}{\mathrm{d}\ln \alpha} \in [-0.214173865, -0.206176031], \qquad \frac{\mathrm{d}\ln \alpha}{\mathrm{d}\ln N} \in [-4.850224325, -4.669103775].$$

The first interval excludes zero before the reciprocal is formed. Under B1–B3, the optical-clock drift result of Filzinger et al. then gives the local diagnostic $|1 + w_0| \leq 6.94 \times 10^{-10}$ [78]. This does not derive that cosmic tangent, does not justify extrapolating it over order-one changes, and does not empty the monotone branch: arbitrarily small positive capacity drift remains viable.

11.3 A common source, and measurement as a completed record

One observer-consensus tower feeds two readout branches. The overlap, record, and repair structure feeds the event and modular branch, while the twelve-port incidence, complete reversible response, and endogenous overlap transport feed the local gauge-type branch, with trace-balanced blocks supplying a separate conditional matter branch. A single self-reading carrier could supply both sets of antecedents. That common source identifies gravity with no gauge boson and fixes no coupling, and it leaves the two gauge constructions distinct: G_{Tan} is reconstructed from a sector category, while G_{packet} is the maximal faithful image of the explicit finite response and matter module. A completed unification requires joint physical realization and a proof that the corresponding current diagrams commute.

Measurement sits at the transition from a private repairable record to a protected public normal form. A record is public when it has survived every authorized comparison and no repair can change it; event projectors, weights, and state updates are defined on the completed algebra-state surface, and state update moves from one public record class to the next. A physical application must identify actual durable records and verify the transaction and readback premises, and the theorem constructs no state space for an interacting continuum field theory.

11.4 Thermodynamics from conditional repair

The four laws form a finite conditional theorem package by short elementary arguments once Axiom 3 is read on transition distributions and the typed source and physical receipts are supplied. Axiom 3 applies separately to states and to transition distributions, and both information projections are solved exactly. The state projection with faithful reference and conserved constraints is the Gibbs exponential family by the information-projection Pythagorean identity, and on one supplied nondegenerate finite spectrum equality of Gibbs distributions identifies the inverse temperature, which is the finite zeroth-law transitivity receipt rather than an equilibration theorem. The transition projection onto the fibre of the complete repaired visible datum is weighted conditional resampling from the same reference, the weighted observation-fiber projector of the consensus construction. That kernel is stochastic, idempotent, reversible, and stationary, fixes every fibre-measurable charge, and contracts relative entropy to the reference, so the second law is a data-processing theorem with the modular form $\Delta S \geq \Delta\langle K\rangle$ on τ-preserving channels and the Landauer erasure bound as a corollary.

The inequality is the mean of a fluctuating entropy production obeying exact integral, pointwise, and level-set fluctuation identities, and the mean is an identity rather than a bound: one repair step's mean entropy production equals the relative entropy from the input state to its repaired image exactly, so it is strictly positive whenever the step changes a strictly positive state and vanishes precisely on the fibre-conditional-reference fixed points. That is a strict single-step orientation off the repaired manifold. The identical kernel is idempotent and therefore produces zero further dissipation after the first projection, so a sustained macroscopic arrow is unclaimed. Detailed balance with a linear Poisson solver on centered currents gives a symmetric positive-semidefinite finite Green–Kubo matrix with an exact cutoff remainder [28, 29]; for each current pair, iterating the idempotent full-fibre time-step gives either zero positive-lag correlation or nonstabilizing partial sums, so a decaying memory tail with a stabilizing sum requires a separately sourced nonidempotent evolution. For a finite joint update the exact bookkeeping split is $\Delta U = \delta Q + \delta W + \operatorname{Re}\operatorname{tr}(\delta\rho\,\delta H)$, the two-term form holding for an ordered fixed-H or fixed-ρ stroke, and the composed repair heat stroke holds H fixed. At every finite regulator the excited Gibbs mass is bounded by $\frac{d-g_0}{g_0}e^{-\beta\Delta}$, the entropy limit $k_B \log g_0$ is a finite corollary of that bound, and finite-step unattainability follows because faithful repair and pinching steps cannot reach the rank-deficient zero-temperature state. Through the central split $K = 2\pi B + Z$ one repair step with a fibre-measurable central charge obeys the cap Clausius inequality $2\pi\Delta\langle B\rangle \leq \Delta S$, which discharges the Einstein branch's finite first-law premise package on the simplex tangent space of this model.

Boundary 11.1. The strict-descent normalizer that settles public facts is a different map and carries no entropy inequality, with an explicit certified two-point counterexample. The pinned source artifact has an exact negative verdict: its state-side resampling action is idempotent while its recurrent transition action has a nonconstant eigenmode with $\lambda = 665437/726948 \in (0, 1)$, so every dynamic intertwiner kills that mode, and its stationary mass is no deterministic pushforward of the state reference. An exact preflight exhausts one declared random-scan grammar, excluding a nonconstant protected observable inside it and leaving adaptive, zero-weight, dilated, and new-source routes untouched. Physical thermodynamics additionally requires source-justified transitions, a shared reference, energy-clock calibration and uniform low-temperature tails on one cofinal family [9, 1].

A constructive finite alternative keeps the full-fibre projector P as equilibrium and uses local heat-bath evolution T. An exact eight-state nonproduct witness preserves a nonconstant public bit and obeys $TP = PT = P$ and the entrywise minorization $T^2 \geq (7/96)P$. On fibre-centred observables, $\|T^n\| \leq (89/96)^{\lfloor n/2\rfloor}$, giving a unique centred Poisson solver and convergent Green–Kubo sums. A merely globally centred protected current retains covariance $2/9$. Thus conserved records coexist with decaying memory under this supplied law; native source selection and physical transport are separate [9, 1].

11.5 What fails together

Authenticated semantic commits and their read-from relation supply the finite events and informational order directly. Independently, the real axis joined to the rank-three source Gram quotient constructs a four-dimensional ambient Lorentzian target carrier and its null-direction sphere; canonical source height enters only through event placement, and intrinsic poset dimension remains a continuum diagnostic. The direct causal-set route uses a common physical refinement passing the faithful-order, density, manifoldlikeness, topology, uniqueness, and curvature rows of Theorem 6.4;

neither event charts nor signature are inserted as independent starting data. On the other branch the carrier exposes twelve response channels grouped by icosahedral symmetry into four bands, complete reversible response closes their public tangent under commutators, and observer agreement requires rechartings to act inside that same response, so compactness and the single fixed line leave the Standard Model gauge Lie type. A separate matter module tests which charges coexist without anomalies, and it turns no finite channel into an observed particle.

The conditional implications share premises, so they fail in groups. Failure of (M4) removes the effective-spacetime and Einstein conclusions together. Failure of (M3) removes the gauge and matter interpretations together. Failure of (M2) removes every statement in laboratory units, including the mass-energy slope and the four-law package in physical form. The finite normal-form, invariant-theory, and algebraic results retain their stated domains under all three. Table 4 summarizes the principal physical interpretation claims made here.

12 Machine verification and reproducibility

Each result maps to an artifact class:

- a Lean library with no admitted propositions in the dependency closure reported here, including formal premise boundaries and countermodels;
- exact-arithmetic code receipts: integer and rational computations, interval certificates with outward rounding, and $\mathbb{F}_2$ rank computations, each emitting a canonical hashed payload;
- simulation receipt bundles pinning the applicable source or module revision, configuration, seed where stochastic, grids, tolerances, and output hashes, with deterministic reserialization distinguished from fresh source replay;
- verifiers that recompute verdicts from clause vectors and reject receipts whose stored verdict disagrees, so a caller cannot assert a truth flag directly;
- adversarial negative controls: every certificate ships with mutations that must be detected, and a control that cannot fail is treated as a defect of the certificate, not as support;
- provenance rules under which machine-readable records classify every quantitative row as a source result, reconstruction implication, diagnostic, prospectively fixed test, or rejected candidate and reject any stronger classification unsupported by the recorded dependencies.

The repository, receipts, schemas, and rebuild instructions are public [1, 2]. The formal and executable stack checks the claims assigned to it and enforces their stated boundaries. Analytic arguments in this manuscript retain their displayed mathematical proofs, and no part of the stack establishes physical realization.

12.1 Artifact map

Table 5 names the artifact family behind each group of results. Repository continuous integration checks the theorem-count floor, rejects admitted proofs in the public library, and rebuilds the finite certificates used here; simulator replay is governed by the simulator receipts and reproduction commands. Exact file paths, manifests, receipts, mutation suites, and negative controls accompany each module in the public repositories [1, 2]. Online Resources 1 to 3 supply compact snapshots of the formal library, the exact certificates, and the finite evidence cited in this table.

Table 5: Principal verification artifacts and their scope.

Results	Principal artifacts
Normal forms, stability, refinement, complexity, and the transactional diamond	The observable-normal-forms library with the component paper's proofs and reductions; local-diamond and Newman arguments proved in this manuscript; the finite transaction-model verifier
Born, Lüders, Tsirelson and its saturation, no-cloning	The event-algebra modules for the basic interface, conditioning, the bound, partition and state maps, and the public record algebra, with per-module axiom audits
Rank-three completion, Coulomb–Green, seam holonomy, and the local face action	The port-Gram repair band, covariance, and frame-quotient modules with a source-pinned receipt and independent verifier; the discrete Coulomb–Green, position-space, and local-face Maxwell modules
Edge-center entropy and central defects	Overlap-cocycle and one-sided reduction proofs in this manuscript; the Einstein-branch entropy module; the identity-channel collar model with its no-go boundary
Gauge Lie type, matrix witness, declared matter-kernel image, and exterior selection	The screen modules for the carrier action, commutant, incidence response, and holonomy bridge; the exact-arithmetic port current certificate; the lattice quotient, trace-balanced kernel, and exterior scan modules
Mass shell, declared internal-clock identity, inertia slope, and the derived action	The mass-shell, declared internal-clock, composite-momentum, proper-time action, port-charge coupling, modular-additivity, and Gibbs-reading modules; the path-Gibbs, Euler–Lagrange, Noether, and enrichment-selection modules; the calibration import with its non-forcing receipt
Coupled matter continuum, interacting Hilbert space and model duration	Uniform-refinement and energy-estimate proofs; full kinetic-metric reduction, completeness and operator-domain proofs; charged self-reading execution and configuration-clock receipts with independent equation and provenance verifiers [9]
Null tomography, small-ball arithmetic, and tensor completion	The Einstein-branch tensor module for null directions, design and decoder, injectivity, and metric ambiguity; analytic constants in this manuscript; explicit null-balance, Ward, Bianchi, and scale premises
Angular-rank theorem, frozen branches, and the dispersion band	The primitive-port prediction and carrier-class band modules; the multipole, universality, prediction, and band receipts with their mutation tests; the custody-bound prediction register
Dark sector, the vacuum coordinate, and the closure roots	The dark-sector, deep-profile, per-cut, collar-premise, and fixed-capacity modules with the matched seen-data diagnostic; the capacity closure and nonidentifiability modules; the interval-certified grain root and the posterior propagation record
Repair gap and the conditional Yang–Mills identification	The Yang–Mills gap, witness, and repair-gap-chain modules; the collar gap certificate and the finite transfer receipt with its test
Exact federation under the canonical law	The level-six icosahedral configuration and seeds, primitive seam arrays, initial and terminal loads, per-schedule V ledgers, quotient hashes, per-carrier response kernels, standalone verifier, and public archive
Closure loop at carrier scale	The event logs of the canonical, integer, overwrite, tetrahedral and octahedral sources, the recovered specifications and invariant vectors, producer-free verifier, and public archive

Table 5 continued

Results	Principal artifacts
Causal poset	The source-record family receipts at $q \leq 55$, the carrier realization receipts and event logs, the standard-library generator that rebuilds the poset from its definitions, checker, and public archive
Electroweak chart and finite local domain	Certified contour and second-sheet outputs with fail-closed producer checks; the analytic theorem of this manuscript; the exact source-order and local-domain receipts with independent verifiers in the simulator repository

13 Empirical meaning and falsifiability

The custody-bound register fixes target definitions, exclusion thresholds, precision floors, exposure classes, and decision rules before eligible comparison [3]. Propositions 10.4 and 10.5 are the prospective discriminators in this paper. Their coefficient rays are fixed separately, their excluded exposure classes are declared, and their dataset-specific comparisons are sealed under the conditions stated in Section 10.6. A qualifying failure rejects the tested physical propagation branch. OPH-wide scope requires a stronger source law that supplies the physical action, readout, frame transport, and exclusivity theorem for that branch.

The gravitational and gauge conclusions have separate empirical boundaries. The Einstein composition requires one physical refinement tower satisfying its event, modular, stress, entropy, continuum, vacuum, and scale premises together. The gauge and matter conclusions require laboratory-current attachment and a continuum operator limit. Failure to construct those maps leaves the corresponding finite theorems intact and removes their claimed physical application. Non-identifiability theorems determine which finite source interfaces cannot supply the missing information.

The dark sector and the gauge-spectrum statement carry their own decision shapes. A world-average charged-lepton measurement more than three standard uncertainties from the balanced-circulant center refutes the equal-rank-two tracial reading of the face-circulant sector. A resolved deep-regime galaxy sample that violates degree-one radial scale covariance or quadrature composition of independent sources removes the uniqueness of the deep law, and a fitted constant that fails to be common across the two channels removes the single-constant reading. For the vacuum coordinate, the scoring is one-sided: exclusion of $(w_0, w_a) = (-1, 0)$ in any direction fails the fixed-capacity branch, while agreement with that point supports nothing, because it is also the ΛCDM null. The Yang–Mills identification fails if the support-visible continuum construction fails, if reflection positivity fails, or if the uniform repair gap is absent; failure of the finite $\mathbb{Z}_2$ diagnostic retracts that computation alone.

14 Discussion and comparison

Operational reconstructions derive quantum structure from information and composition principles [19, 20, 21]. OPH makes a narrower statement at the quantum stage: a completed finite record, once represented by an algebra-state pair, carries the standard event, conditioning, expectation, and correlation identities, with the observer-indexed normal form and its transaction contract as the distinctive input. Those reconstructions derive the quantum state space itself from operational axioms, while OPH takes the finite algebra-state representation as an explicit input and contributes the completed-record surface on which the identities operate together with the consensus theorem that defines completion. Four neighbors calibrate the record story. Quantum Darwinism locates classical objectivity in redundant environmental records [49], and the completed-consensus record plays the same public role with redundancy replaced by protected overlap agreement and an explicit repair dynamics. The cellular-automaton program shares the finite substrate [50], while OPH locates the quantum identities on the completed record surface rather than in an underlying ontological basis. Wigner's-friend extensions constrain agents reasoning about one another's unfinished measurements [51], whose premises concern pre-consensus perspectives that the completed surface does not represent. Relational quantum mechanics treats physical facts as relational [47] and the thermal-time program reads time from modular structure [48]; OPH attaches public facts to protected overlap records and consumes normalized modular flow only on a typed common tower, with its local-diamond proof neighboring asynchronous agreement [44, 45].

The gravitational branch belongs to the family of thermodynamic and entanglement-based routes to geometry [58, 59, 62], with Bisognano–Wichmann modular flow and half-sided modular inclusions as established analytic ingredients [55, 56, 57]. The contribution of Theorem 7.2 is the explicit composition contract, with event, modular, stress, entropy, asymptotic, vacuum, and scale premises in one statement and the finite null ambiguity identifying exactly where a metric-proportional term enters. The route from a null-projected balance and the Bianchi identity to the full equation, with Λ as an integration constant, is familiar from trace-free and unimodular formulations [60, 61]; what is added here is the finite nine-direction design with exact determinant and decoder constants and the premise contract (G1)–(G6) stated as one implication.

The gauge section contains two reconstruction routes and one conditional global-form witness. Doplicher–Roberts/Tannaka reconstruction supplies the structural sector group [66, 67]; independently, complete A1 response and endogenous A2 transport force the local Standard Model gauge Lie algebra; and the explicit matrix map with the exterior-module scan gives the conditional packet group. Equality of the sector group and the packet group is a separate commuting-square premise, and that separation keeps the categorical reconstruction, the local Lie-type theorem, and the physical global group apart.

A useful comparison of unification proposals asks how much physical structure follows from each declared input and which observations distinguish the resulting models. OPH offers a concrete case through its protected-record construction, gauge Lie-type constraint and coupled matter example, with proofs and executable checks attached to individual implications. Formal verification measures deductive reliability; predictive success requires physical realization and eligible measurements. Neither theorem counts nor numerical coincidences define a probability that an architecture describes nature. The realization premises of Section 13 specify that scientific test for OPH.

15 Conclusion

The finite architecture relates records, spatial carriers and matter through explicit mathematical interfaces. Its normal-form theorems control agreement, stability and refinement; finite effect valuations yield the probability rule for quantum records; carrier completion gives a three-dimensional position readback; and complete response with endogenous transport forces $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. On the declared matter menu, exhaustive selection yields an anomaly-free fifteen-state generation with common $\mathbb{Z}_6$ kernel.

A supplied charged-scalar/Maxwell action connects these mathematical objects to a controlled nonlinear continuum trajectory and an interacting quantum state space. Uniform refinement gives first-order convergence of its real quartic-wave sector. On the fixed mesh, the complete coupled metric gives a unique self-adjoint Hamiltonian under the declared quantization prescription. Bounded self-reading patches execute the charged model, while its changing configuration determines model duration at fixed energy. These constructions exhibit calculable fields, states and equations; a source-selected action and one common physical realization are additional requirements.

The gravitational branch assumes a common physical refinement tower with the stated stress, entropy, continuum and scale data. Null tomography, Ward conservation and the Bianchi identity then yield the Einstein equation up to a cosmological constant. Constructing that tower with the gauge and matter readouts is a separate requirement. Modular charge carrying no electromagnetic readout classifies a non-luminous gravitating sector whose deep-regime law is unique under scale covariance and quadrature composition, carrying one constant into flat rotation curves and $v^4 = GM_b a_0$; reading the dark-energy density as $\rho_{\rm DE} = \kappa/N$ sends fixed capacity to $(w_0, w_a) = (-1, 0)$; and on the support-visible compact-gauge branch the continuum Yang–Mills Hamiltonian gap is the limiting repair gap under the named certificates.

The closure formulas are retrospective diagnostics of declared maps. The prospective content is the frozen dispersion surface, where $C_4 < 0$ with $B_0/C_4^2 \geq 10/21$ and B_6/B_0 inside $[-16/135, 16/75]$ excludes the whole declared carrier class at once when a resolved measurement falls outside it.

The finite theorems establish implications on their stated domains; the quantitative formulas are diagnostics or prospectively fixed tests. The five maps of Section 1.3 are consumed rather than constructed, so these results do not establish that the architecture describes our universe. A physical reconstruction requires one realization preserving the proved relations across records, fields, states and scales. The closure hypothesis of Section 1.5 proposes that such a self-consistent structure can determine its own physical content; the conditional theorems specify what that proposal must realize.

Reproducibility statement

All theorems, certificates, simulation configurations, receipts, and the formal library are public in the project repositories [1, 2]. The paper release manifest records the release identifier (r2040) together with PDF hashes and sizes. Computational receipt bundles pin the applicable source or module revision, configuration, seed where stochastic, and output hashes. Clean-checkout rebuild instructions and negative controls accompany the artifacts. Machine-readable claim and prediction provenance is available with the artifacts [3].

The Lean source tree, pinned toolchain, exact verifiers, manifests, receipts, negative controls, tests, the level-six exact-federation archive, the closure-loop archive, the causal-poset archive, and local-domain evidence are identified in Table 5. The complete simulation source is maintained in the public simulator repository at the evidence-producing revisions [2].

This manuscript synthesizes and strengthens results whose longer proofs also appear in public component preprints on observable normal forms, finite event algebras, consensus, Einstein reconstruction, and gauge structure [5, 6, 7, 10, 11]. The component relationship is disclosed here so that overlap is visible.

Four Online Resources accompany this article. Online Resource 1 contains the tracked Lean source tree with its pinned toolchain and dependency manifests. Online Resource 2 contains the exact-arithmetic certificate closure with its receipts, reference manifests, negative controls, and tests. Online Resource 3 contains the hash-bound finite evidence for the exact-federation, closure-loop, causal-poset, and local-domain computations reported here. Online Resource 4 contains the component preprints named above together with their sources. Each archive carries its own manifest, license, and reproduction instructions, and each is a snapshot of material that is also public in the repositories [1, 2].

Statements and Declarations

Funding. The authors declare that no external grants or dedicated third-party research funding were received for the preparation of this manuscript.
Author affiliations. Bernhard Mueller is affiliated with Pragma Research Inc. and with the Information Physics Institute. Alexander Osika is affiliated with EtherWorks. Jinwook Kim is affiliated with Oraclizer Labs.
Author contributions. Bernhard Mueller developed the OPH concept and mathematical synthesis, wrote the main manuscript, and coordinated the formal and computational evidence. Alexander Osika proposed that the screen pixel ratio and the correctable-record capacity are derivable rather than fitted, supplied the detuning fixed-point equation that determines the pixel ratio, and originated the echosahedral patch-hardware design on which the screen-microphysics branch is built, in addition to the physical-realization and hardware boundary, prototype framing, and manuscript review. Mario Poneder contributed the finite port-current construction and verifier, gauge and matter certificates, and proof review of the edge-entropy and Einstein-normalization branches. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype-engineering review. Ben Cassie contributed implementation-claim and physical-evidence review. Peter Nguyen contributed proof auditing, branch-boundary analysis, compact-gauge and global-quotient analysis, and manuscript consistency review. Jinwook Kim developed the mathematical constructions, proofs, and counterexamples for the mechanism-variant comparison kernel and conditional example, protected-behavior first-hit obstruction profiles and finite-state transport, source-derived phase-effect selection, canonical adaptive-repair endpoint stratification, cumulative attempt-capacity bounds and sharpness results, and input-independent fixed computation-federation progress and execution. He constructed a provably fair emission-order round-robin scheduler with a linear exact-output attempt bound, established order-sharp quadratic accepted-step bounds for the actual compiler with the exponential fanout control, and proved the historical weak-relation fair-stuttering no-go and the weak-fair delay versus bounded-waste horizon separation. He formalized these results in Lean, connected the repair and execution results to Tower endpoints, and contributed their manuscript, claim-registry, ledger, and reproducible-build surfaces. David Matscheko contributed proof review of consensus repair, observable normal forms, modular and Einstein algebra, edge entropy, hypercharge, and the $\mathbb{Z}_6$ quotient. Jonathan Hill contributed Lean formalization and proof auditing of observable normal forms, refinement and repair results, complexity classifications, and artifact coverage. William T. Glynn contributed simulator development, computational implementation, reproducible-build infrastructure, and release-validation work in the main OPH research repository. Maarten Antonie Visser contributed consensus-protocol review, physics feedback on the foundations and emergent-spacetime framing, and the source-derived causal-poset reading of authenticated provenance that the event-precedence layer implements. Kale Arnav Anirudha contributed consensus-paper material, asynchronous-agreement analysis, repair-map definitions, and formal review of the protocol surface. Brieuc de La Fournière contributed finite Lean constructions on the source-derived causal-order and carrier-symmetry surfaces, including the cover-relation provenance compiler and the antipodal port-action identification, and external comparison controls for the charged-lepton branch. All authors reviewed and approved the manuscript. All authors agree to be accountable for all aspects of the work.
Corresponding author. Correspondence should be addressed to Bernhard Mueller.
Email: `bernhard@floatingpragma.ai`.
Data availability. The finite witnesses, simulation configurations, receipts, scientific claim registries, and release manifests supporting this study are available in the public OPH and simulation repositories [1, 2]. Individual receipts and their bundle documentation provide the source and out-

put provenance for computational claims. The artifact map identifies the level-six exact-federation archive, the closure-loop archive, the causal-poset archive, and the local-domain evidence cited in the manuscript. Online Resource 3 supplies that evidence as a hash-bound archive.

Code availability. The Lean sources, exact-arithmetic verifiers, finite scans, simulation code, tests, and rebuild instructions are available in the same public repositories [1, 2]. The computational claims in this manuscript identify their principal files in Section 12. Online Resources 1 and 2 supply the formal and exact-computation sources as compact snapshots, and Online Resource 4 supplies the component preprints.

Ethics approval and consent to participate. Not applicable. The study involved no human participants, human data, or animals.

Consent for publication. Not applicable.

Competing interests. Bernhard Mueller is affiliated with Pragma Research Inc. and with the Information Physics Institute. Alexander Osika is affiliated with EtherWorks. Jinwook Kim is affiliated with Oraclizer Labs. The authors participate in OPH-related research, software, simulation, or prototype-development programs and may receive professional or reputational benefit from this work. These relationships are disclosed. The authors declare no other competing interests.

References

[1] FloatingPragma (2026). Observer Patch Holography: Lean library, exact code receipts, finite certificates, and scientific claim registries. `https://github.com/FloatingPragma/observer-patch-holography`

[2] Mueller, B. (2026). OPH physics simulator and reproduction source. `https://github.com/muellerberndt/oph-physics-sim`. Exact-federation archive: `https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/exact_federation_L6_canonical_20260909`. Closure-loop archive: `https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/closure_loop`. Causal-poset archive: `https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/source_net_causal_poset`. Active refinement-readout revision: `https://github.com/muellerberndt/oph-physics-sim/tree/036b12b608a9ca4c19d69f5e951b3b83a7a51cd8`. Finite local-domain revision: `https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506`. Finite local-domain data: `https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506/data/local_domain`.

[3] FloatingPragma (2026). Precommitted comparison certificates, custody records, and the frozen prediction register. Immutable source receipt: `https://github.com/FloatingPragma/observer-patch-holography/blob/66176656dc1143f9ec50ba1a6e409c403545857f/code/a5_fingerprint/runtime/spin_six_primitive_port_prediction_receipt.json`. Append-only custody packet: `https://github.com/FloatingPragma/oph-meta/tree/8cc5261653e37cbca0e6017fcc95a9fe7f649963/falsification/frozen_targets/fz11_2026-07-31`. Source-seam edge receipt: `https://github.com/FloatingPragma/observer-patch-holography/blob/bc5595f8dbb2d2886e2a64ddf447f69fbb00eb3f/code/a5_fingerprint/runtime/seam_current_edge_prediction_receipt.json`. Source-seam custody and decision-rule clarification: `https://github.com/FloatingPragma/oph-meta/tree/25da61a800226e0232336ccc86de8dec7d6b51c6/falsification/frozen_targets/fz12_2026-08-02`. Koide custody packet: `https://github.com/FloatingPragma/oph-meta/tree/df097d8fe7c38d0

08a1ba7827f7d573286ae2012/falsification/frozen_targets/fz10_2026-07-28. Live register: https://github.com/FloatingPragma/observer-patch-holography/blob/main/claims/frozen_prediction_register.json.

[4] FloatingPragma (2026). *The OPH Axiom Reference.* https://github.com/FloatingPragma/observer-patch-holography/blob/ff1981be069d3e5fa70d953e27b3786dd4c2b99d/docs/AXIOM_REFERENCE.md

[5] Mueller, B., Kim, J., Matscheko, D., and Hill, J. (2026). *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*. Public manuscript, distributed as extra/observable_normal_forms.pdf in the same repository revision as this paper. https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/observable_normal_forms.pdf

[6] Mueller, B. (2026). *Verified Projection-Event Calculus in Lean 4: Bundled Arbitrary-Partition Pinching, Lüders Retractions, and CHSH Interoperability*. Public manuscript and formal artifact. https://github.com/FloatingPragma/observer-patch-holography/blob/ff1981be069d3e5fa70d953e27b3786dd4c2b99d/extra/machine_checked_finite_event_algebras.pdf

[7] Mueller, B., Xue, K., Kim, J., Anirudha, K. A., Matscheko, D., and Hill, J. (2026). *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/ff1981be069d3e5fa70d953e27b3786dd4c2b99d/paper/reality_as_consensus_protocol.pdf

[8] Mueller, B., Osika, A., Xue, K., Cassie, B., and de La Fournière, B. (2026). *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in Observer-Patch Holography*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/screen_microphysics_and_observer_synchronization.pdf

[9] Mueller, B., Osika, A., Poneder, M., Xue, K., Cassie, B., Nguyen, P., Kim, J., Matscheko, D., Hill, J., Glynn, W. T., Visser, M. A., Anirudha, K. A., and de La Fournière, B. (2026). *From Observer Consensus to Standard Physics*. Public manuscript. https://wkaxfdgxoqmghwgshymt.supabase.co/storage/v1/object/public/papers/from_observer_consensus_to_standard_physics.pdf

[10] Mueller, B., Osika, A., Poneder, M., Xue, K., Nguyen, P., Visser, M. A., Matscheko, D., and de La Fournière, B. (2026). *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[11] Mueller, B., Osika, A., Poneder, M., Xue, K., Nguyen, P., and Matscheko, D. (2026). *Deriving Standard Model Gauge Structure from Observer Overlap Consistency*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/ff1981be069d3e5fa70d953e27b3786dd4c2b99d/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[12] Mueller, B., Osika, A., Poneder, M., and Xue, K. (2026). *Deriving the Particle Zoo from Observer Consistency*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/7260dc7c4ac05f84c021167f1e095a58998573f1/paper/deriving_the_particle_zoo_from_observer_consistency.pdf

[13] Mueller, B. (2026). *The Positive-Chamber Koide Identity for Icosahedral Face Circulants*. Public manuscript and formal artifact. https://github.com/FloatingPragma/observer-patch-holography/blob/7260dc7c4ac05f84c021167f1e095a58998573f1/extra/koide_identity_from_positive_c3_face_circulants.pdf

[14] Mueller, B. (2026). *The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/7260dc7c4ac05f84c021167f1e095a58998573f1/extra/de_sitter_time_advance_sign_from_fixed_screen_capacity.pdf

[15] Mueller, B. and Matscheko, D. (2026). *Observer-Patch Holography and the Dark Sector: Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/7260dc7c4ac05f84c021167f1e095a58998573f1/cosmology/oph_dark_matter_paper.pdf

[16] Mueller, B. (2026). *Observer-Patch Holography Cosmological Vacuum and Structure Formation: Finite Transfer Conditions, Fluctuation Ensembles, and Proto-Objects*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/7260dc7c4ac05f84c021167f1e095a58998573f1/cosmology/oph_cosmological_vacuum_and_structure_formation.pdf

[17] Mueller, B. and Hill, J. (2026). *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics: A Support-Visible Route to the Clay Problem*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/7260dc7c4ac05f84c021167f1e095a58998573f1/extra/yang_mills_gap_clay_problem.pdf

[18] Jaffe, A. and Witten, E. (2000). *Quantum Yang–Mills Theory*. Official problem description, Clay Mathematics Institute Millennium Prize Problems. https://www.claymath.org/millennium/yang-mills-the-maths-gap/

[19] Hardy, L. (2001). "Quantum theory from five reasonable axioms." arXiv:quant-ph/0101012. https://arxiv.org/abs/quant-ph/0101012

[20] Chiribella, G., D'Ariano, G. M., and Perinotti, P. (2011). "Informational derivation of quantum theory." *Physical Review A* **84**, 012311. https://doi.org/10.1103/PhysRevA.84.012311

[21] Masanes, L., and Müller, M. P. (2011). "A derivation of quantum theory from physical requirements." *New Journal of Physics* **13**, 063001. https://doi.org/10.1088/1367-2630/13/6/063001

[22] Born, M. (1926). "Zur Quantenmechanik der Stoßvorgänge." *Zeitschrift für Physik* **37**, 863–867. https://doi.org/10.1007/BF01397477

[23] Lüders, G. (1950). "Über die Zustandsänderung durch den Meßprozeß." *Annalen der Physik* **443**, 322–328. https://doi.org/10.1002/andp.19504430510

[24] Clauser, J. F., Horne, M. A., Shimony, A., and Holt, R. A. (1969). "Proposed experiment to test local hidden-variable theories." *Physical Review Letters* **23**(15), 880–884. https://doi.org/10.1103/PhysRevLett.23.880

[25] Cirel'son, B. S. (1980). "Quantum generalizations of Bell's inequality." *Letters in Mathematical Physics* **4**(2), 93–100. https://doi.org/10.1007/BF00417500

[26] Umegaki, H. (1954). "Conditional expectation in an operator algebra, I." *Tôhoku Mathematical Journal* **6**(2–3), 177–181. https://doi.org/10.2748/tmj/1178245177

[27] Takesaki, M. (1972). "Conditional expectations in von Neumann algebras." *Journal of Functional Analysis* **9**(3), 306–321. https://doi.org/10.1016/0022-1236(72)90004-3

[28] Onsager, L. (1931). "Reciprocal relations in irreversible processes. I." *Physical Review* **37**, 405–426. https://doi.org/10.1103/PhysRev.37.405

[29] Green, M. S. (1954). "Markoff random processes and the statistical mechanics of time-dependent phenomena. II." *Journal of Chemical Physics* **22**, 398–413. https://doi.org/10.1063/1.1740082 Kubo, R. (1957). "Statistical-mechanical theory of irreversible processes. I." *Journal of the Physical Society of Japan* **12**, 570–586. https://doi.org/10.1143/JPSJ.12.570

[30] Bombelli, L., Lee, J., Meyer, D., and Sorkin, R. D. (1987). "Space-time as a causal set." *Physical Review Letters* **59**, 521–524. https://doi.org/10.1103/PhysRevLett.59.521

[31] Surya, S. (2019). "The causal set approach to quantum gravity." *Living Reviews in Relativity* **22**, 5. https://doi.org/10.1007/s41114-019-0023-1

[32] Bombelli, L., and Meyer, D. A. (1989). "The origin of Lorentzian geometry." *Physics Letters A* **141**, 226–228. https://doi.org/10.1016/0375-9601(89)90474-X

[33] Brightwell, G., and Gregory, R. (1991). "Structure of random discrete spacetime." *Physical Review Letters* **66**, 260–263. https://doi.org/10.1103/PhysRevLett.66.260

[34] Major, S., Rideout, D., and Surya, S. (2007). "On recovering continuum topology from a causal set." *Journal of Mathematical Physics* **48**, 032501. https://doi.org/10.1063/1.2435599

[35] Major, S., Rideout, D., and Surya, S. (2009). "Stable homology as an indicator of manifoldlikeness in causal set theory." *Classical and Quantum Gravity* **26**, 175008. https://doi.org/10.1088/0264-9381/26/17/175008

[36] Glaser, L., and Surya, S. (2013). "Towards a definition of locality in a manifoldlike causal set." *Physical Review D* **88**, 124026. https://doi.org/10.1103/PhysRevD.88.124026

[37] Reid, D. D. (2003). "Manifold dimension of a causal set: Tests in conformally flat spacetimes." *Physical Review D* **67**, 024034. https://doi.org/10.1103/PhysRevD.67.024034

[38] Benincasa, D. M. T., and Dowker, F. (2010). "The scalar curvature of a causal set." *Physical Review Letters* **104**, 181301. https://doi.org/10.1103/PhysRevLett.104.181301

[39] Dowker, F., and Glaser, L. (2013). "Causal set d'Alembertians for various dimensions." *Classical and Quantum Gravity* **30**, 195016. https://doi.org/10.1088/0264-9381/30/19/195016

[40] Bombelli, L., Henson, J., and Sorkin, R. D. (2009). "Discreteness without symmetry breaking: A theorem." *Modern Physics Letters A* **24**, 2579–2587. https://doi.org/10.1142/S0217732309031958

[41] Hawking, S. W., King, A. R., and McCarthy, P. J. (1976). "A new topology for curved space–time which incorporates the causal, differential, and conformal structures." *Journal of Mathematical Physics* **17**, 174–181. https://doi.org/10.1063/1.522874

[42] Malament, D. B. (1977). "The class of continuous timelike curves determines the topology of spacetime." *Journal of Mathematical Physics* **18**, 1399–1404. https://doi.org/10.1063/1.523436

[43] Lamport, L. (1978). "Time, clocks, and the ordering of events in a distributed system." *Communications of the ACM* **21**(7), 558–565. https://doi.org/10.1145/359545.359563

[44] Lamport, L., Shostak, R., and Pease, M. (1982). "The Byzantine generals problem." *ACM Transactions on Programming Languages and Systems* **4**(3), 382–401. https://doi.org/10.1145/357172.357176

[45] Fischer, M. J., Lynch, N. A., and Paterson, M. S. (1985). "Impossibility of distributed consensus with one faulty process." *Journal of the ACM* **32**(2), 374–382. https://doi.org/10.1145/3149.214121

[46] Newman, M. H. A. (1942). "On theories with a combinatorial definition of equivalence." *Annals of Mathematics* **43**(2), 223–243. https://doi.org/10.2307/1968867

[47] Rovelli, C. (1996). "Relational quantum mechanics." *International Journal of Theoretical Physics* **35**, 1637–1678. https://doi.org/10.1007/BF02302261

[48] Connes, A., and Rovelli, C. (1994). "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories." *Classical and Quantum Gravity* **11**, 2899–2917. https://doi.org/10.1088/0264-9381/11/12/007

[49] Zurek, W. H. (2003). "Decoherence, einselection, and the quantum origins of the classical." *Reviews of Modern Physics* **75**, 715–775. https://doi.org/10.1103/RevModPhys.75.715

[50] 't Hooft, G. (2016). *The Cellular Automaton Interpretation of Quantum Mechanics*. Fundamental Theories of Physics 185. Springer, Cham. https://doi.org/10.1007/978-3-319-41285-6

[51] Frauchiger, D., and Renner, R. (2018). "Quantum theory cannot consistently describe the use of itself." *Nature Communications* **9**, 3711. https://doi.org/10.1038/s41467-018-05739-8

[52] Ratcliffe, J. G. (2019). *Foundations of Hyperbolic Manifolds*, 3rd ed. Graduate Texts in Mathematics 149. Springer, Cham. https://doi.org/10.1007/978-3-030-31597-9

[53] Serre, J.-P. (1977). *Linear Representations of Finite Groups*. Graduate Texts in Mathematics 42. Springer, New York. https://doi.org/10.1007/978-1-4684-9458-7

[54] Cohan, N. V. (1958). "The spherical harmonics with the symmetry of the icosahedral group." *Proceedings of the Cambridge Philosophical Society* **54**(1), 28–38. https://doi.org/10.1017/S0305004100033156

[55] Bisognano, J. J., and Wichmann, E. H. (1975). "On the duality condition for a Hermitian scalar field." *Journal of Mathematical Physics* **16**, 985–1007. https://doi.org/10.1063/1.522605

[56] Bisognano, J. J., and Wichmann, E. H. (1976). "On the duality condition for quantum fields." *Journal of Mathematical Physics* **17**, 303–321. https://doi.org/10.1063/1.522898

[57] Wiesbrock, H.-W. (1993). "Half-sided modular inclusions of von Neumann algebras." *Communications in Mathematical Physics* **157**, 83–92. https://doi.org/10.1007/BF02098019

[58] Jacobson, T. (1995). "Thermodynamics of spacetime: The Einstein equation of state." *Physical Review Letters* **75**, 1260–1263. https://doi.org/10.1103/PhysRevLett.75.1260

[59] Jacobson, T. (2016). "Entanglement equilibrium and the Einstein equation." *Physical Review Letters* **116**, 201101. https://doi.org/10.1103/PhysRevLett.116.201101

[60] van der Bij, J. J., van Dam, H., and Ng, Y. J. (1982). "The exchange of massless spin-two particles." *Physica A* **116**, 307–320. https://doi.org/10.1016/0378-4371(82)90247-3

[61] Ellis, G. F. R., van Elst, H., Murugan, J., and Uzan, J.-P. (2011). "On the trace-free Einstein equations as a viable alternative to general relativity." *Classical and Quantum Gravity* **28**, 225007. https://doi.org/10.1088/0264-9381/28/22/225007

[62] Van Raamsdonk, M. (2010). "Building up spacetime with quantum entanglement." *General Relativity and Gravitation* **42**, 2323–2329. https://doi.org/10.1007/s10714-010-1034-0

[63] Donnelly, W. (2012). "Decomposition of entanglement entropy in lattice gauge theory." *Physical Review D* **85**, 085004. https://doi.org/10.1103/PhysRevD.85.085004

[64] Casini, H., Huerta, M., and Rosabal, J. A. (2014). "Remarks on entanglement entropy for gauge fields." *Physical Review D* **89**, 085012. https://doi.org/10.1103/PhysRevD.89.085012

[65] Hall, B. C. (2015). *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Graduate Texts in Mathematics 222. Springer, Cham. https://doi.org/10.1007/978-3-319-13467-3

[66] Doplicher, S., and Roberts, J. E. (1989). "A new duality theory for compact groups." *Inventiones Mathematicae* **98**, 157–218. https://doi.org/10.1007/BF01388849

[67] Doplicher, S., and Roberts, J. E. (1990). "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics." *Communications in Mathematical Physics* **131**, 51–107. https://doi.org/10.1007/BF02097680

[68] Georgi, H., and Glashow, S. L. (1974). "Unity of all elementary-particle forces." *Physical Review Letters* **32**(8), 438–441. https://doi.org/10.1103/PhysRevLett.32.438

[69] Bouchiat, C., Iliopoulos, J., and Meyer, Ph. (1972). "An anomaly-free version of Weinberg's model." *Physics Letters B* **38**(7), 519–523. https://doi.org/10.1016/0370-2693(72)90532-1

[70] Tong, D. (2017). "Line operators in the Standard Model." *Journal of High Energy Physics* **07**, 104. https://doi.org/10.1007/JHEP07(2017)104

[71] Witten, E. (1982). "An SU(2) anomaly." *Physics Letters B* **117**, 324–328. https://doi.org/10.1016/0370-2693(82)90728-6

[72] Kostelecký, V. A., and Mewes, M. (2009). "Electrodynamics with Lorentz-violating operators of arbitrary dimension." *Physical Review D* **80**, 015020. https://doi.org/10.1103/PhysRevD.80.015020

[73] Kostelecký, V. A., and Russell, N. (2026). "Data tables for Lorentz and CPT violation." 2026 edition, arXiv:0801.0287. https://arxiv.org/abs/0801.0287. Original journal article: https://doi.org/10.1103/RevModPhys.83.11

[74] Pierre Auger Collaboration (2022). "Testing effects of Lorentz invariance violation in the propagation of astroparticles with the Pierre Auger Observatory." *Journal of Cosmology and Astroparticle Physics* **01**, 023. https://doi.org/10.1088/1475-7516/2022/01/023

[75] Xi, Y., and Shu, F.-W. (2025). "Constraints on Lorentz invariance violation from GRB 221009A using the DisCan method." *Chinese Physics C* **49**, 125101. https://doi.org/10.1088/1674-1137/adfa01

[76] LHAASO Collaboration (2024). "Stringent tests of Lorentz invariance violation from LHAASO observations of GRB 221009A." *Physical Review Letters* **133**, 071501. https://doi.org/10.1103/PhysRevLett.133.071501

[77] DESI Collaboration (2025). "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints." arXiv:2503.14738. https://arxiv.org/abs/2503.14738. Official chains and checksum manifest: https://data.desi.lbl.gov/public/papers/y3/bao-cosmo-params/README.html

[78] Filzinger, M., et al. (2023). "Improved Limits on the Coupling of Ultralight Bosonic Dark Matter to Photons from Optical Atomic Clock Comparisons." *Physical Review Letters* **130**, 253001. https://doi.org/10.1103/PhysRevLett.130.253001

[79] Mohr, P. J., Newell, D. B., Taylor, B. N., and Tiesinga, E. (2025). "CODATA recommended values of the fundamental physical constants: 2022." *Reviews of Modern Physics* **97**, 025002. https://doi.org/10.1103/RevModPhys.97.025002

[80] Takahashi, F., et al. (Particle Data Group) (2026). "Review of Particle Physics." *International Journal of Modern Physics A* **41**, 2630011. https://doi.org/10.1142/S0217751X26300115. 2026 particle listings: https://pdg.lbl.gov/2026/listings/particle_properties.html

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-5.6-Sol, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.